

AKADEMIKA

ACT-01, ACT-02, ACT-03, ACT-04

**EXPERIMENTAL & SERVICE
MANUAL**

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ANALOG COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. This means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training product. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook at hand for any further help.

Set all accessories in order on unpacking, and check integrity with respect to its checklist. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In ACT-01, ACT-02, ACT-03 and ACT-04 kits, signals are to be monitored with an oscilloscope, the scope input channel should be a.c. coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

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These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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- VCO Signals
- Audio Preamplifier
- Sweep Generator
- Spectrum Detector (RF Detector)
- Switch Faults

ACT-02

- Balanced Modulator
- Ring Modulator
- Ceramic Filter
- Tunable Filter
- Low Pass Filter
- Switch Faults

ACT-03

- Frequency Converter
- 2 stage Selective Amplifier
- AM-Diode Detector
- Automatic Gain Control
- Switch Faults

ACT-04

- Phase Modulator
- Limiter
- Automatic Frequency Control
- Foster Seeley & Ratio Detector
- Product Detector/Quadrature Detector
- Switch Faults

TECHNICAL SPECIFICATIONS

ACT-01

On Board Function Generator

Waveform	: Sine, Triangular, and Square.
Amplitude	: 0-2Vpp
Frequency Range	: 100Hz to 10 KHz.
Synchronous Square Wave	: Frequency x4 and /4 of the main Signal, Adjustable Amplitude

Voltage Control Oscillator

a) VCO1 Carrier Frequency generator	
Frequency Range	: i) 400 KHz to 1500 KHz ii) 400 KHz to 500 KHz
Amplitude	: Variable 0-2Vpp.
Out Impedance	: 50 Ω
Input to VCO1	: Modulating Signal, Marker, AFC (Automatic Frequency Control)
b) VCO2 Carrier Frequency Generator	
Frequency Range	: 400 KHz to 1500 KHz.
Amplitude	: Variable: 0 to 2Vpp.
Out Impedance	: 50 Ω
Input to VCO1	: Modulating signal.

Audio Preamplifier

Audio Input	: Microphone
Input Impedance	: 600 Ω .
Voltage Gain	: 1 to 100.
RC Filter	: 3.4 KHz cutoff.
Maximum Output Voltage	: 5V.

Sweep Generator

Sweep Frequency	: 10Hz.
Sweep Depth	: Adjustable
Output for Oscilloscope	: X-axis.

Spectrum Detector

Minimum Input	: 100mvpp.
Input Level	: Adjustable
Interconnection	: 2mm Banana Socket.
Power Supply	: GND, +5V, +12V, -12V.

Interconnection

With The Help Of 2mm Connecting Chords Provided.

Test Points

Various Test Points Are Provided Onboard To Observe Signals On Oscilloscope.

Switch Faults

Switch Fault Section Provides 4 Switches. These Switches Can Be Used To Simulate Fault Conditions In Various Parts Of The Circuit.

Deliverables	: Mini Links	: 06 Nos.
	Short Links	: 06 Nos
	Long Links	: 06 Nos
	Microphone	: 01 No
	Power Supply	: 01 No.
	1.5 V Battery (Pencil cell)	
	AA Size	: 01 No
	ACT-01, ACT-02, ACT-03, ACT-04	
	Power Card	: 01 No
	Jumper Caps	: 10 No
	Power Chord	: 01 No
	Screw Driver	: 01 No

ACT-02**Balance Modulators****2 Nos Balance Modulator for DSB/SSB Amplitude Modulation**

Carrier Input	: 1 Khz – 1000 Khz
Modulating Input	: 0.1 Khz – 100 Khz
Carrier Null	: Adjustable.
Output Amplitude	: Adjustable.

Ring Modulator**Diode Balance Modulator with AM DSB Carrier (USB & LSB)**

Carrier Input	: 4 Khz – 20 Khz
Modulating Input	: 0.1 Khz – 4 Khz

Ceramic Filter

Central Frequency	: 455 Khz.
Bandwidth	: 3 ± 1 Khz (Adjustable)

Tunable Filter

Central Frequency	: Adjustable From 2 Khz To 8khz.
Detector Type	: Diode Envelope.

Low Pass Active Filter

Cutoff Frequency : 3.4 Khz.

Interconnections

With The Help of 2mm Connecting Chords Provided.

Test Points

Various Test Points Are Provided Onboard To Observe Signals On Oscilloscope.

Switch Faults

Switch Fault Section Provides 4 Switches. These Switches Can Be Used To Simulate Fault Conditions In Various Parts Of The Circuit.

Power Supply : GND, +5V, +12V, -12V.

ACT-03

Frequency Converter (Mixer)

Dual Gate MOSFET Used To Mix-up RF and LO Frequency.

Inputs	: Local Oscillator and RF Signal.
Output Intermediate Frequency	: 455 Khz Adjustable.
Fi Filter	: Dual Tuned LC.

1st And 2nd Stage Amplifier

Central Frequency	: 455khz Adjustable.
Load Impedance	: Variable R-L-C.
Gain	: 34db.

Impedance Matching With Transformer and Autotransformer

Diode Envelope Detector

Detection of the Positive or Negative Envelope with Variable R-C Filter

Automatic Gain Control

Output can be controlled in case of maximum input signal.

Interconnections

With the help of 2mm connecting chords provided.

Test Points

Various test points are provided onboard to observe signals on oscilloscope.

Switch Faults

Switch fault section provides 4 switches. These switches can be used to simulate fault conditions in various parts of the circuit.

Power Supply : GND, +5V, +12V, -12V.

ACT-04

Phase Modulator

Modulation Technique : FET Amplifier with Resonant Load.

Operating Frequency : 400 Khz To 500 Khz

Limiter

Operating Frequency : 455 KHZ

Input Amplitude : From 0.5 to 5Vpp.

Automatic Frequency Control

Frequency Of FM Modulator Can Be Controlled With Integrative Effect.

Foster-Seeley Demodulator

Operating Frequency : 400 Khz To 500 Khz

Input Amplitude : 1 Vpp

Ratio Detector

Operating Frequency : 400 Khz To 500 Khz

Input Amplitude : 1Vpp.

Phase Detector and FM Quadrature Detector

Phase Shifter, Product Detector and RC Filter.

Operating Frequency : 400 Khz To 500 Khz

Input Amplitude : 1Vpp

Interconnections

With The Help Of 2mm Connecting Chords Provided.

Test Points

Various Test Points Are Provided Onboard To Observe Signals On Oscilloscope.

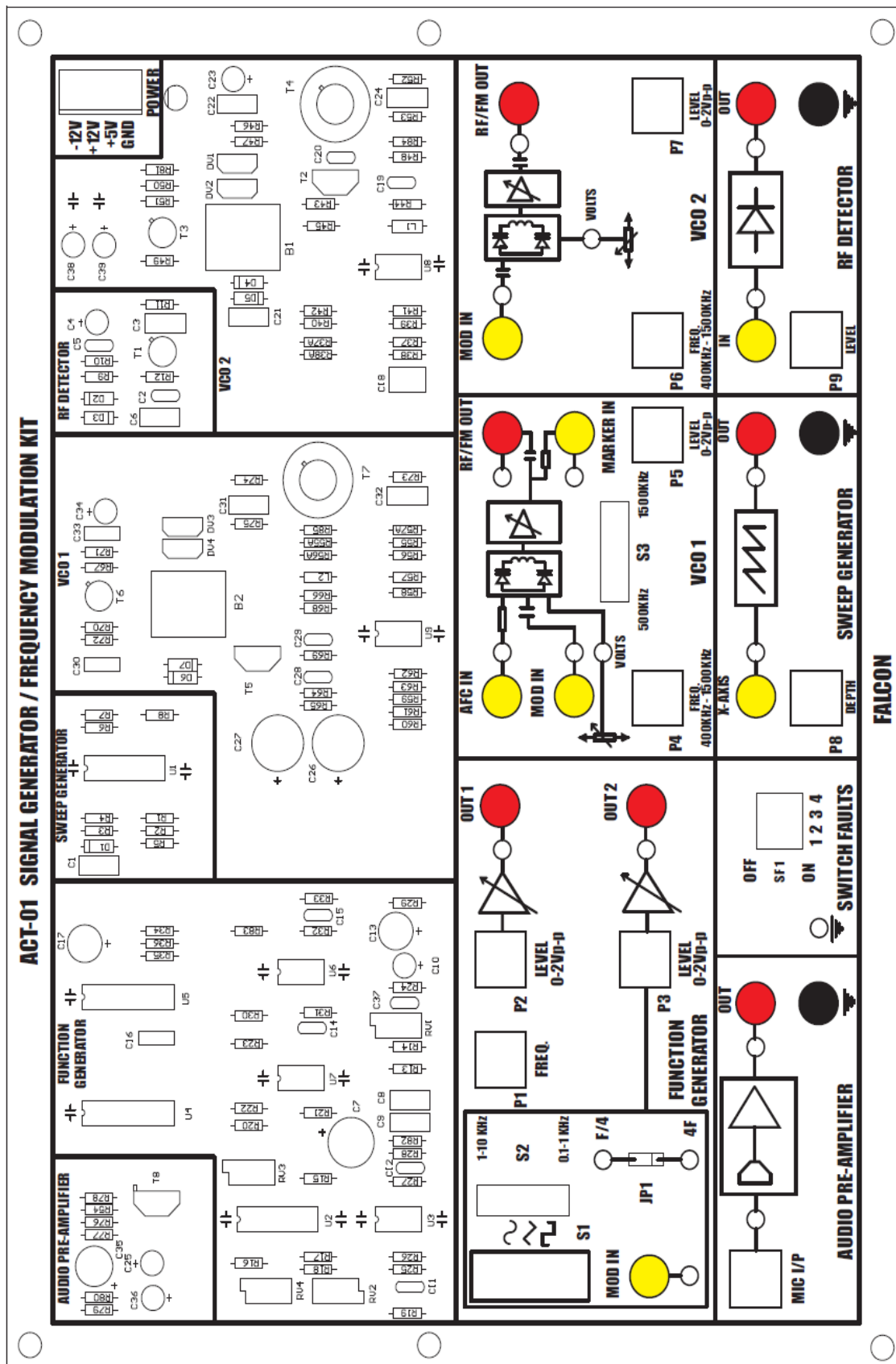
Switch Faults

Switch Fault Section Provides 4 Switches. These Switches Can Be Used To Simulate Fault Conditions In Various Parts Of The Circuit.

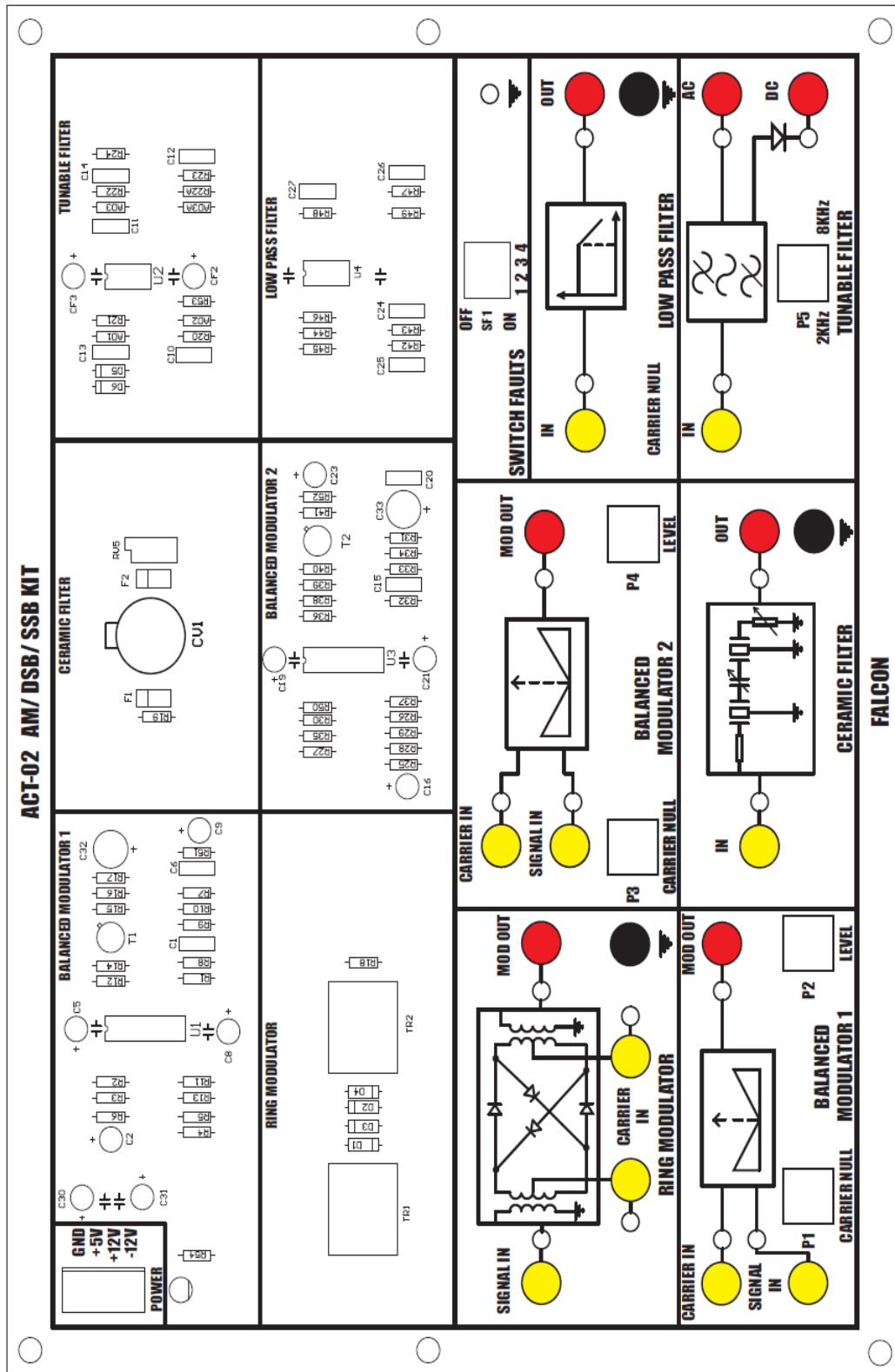
Power Supply : GND, +5V, +12V, -12V.

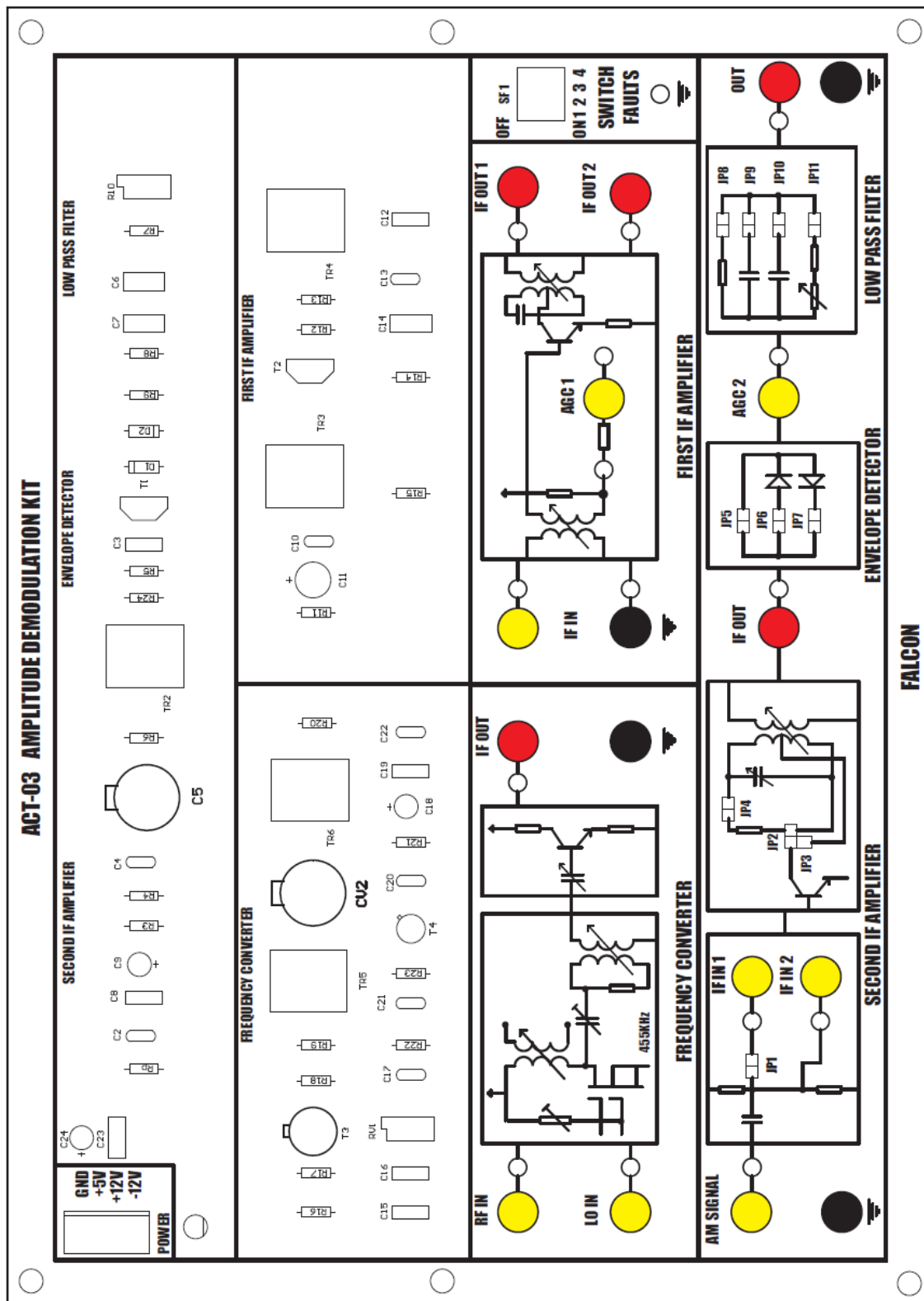
FUNCTIONAL BLOCK

ACT-01

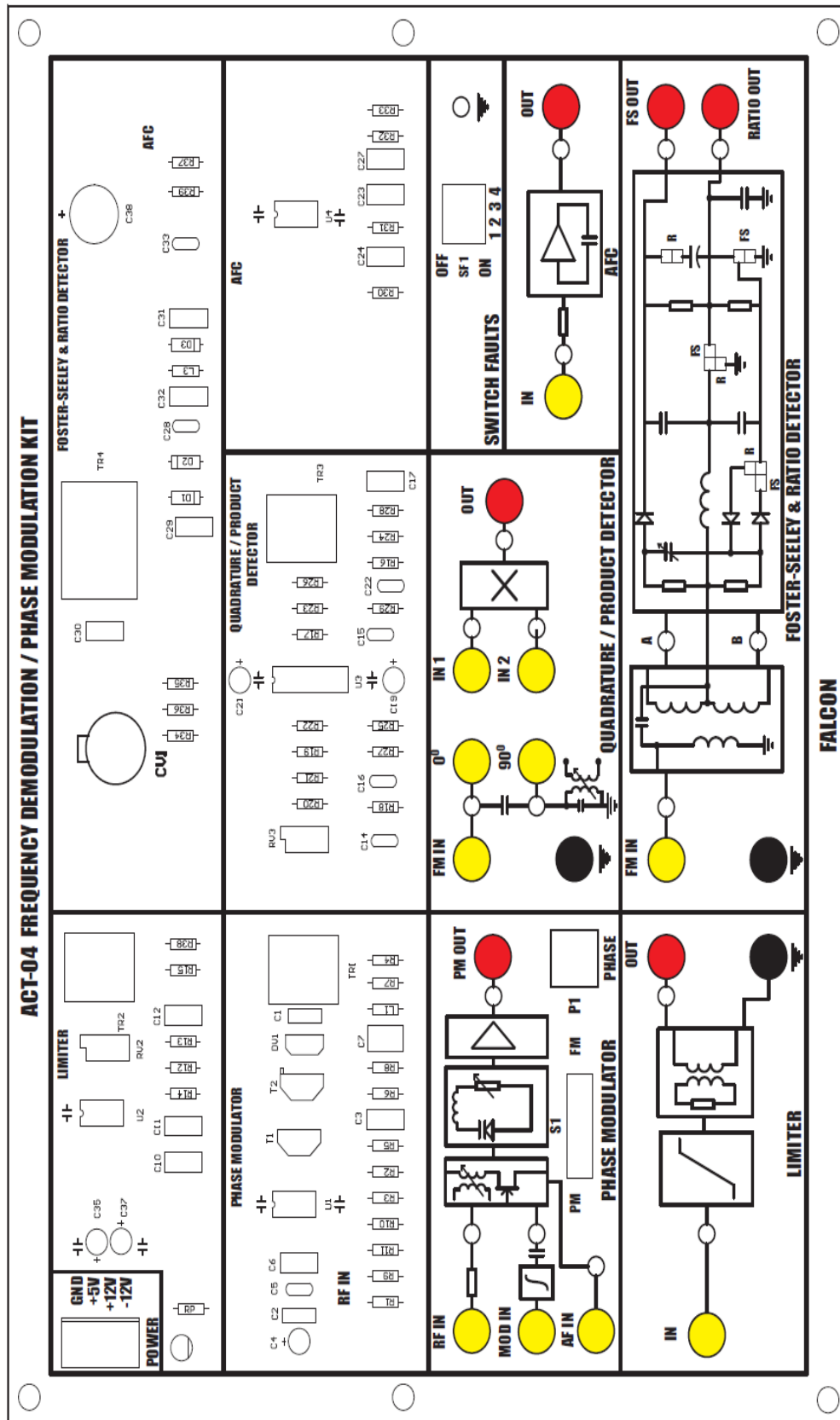


ACT-02



ACT-03

ACT-04



INTRODUCTION

ACT-01

1.1 Function Generator

The integrated circuit U1 (ICL8038) generates sine (pin 2) and triangular waveform (pin 3). The capacitors which can be connected to pin 10 enable to select the range (1 or 10 KHz max). The frequency is varied with the pot RV1 FREQ. (RV5 limits the inferior frequency). The trimmers RV3 and RV4 adjust the symmetry and the form of the sine signal.

With the jumpers JP1 and JP2, the sine and the triangular signals are applied to the amplifier U2 (CA3140) and then to the output. The pot RV6 varies the signal amplitude.

With JP3 the square signal can be applied to the output amplifier.

Frequency range can be selected with help of switch Sw1 and can be observed at OUT post.

1.2 Sweep Generator

The integrated circuits IC1A/B generate a saw-tooth signal whose frequency is about 10 Hz and amplitude about 10 Vpp. This signal is available with fixed amplitude across point 1 (X AXIS), and with variable amplitude across point 2 (OUT).

1.3 RF Detector

The circuit consists in a transistor amplifier followed by a peak diode detector and a low pass output filter RC.

1.4 VCO1 and VCO2

VCO1

The VCO is based on an Hartley oscillator consisting of: transistor BF241; primary of the transformer in parallel to the varicap diodes; reaction between emitter and collector of the transistor carried out with a capacitor in series to a resistor.

The signal is taken from the secondary of the transformer and amplified by a 2-transistor stage. The trimmer RV13 adjusts the amplitude of the signal from 0 to 2Vpp. across the input MARKER IN it is possible to add an external signal to the signal generated by the VCO to be used as frequency reference.

The frequency of the VCO is varied with the trimmer RV12, which supplies the bias voltage to the Varicaps (from the low frequency amplifier IC9LM741). A switch selects the range (500 or 1500 kHz max.). A low frequency signal for FM modulation of the VCO, and a D.C. voltage used to carry out an automatic frequency circuit can be applied across the input of an amplifier.

VCO2

The circuit is analogous to the one of the VCO1.

1.5 Audio Preamplifier

MIC. IN Post (13) gives Audio signal to base of transistor BC548. Transistor BC548 & other circuitry will amplify it and output is available at post (14).

ACT-02

2.1 Balanced Modulator

The integrated circuit LM1496 consists in a quadruple differential amplifier, driven by a further differential stage. The output signal consists in a constant which multiplies the product of the two input signals CARRIER and SIGNAL. The trimmer NULL CARRIER enables the balancing or unbalancing of the circuit: in the first case the output is exactly the product of the input signals, and the circuit operates as balanced modulator; in the second case the output contains a fixed component of the signal CARRIER and the circuit operates as amplitude, which is taken from the transistor emitter.

2.2 Ring Modulator

The carrier signal is applied between the central sockets of the two transformers (points 6-7), the modulating signal on the primary of the first transformer (points 4-5). The jumpers J1-J2 enable the insertion of a couple of diodes in order to switch from a balanced modulator (2 diodes) to a ring modulator operation, to simulate a non proper operation of the circuit.

2.3 Ceramic Filter

The filter consists of 2 sections coupled between them with a capacity RV5 is the (adjustable) load resistor.

2.4 Balanced Modulator II

The circuit is analogous to the one of the BALANCED MODULATOR I

2.5 Tunable Filter

From the principle diagram you can note that the filter essentially consists of 3 sections: a selector circuit L-C; an operational amplifier; a diode amplitude detector. The inductance L of the selector circuit can be simulated with the circuit, where RV6 enables the variation of the value of the simulated inductance and consequently the central frequency of the same filter.

2.6 Low Pass Filter

This is a classic low pass active filter type VCVS (Voltage-Controlled Voltage Source), with cutting frequency of 3400Hz. As the filter is of the 4th order, its slope is 24dB/octave (80dB/decade).

ACT-03

3.1 Frequency Converter

The mixer is composed of three sections : a MOSFET, carrying out the interaction between the LO (Local Oscillator) and the RF signal; a low pass filter centered at

455 kHz, consisting of two tuned circuits coupled between them by a variable capacitor; an output amplifier stage.

The coils of the tuned circuits can be accessed from the rear side and enable the calibration of the filter also at frequencies different from 455 kHz (between 400 and 500 kHz about). The coupling capacity enables to widen or reduce the band pass of the same filter.

3.2 First If Stage – Second If Stage

With the jumpers and the connections, the two circuits can be configured in different ways, to carry out the experiments described in experimental manual.

ACT-04

4.1 Phase Modulator

The actual modulator consists of a FET amplifier with resonant load; the low frequency modulating signal varies, with the Varicap diode, the central frequency of the circuit (455KHz when the modulating signal is lacking). This introduces a phase variation in the RF signal applied to the input. The operational amplifier, when inserted integrates the low modulator.

4.2 Limiter

The integrated circuit CA3028 has a differential amplifier (T1 and T2) inside powered by a constant current source (T3).

In the used configuration, the integrator carries out an amplifier with emitter coupling, consisting of an emitter follower (T1) with a common base (T2) in cascade. The output is taken on the collector of T2, with a circuit tuned at 455KHz and a transformer. The circuit offers very good performances as amplitude limiters, as the saturation of the transistors T1 and T2 is avoided by the action of the constant source T3.

4.3 Automatic Frequency Control – AFC

It is an integrator circuit, used in the automatic frequency control to integrate the continuous voltage proportional to the frequency error.

4.4 Foster-Seeley & Ratio Detector

With the jumpers, the circuit can be configured so as to select a Foster-Seeley Detector or a Ratio Detector (see the exercises in operational manual).

4.5 Product Detector

Which includes: a phase shifter circuit; a product detector (LM1496); a low pass filter type RC across the output of the detector

The product detector multiplies the two input signals between them and can be used alone, to demodulate the signals modulated in phase.

In combination with the shifter circuit there is a Quadrature Detector, which can be used to demodulate the signals modulated in frequency.

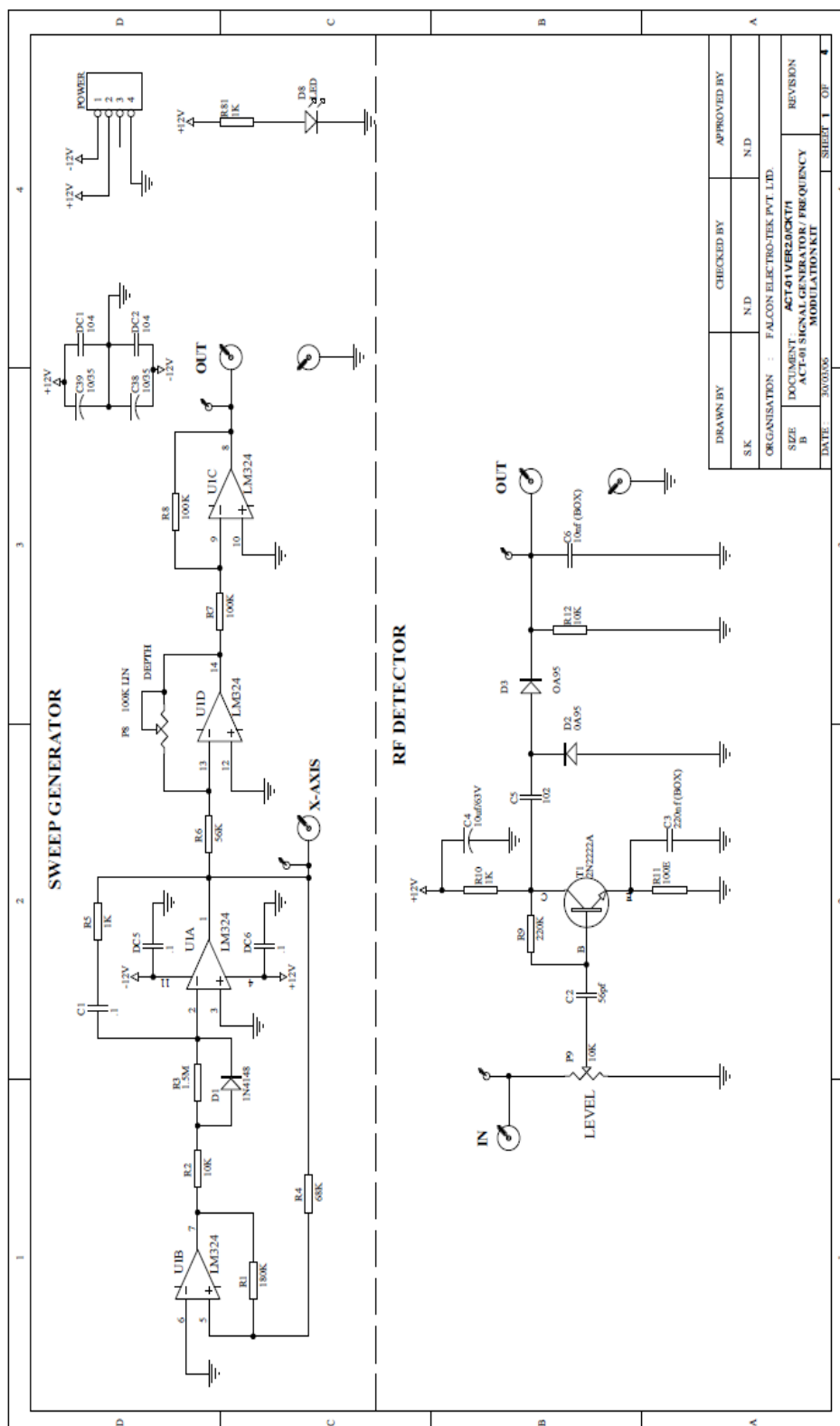
SWITCH FAULTS

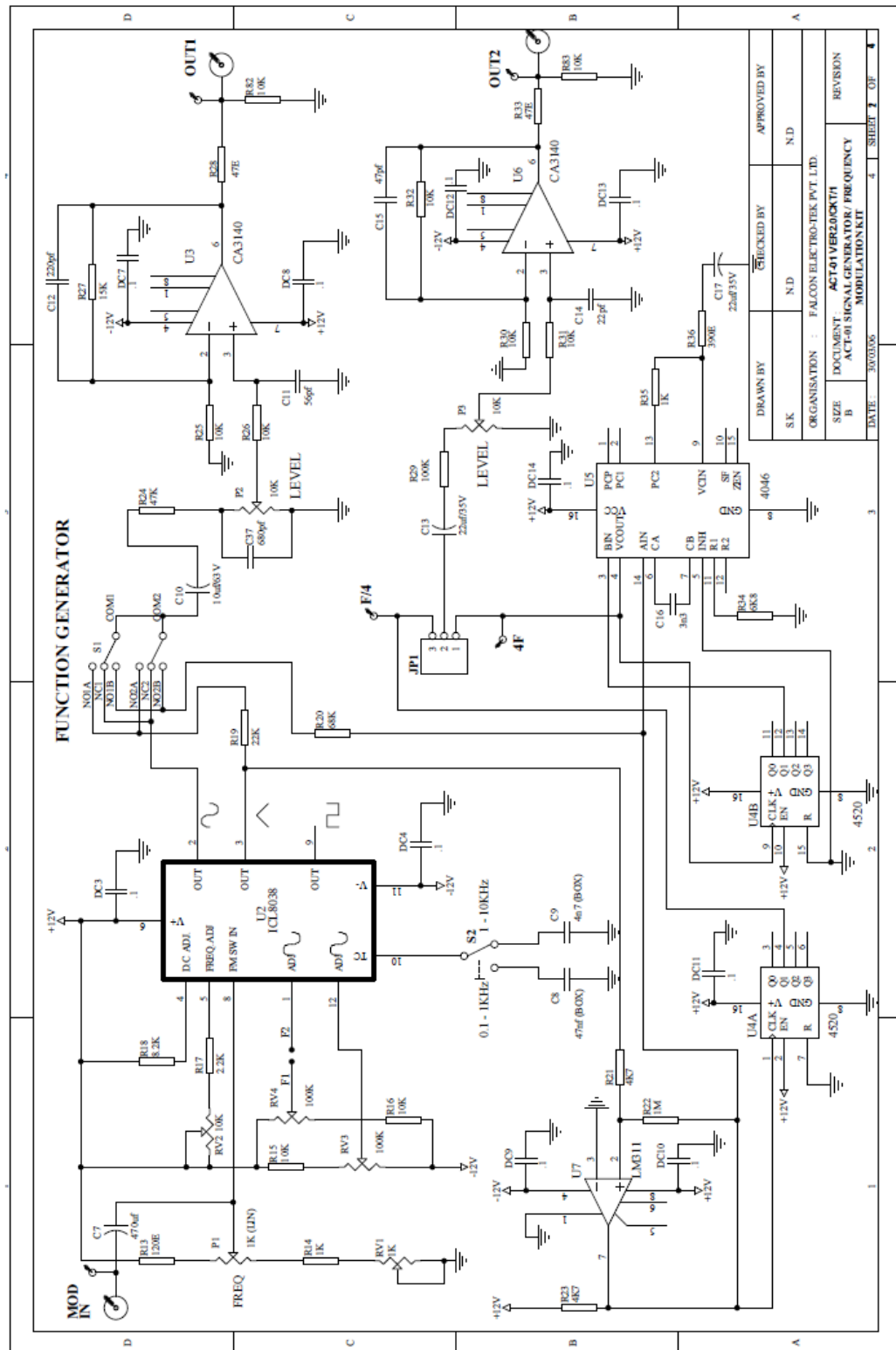
Switch fault section provides 4 switches each on ACT-01, ACT-02, ACT-03, and ACT-04. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the ACT kits are fully operational all switches should be set to OFF position before use.

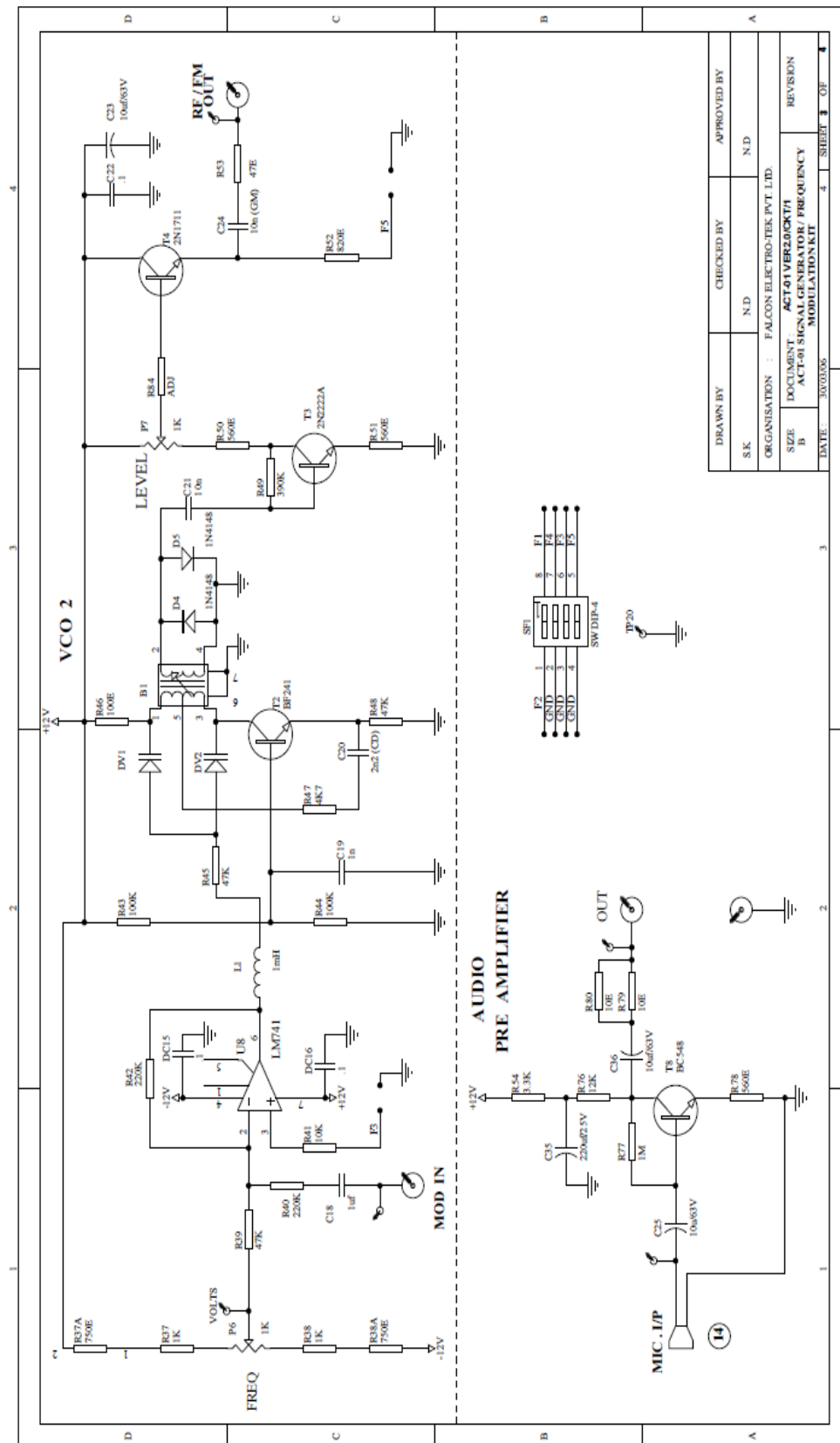
- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 1 of SF 1 in ACT-02 Switch fault section to ON position. This will open D4 from ring modulator; Ring modulator operates in three-diode configuration.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 3 of SF 1 in ACT-02 Switch fault section to ON position. This will open R30 from gain adjust pins of balance modulator 2. Balance modulator2 maximum output reduced to 100 mV from 2V.
- Put switch 4 of SF 1 in ACT-02 Switch fault section to ON position. This will open filter capacitor C24 from low pass filter, filter output gets distorted.
- Put switch 1 of SF 1 in ACT-03 Switch fault section to ON position. This will open C17 from FET from frequency converter section. Frequency converter output gets reduced this will disturb the next section output.
- Put switch 2 of SF 1 in ACT-03 Switch fault section to ON position. This will open C7 from 2nd IF section low pass filter, filter output gets distorted.
- Put switch 3 of SF 1 in ACT-03 Switch fault section to ON position. This will open R41 Bias resistor of Transistor T2 from GND side in 1st IF section. 1st IF section output reduces.
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600KHz.
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.
- Put switch 2 of SF 1 in ACT-04 Switch fault section to ON position. This will open R38 from Limiter. Limiter action is disabled.
- Put switch 3 of SF 1 in ACT-04 Switch fault section to ON position. This will open TR3 from Product detector, 90° output disturbed Put switch 4 of SF 1 in ACT-04 Switch fault section to ON position. This will open DV1's GND side in phase modulator, no variation at phase modulator output and amplitude is very low.

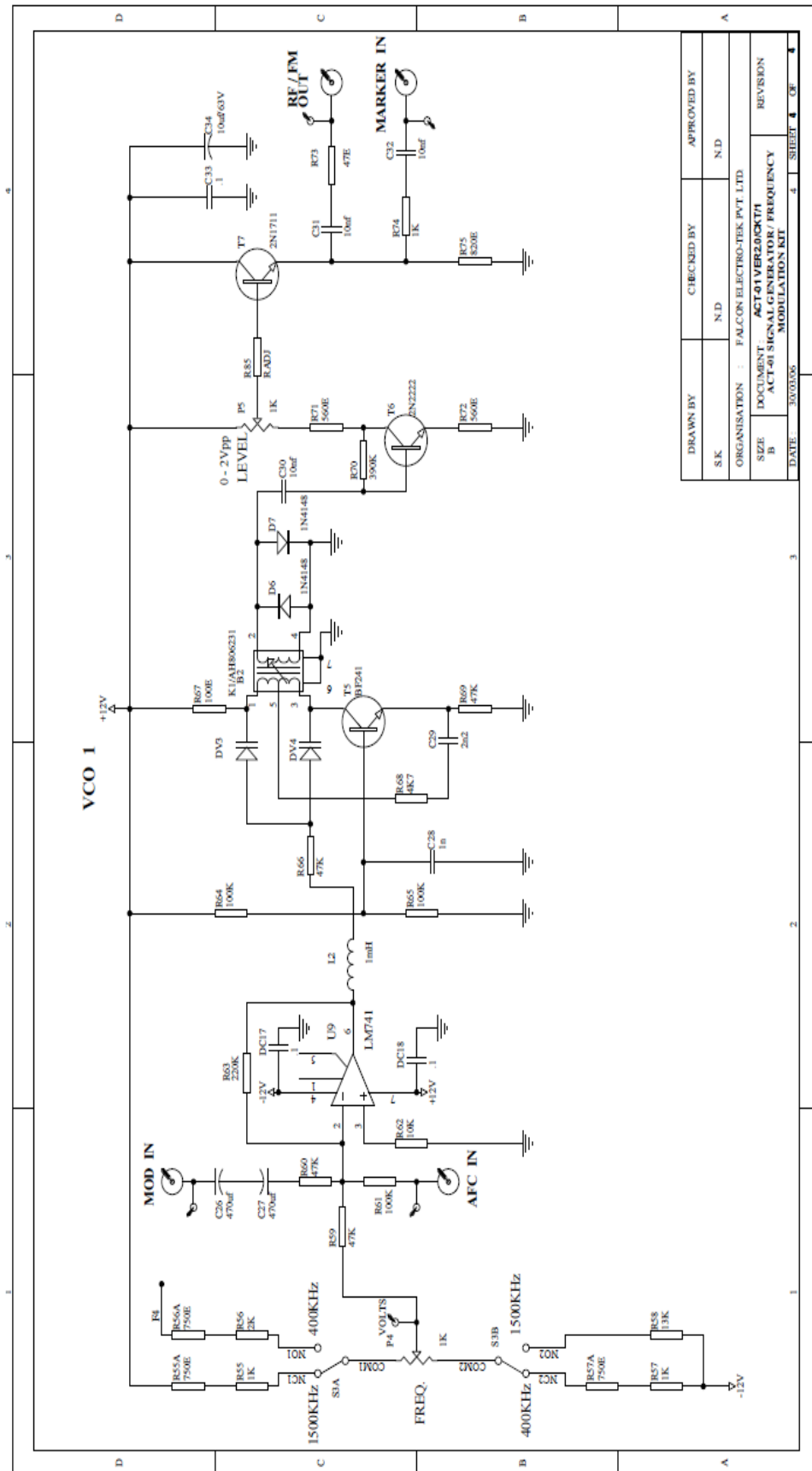
CIRCUIT DIAGRAM

ACT-01

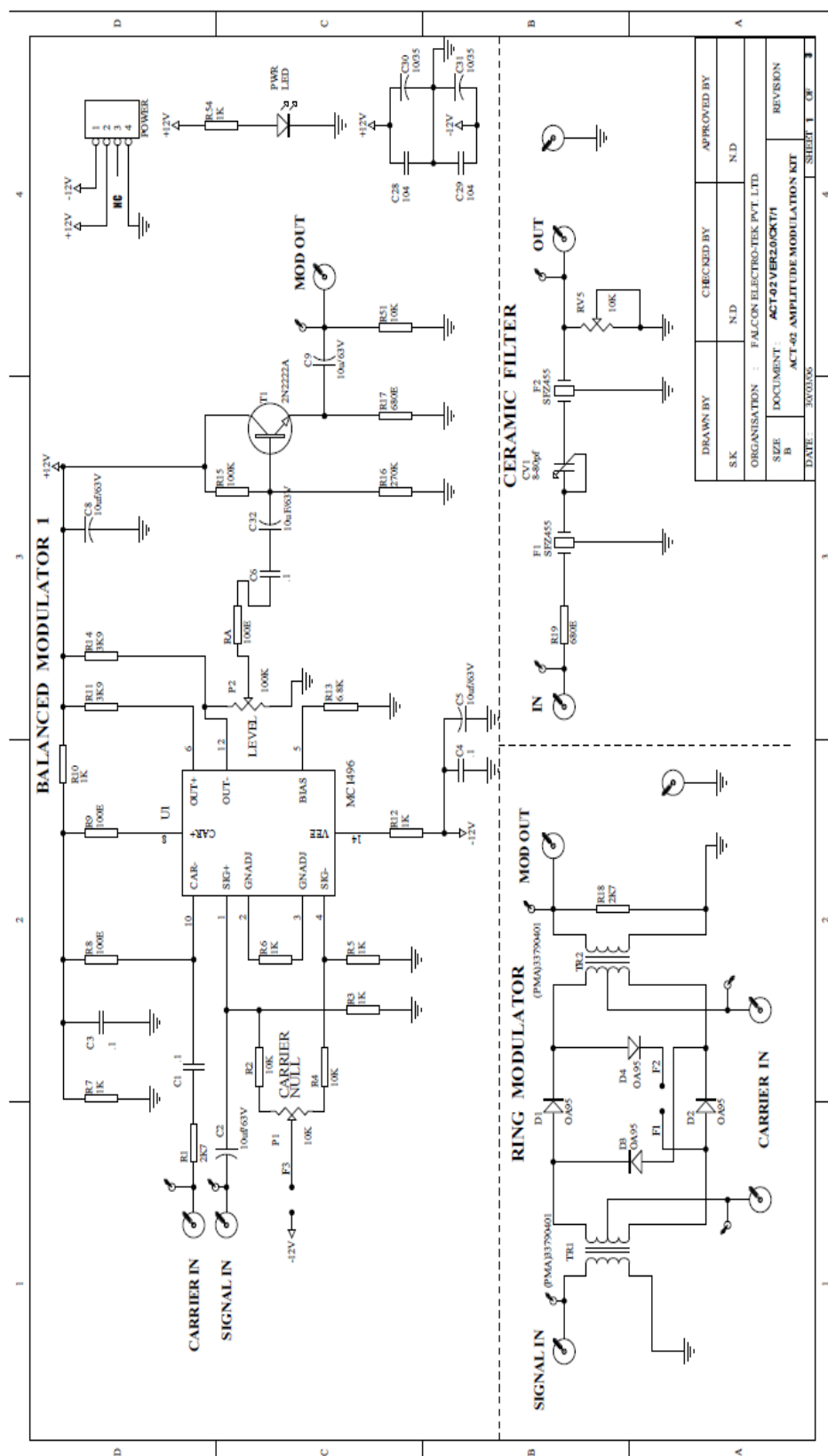


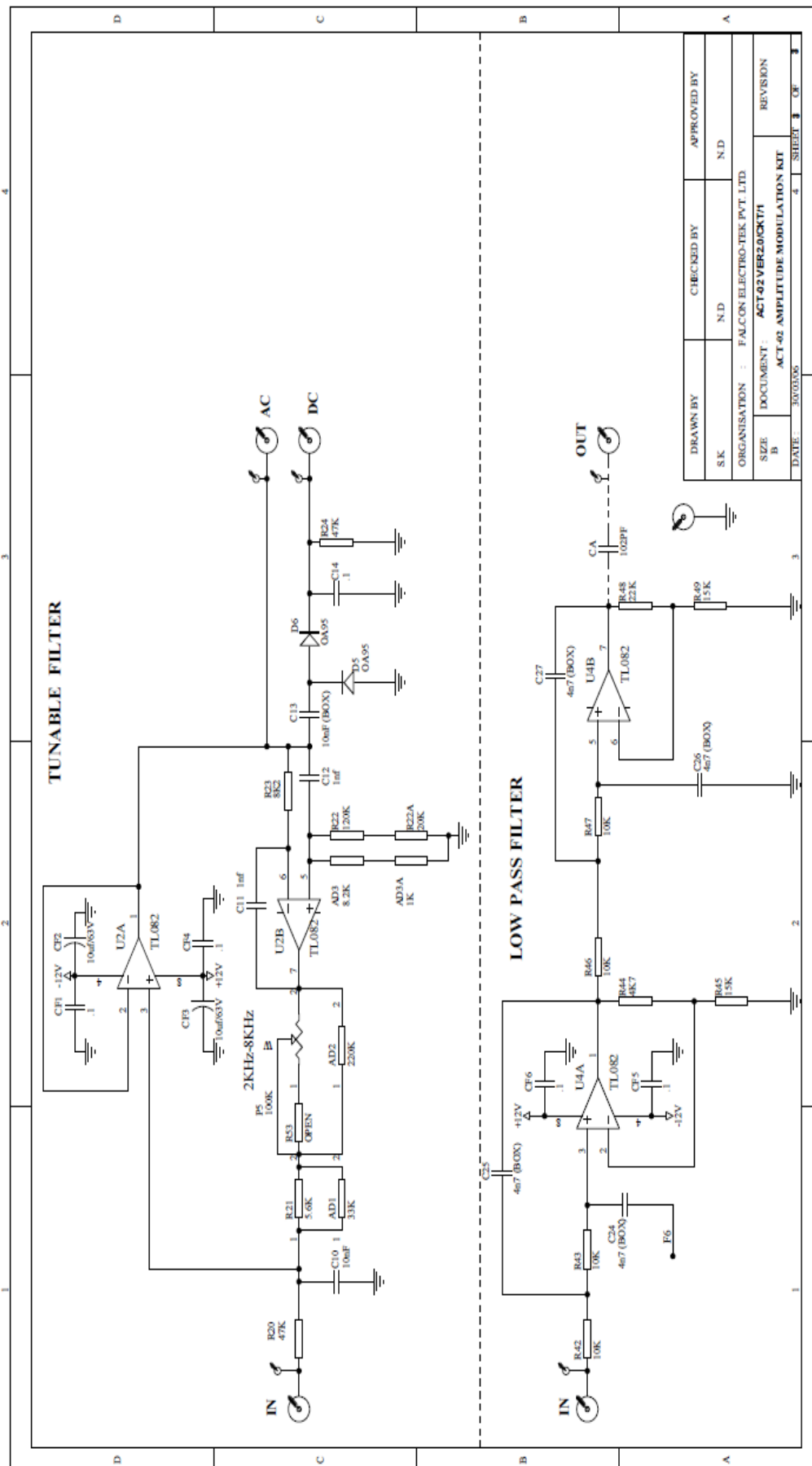


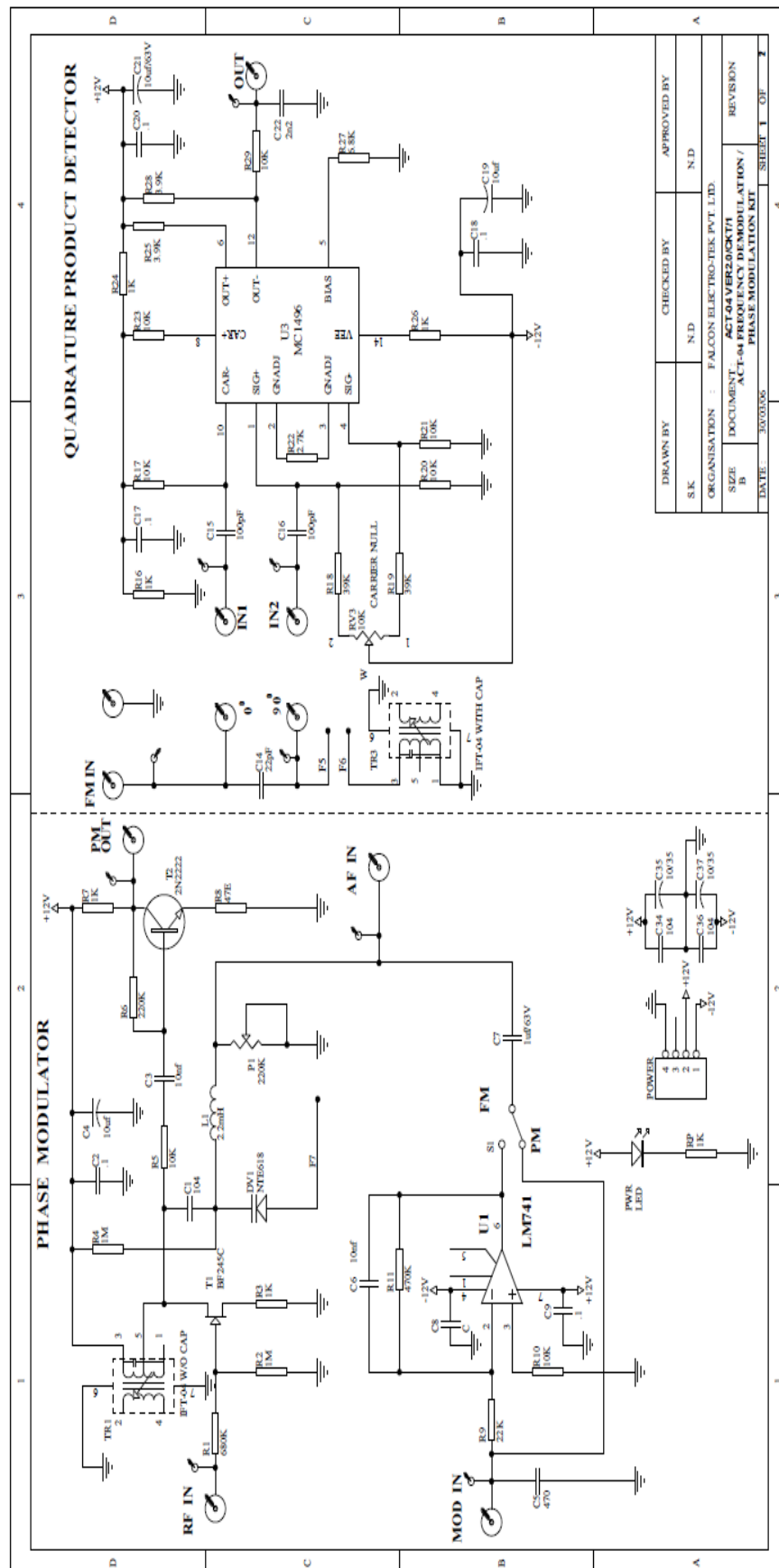


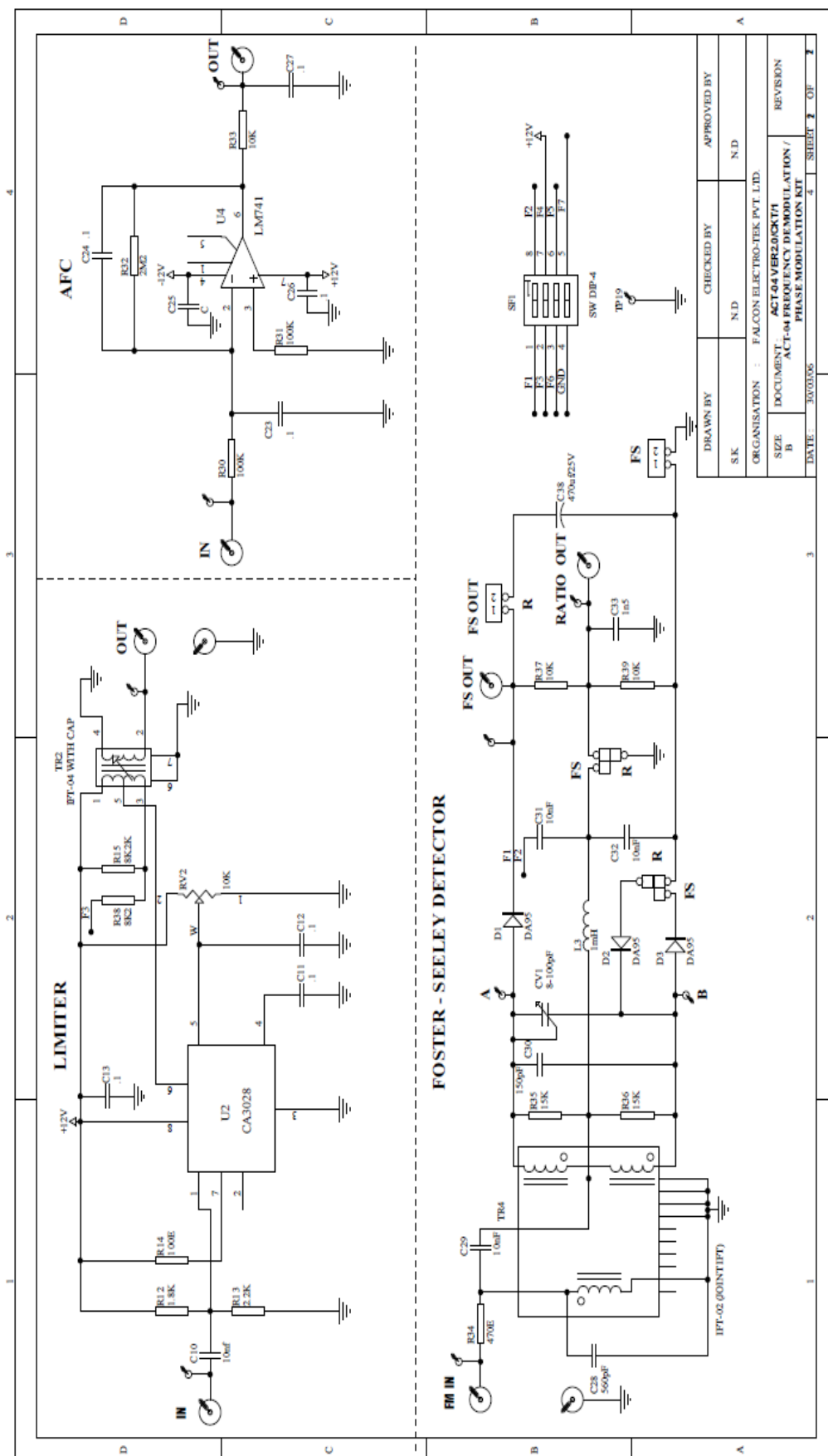


ACT-02

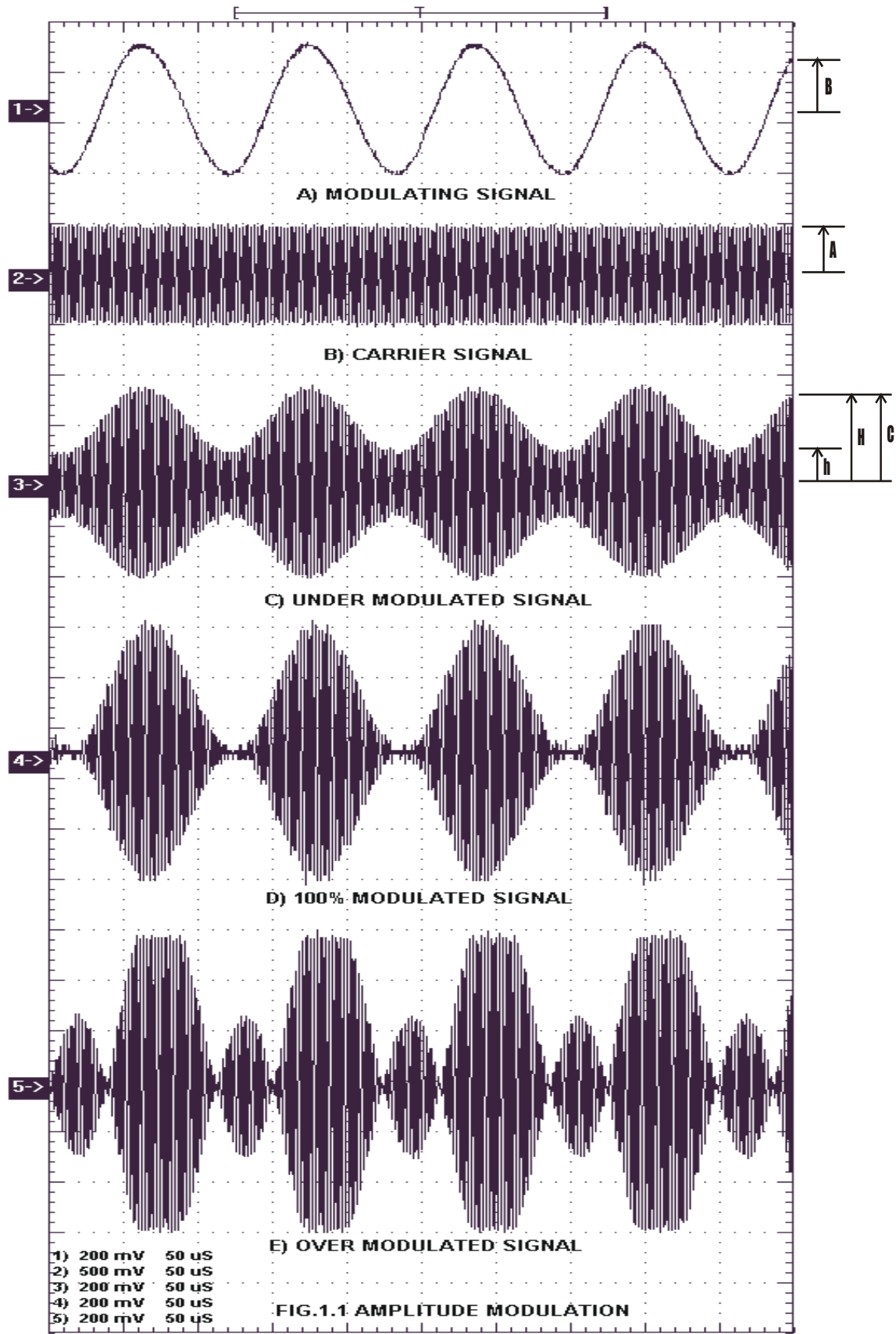




ACT-04



EXPERIMENT NO.1



EXPERIMENT NO.1

NAME:-

Study of Amplitude Modulation

OBJECTIVE:-

- A To study the operation of Amplitude Modulator.
- B To calculate the modulation index of Amplitude modulated wave.
- C To observe the linearity curve of the modulator.
- D To observe the spectrum of AM signal.

THEORY:-

Amplitude Modulation

It is a type of modulation in which the amplitude of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal.

Consider a sine wave signal $v_m(t)$ with frequency f (fig. 1.1)

$$V_m(t) = B \cdot \sin(2\pi f \cdot t)$$

And another sine wave signal $V_c(t)$ with frequency F .

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

In this signal $v_m(t)$ is called modulating signal and the signal $v_c(t)$ is called carrier signal.

Vary the amplitude of the carrier $v_c(t)$ after adding the modulating signal $v_m(t)$ to A.

You obtain a signal $v_M(t)$ amplitude modulated, which can be expressed by:

$$v_M(t) = [A + k \cdot B \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t) = A \cdot [1 + m \cdot \sin(2\pi f \cdot t)] \cdot \sin(2\pi F \cdot t)$$

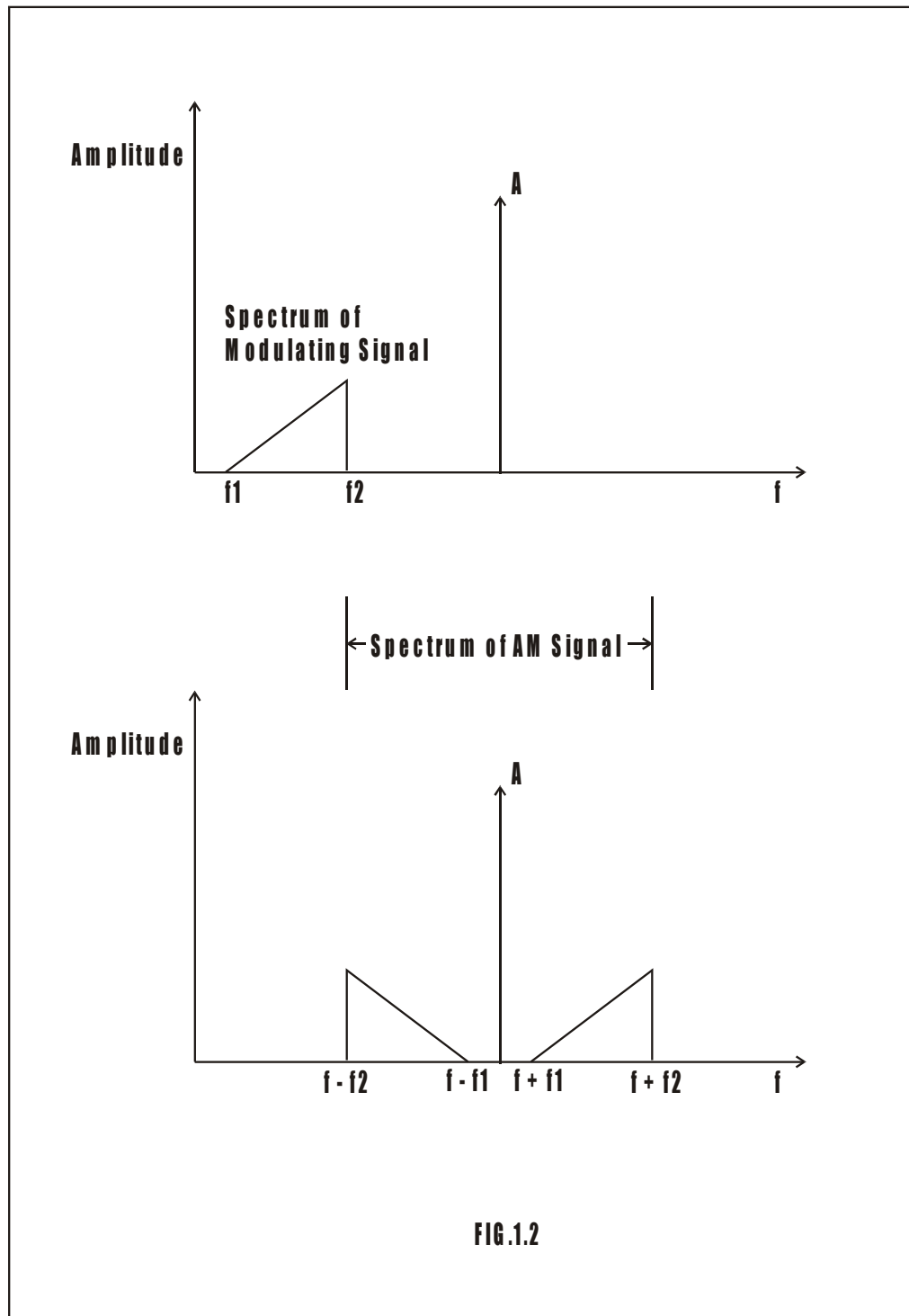
With k = constant of proportionality.

Percentage modulation signal is defined as the value:

$$m = \frac{K \cdot B}{A} \cdot 100$$

With reference to fig. 1.1c, the modulation index m can be calculated in this way

$$m = \frac{H - h}{H + h} \cdot 100\%$$



Spectrum of the Modulated Signal

With simple trigonometric passages, the relation expressing the modulated signal v_M becomes:

$$v_M(t) = A \sin(2\pi F \cdot t) + m \frac{A}{2} \cos[(2\pi(F-f) \cdot t)] - m \frac{A}{2} \cos[(2\pi(F+f) \cdot t)]$$

From which we can conclude that the signal modulated by amplitude of a sine wave consists of three sine components:

$$A \sin(2\pi F \cdot t)$$

carrier

$$m \frac{A}{2} \cos[(2\pi(F-f)) \cdot t] \quad \text{lower side band}$$

$$m \frac{A}{2} \cos[(2\pi(F+f)) \cdot t] \quad \text{upper side band.}$$

Power of the Modulated Signal

The total power of an AM signal is the sum of power contributed by carrier and the two bands.

Considering sine wave as modulating signal and a load resistance R , the different Components supply the following powers:

$$P_C = A^2 / 2 \cdot R \quad \text{power associated to the carrier}$$

$$P_L = (m \cdot A)^2 / 8 \cdot R \quad \text{power associated to the lower side band.}$$

$$P_U = (m \cdot A)^2 / 8 \cdot R \quad \text{power associated to the lower side band.}$$

It is important to note that:

- The power associated to the carrier is fixed and does not depend on the modulation.
- The power associated to each side band depends on the index of modulation, and reaches maximum at 25% of the power of the carrier (50% the two side bands together).

Non-Sinusoidal Modulating Signals Spectrum

Consider a modulating signal not constituted by a single sine wave, such as a generic signal with frequency spectrum ranging between f_1 and f_2 . With the amplitude modulation this spectrum is moved over and under the carrier (fig. 1.2)

It is evident that larger the spectrum of the modulating signal, larger is the band B_W occupied by the modulating signal. The resulted B_W is equal to the double of the modulating signal:

$$B_W = 2 \cdot f_2.$$

Amplitude Modulation Generation

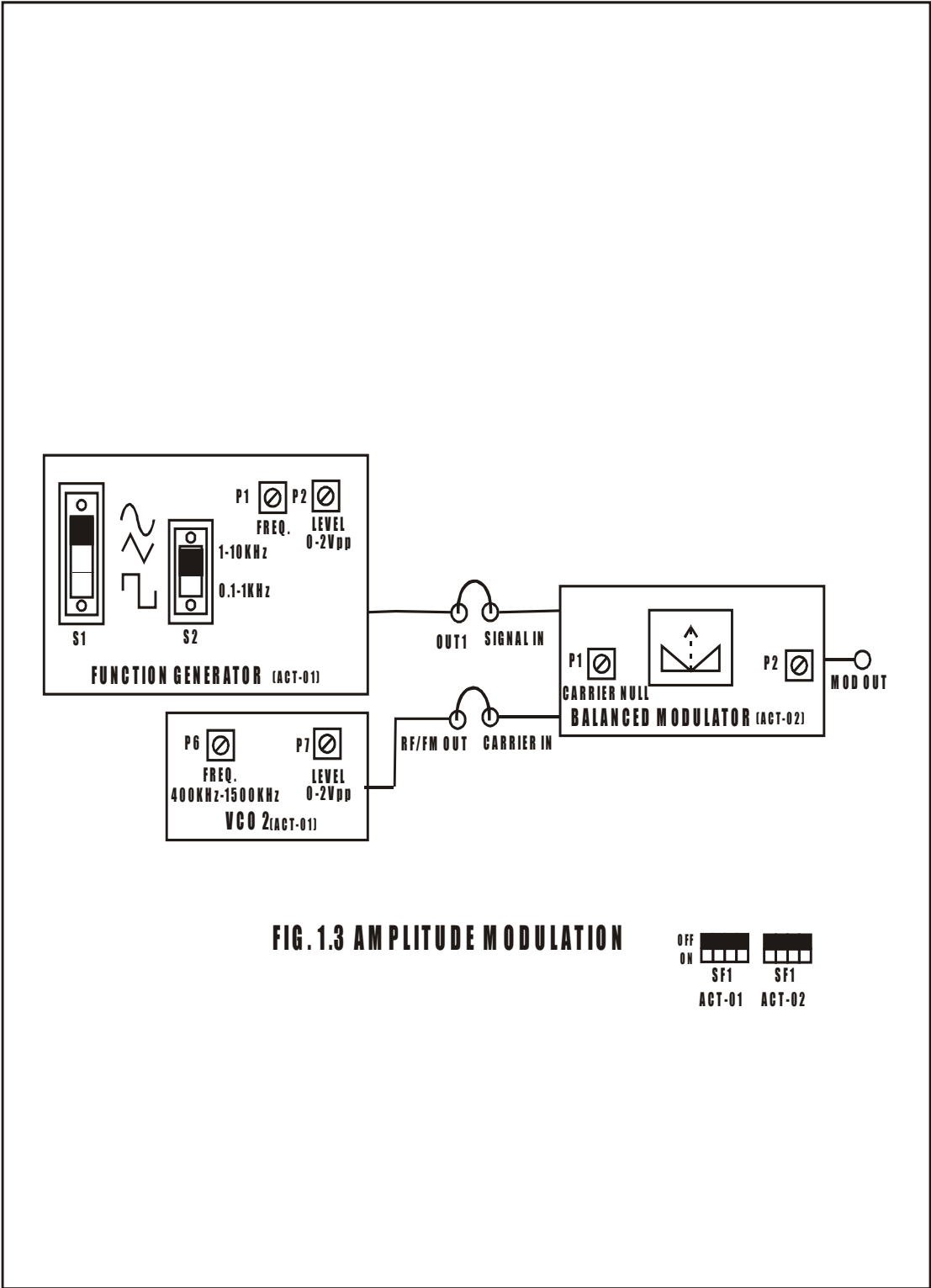
The circuits used to generate an amplitude modulation must vary the amplitude of a high frequency signal (carrier) as function of the amplitude of a low frequency signal (modulating signal).

In an AM transmitter we speak of

- High-level modulation, if the modulation is carried out directly in the last power stage, which is generally an amplifier in class C.
- Low frequency modulation, when the modulation is carried out by stages, which are before the final power amplifier.

Semiconductor devices can be used in case of low power and valve ones when the required power is high.

The circuit used in the exercise shows that, amplitude modulated signal is generated by a differential amplifier whose gain is varied by the modulating signal. The integrated circuit used in the above case, can also be used to generate the amplitude modulation with suppressed carrier, which is the objective of another exercise.



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The circuit used in the exercise shows that, amplitude modulated signal is generated by a differential amplifier whose gain is varied by the modulating signal. The integrated circuit used in the above case, can also be used to generate the amplitude modulation with suppressed carrier, which is the objective of another exercise.

A. To study the operation of an Amplitude Modulator

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01 & ACT-02 switch ON the power supply.
- Refer to the fig. 1.3 & Carry out the following connections and settings.
- Connect output of generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Keep switch **S1** to sine wave position (ACT-01).
- Keep switch **S2** in lower position (ACT-01).
- Using pot **P1** keep sine wave signal frequency at **1KHz**, and using pot **P2** keep sine wave signal amplitude at **0.5Vpp** (ACT-01).
- Using pot **P6** keep VCO2 frequency at **450KHz** and using pot **P7** keep amplitude at **1Vpp** (ACT-01).
- Using pot **P1** keep Balanced Modulator 1 **CARRIER NULL** completely rotated clockwise or counter clockwise, so as to “unbalance” the modulator and to obtain an AM signal with not suppressed carrier across the output; using pot **P2** keep **LEVEL** in fully clockwise.

- Connect the oscilloscope to the inputs of the balanced modulator 1 and detect the modulating signal and the carrier signal (fig.1.1a/b)
- Move the probe from input to output of the modulator, where a signal modulated in amplitude is detected (fig.1.1c). Note that the modulated signal envelope corresponds to the waveform of the modulating signal.
- Vary the amplitude of the modulating signal and check the 3 following conditions:
Modulation percentage lower than 100% (fig. 1.1c), equal to 100% (fig. 1.1d), superior to 100% (over modulation, fig. 1.1e)
- Vary the frequency and waveform of the modulating signal, and check the corresponding variations of the modulated signal.
- Vary the amplitude of the modulating signal and note that the modulated signal can result saturated or over modulated.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.

B. To calculate the Modulation index of Amplitude Modulated wave

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

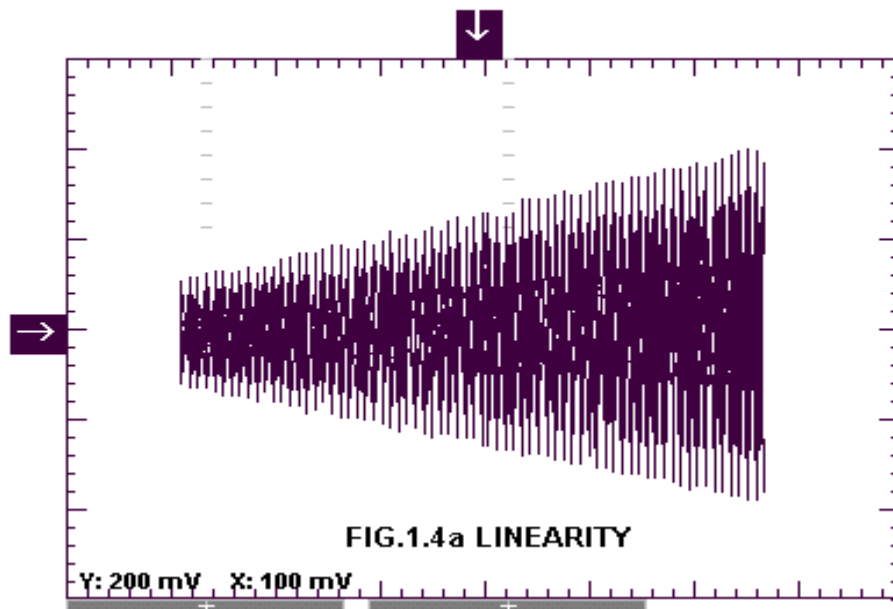
PROCEDURE:-

- Perform the operation as done in Procedure **(A)** and obtain the AM modulated wave as shown in fig. (1.1).
- Using an oscilloscope at Balance Modulator measure from the waveform.
 - The amplitude **B** of the modulating signal (post **SIGNAL IN** of ACT-02)
 - The amplitudes **H** and **h** of the modulated signal, and the amplitude **C** of the envelope of the modulated signal (post **MOD OUT** of ACT-02)
- Calculate the constant **k** of the modulator, equal to: $k = C/B$ You find a value a little over 1
- Calculate the amplitude A of the carrier, equal to:

$$A = \frac{H+h}{2}$$

Calculate the percentage index of modulation m, equal to:

$$m = \frac{H-h}{H+h} \cdot 100\%$$



C. To Observe the Linearity Curve of the Modulator

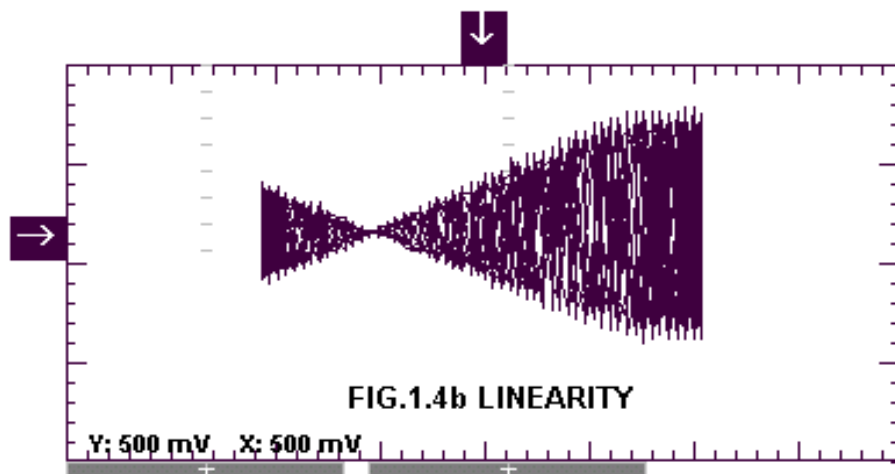
EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Perform the operation as done in Procedure (A)
- Keep P1 fully anti-clockwise.
- Set the oscilloscope in X-Y mode. Connect the modulating signal post **SIGNAL** of ACT-02 to the X-axis of the CRO and the modulated signal post **MOD OUT** of ACT-02 to the Y-axis of the CRO.
- Observe on the diagram there is a trapezium similar to the one of fig. 1.4a; this indicates the behavior of the envelope of the modulated signal as function of the modulating signal amplitude. The display method posts out the eventual non-linearity or distortions of the modulated signal. Increase the amplitude of the modulating signal and observe how the saturation and the over modulation of the modulated signal are displayed (fig. 1.4b)



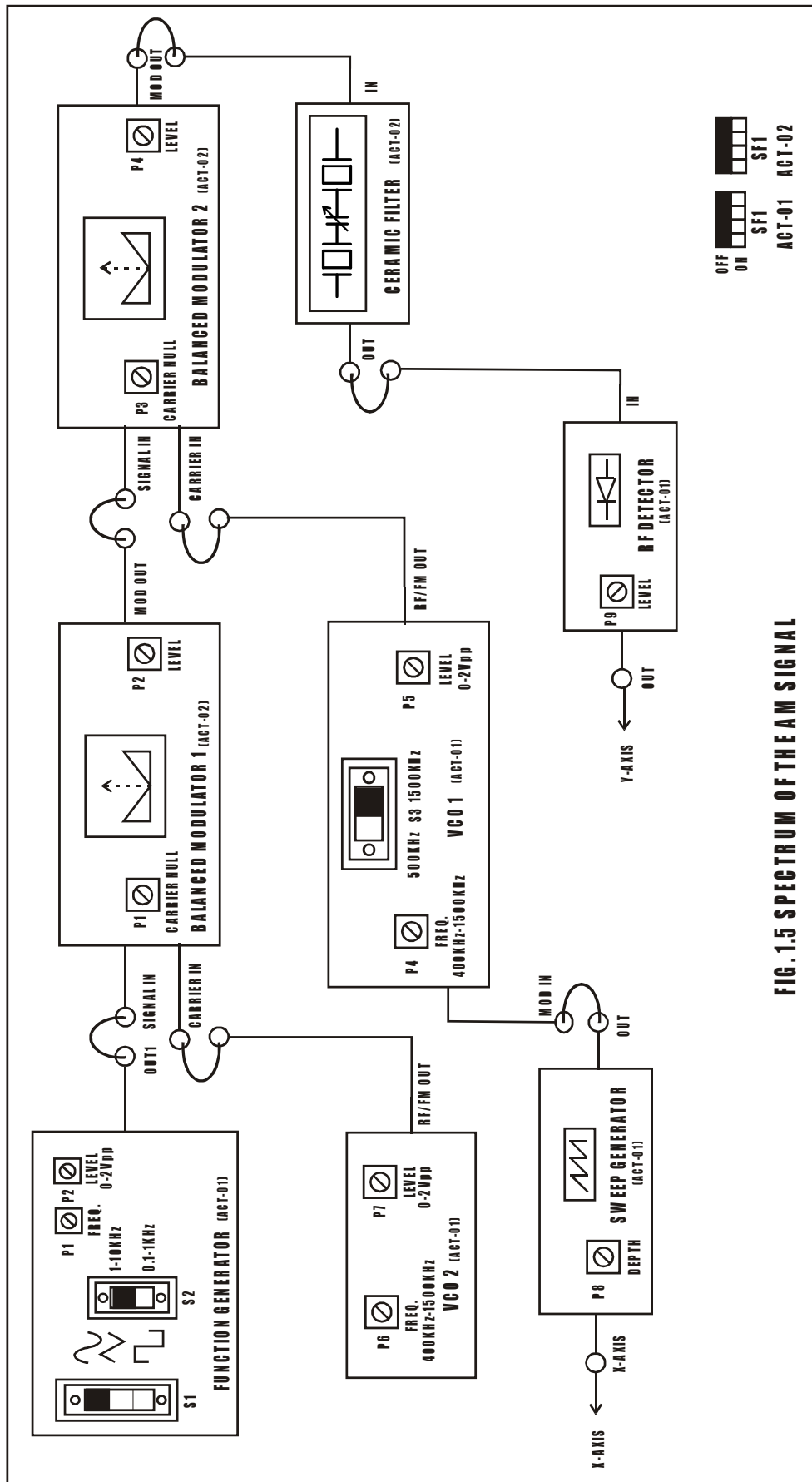


FIG. 1.5 SPECTRUM OF THE AM SIGNAL

D. To Observe the Spectrum of the AM Signal

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

Refer to the “Appendix A” for the explanation of the operating principle of the spectrum analyzer.

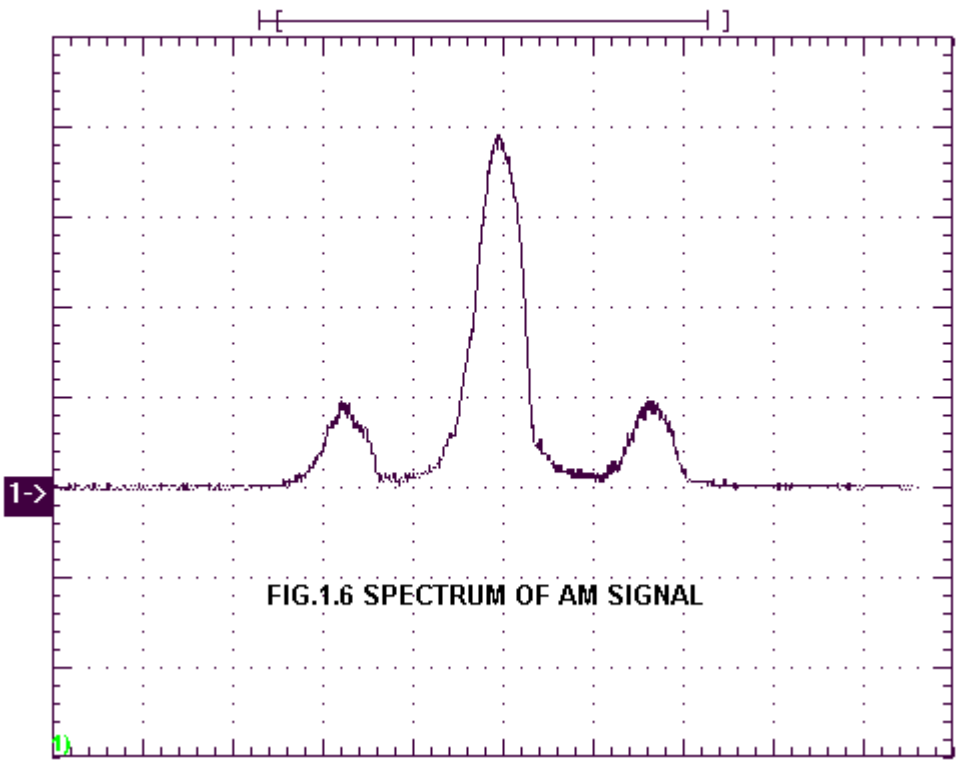
- Connect power supply with proper polarity to the kits ACT-01 & ACT-02 switch it ON.
- Refer to the fig.1.5 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **10 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** at **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **900 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
 - Using pot **P6** keep VCO2 frequency at **450 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **1Vpp**.
 - Keep Sweep **DEPTH** almost completely counter clockwise pot **P8**.
 - Keep RF Detector **LEVEL** completely clockwise pot **P9**.
- In ACT-02
 - Keep BALANCED MODULATOR1 **CARRIER NULL** rotated completely clockwise using pot **P1**.
 - Keep BALANCED MODULATOR1 **LEVEL** maximum completely clockwise using pot **P2**.
 - Keep BALANCED MODULATOR2 **CARRIER NULL** in central position, so that the circuit operates as frequency converter (balanced modulator with suppressed carrier) using pot **P3**.
 - Keep BALANCED MODULATOR2 **LEVEL** maximum completely clockwise using pot **P4**.
- Connect output of function generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of Balance modulator 2 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 2 (ACT-02) **CARRIER IN** post.

- Connect output of Balance modulator 2 (ACT-02) **MOD OUT** post to the input of Ceramic filter (ACT-02) **IN** post.
- Connect output of Ceramic filter (ACT-02) **OUT** post to the input of RF detector (ACT-01) **IN** post.
- Connect modulation in of VCO1 of (ACT-01) **MOD IN** post to the out of sweep generator circuit of (ACT-01) **OUT** post.
- Set the oscilloscopes in X-Y mode. Connect the sweep generator **X-AXIS** post of (ACT-01) to the X-axis, and the detected signal from RF detector **OUT** post of (ACT-01) to the Y-axis.
- Keep volt/div of CRO **X-axis** 0.5V and **Y-axis** 0.5V.
- Vary the frequency of the carrier (VCO2) until the oscilloscope shows a representation similar to the one of fig. 1.6. To obtain the best wave adjust the deviation of the sweep generator **DEPTH** using pot **P8** (ACT-01); the pass band of the ceramic filter (variable capacity); and the **CARRIER NULL** using pot **P3** (ACT-02) of the balanced modulator 2.
- The waveform represents the spectrum of the AM signal, consisting in the carrier and the 2 side bands.
- Vary frequency, amplitude and waveform of the modulating signal and examine the variations on the spectrum as shown in fig. 1.6.

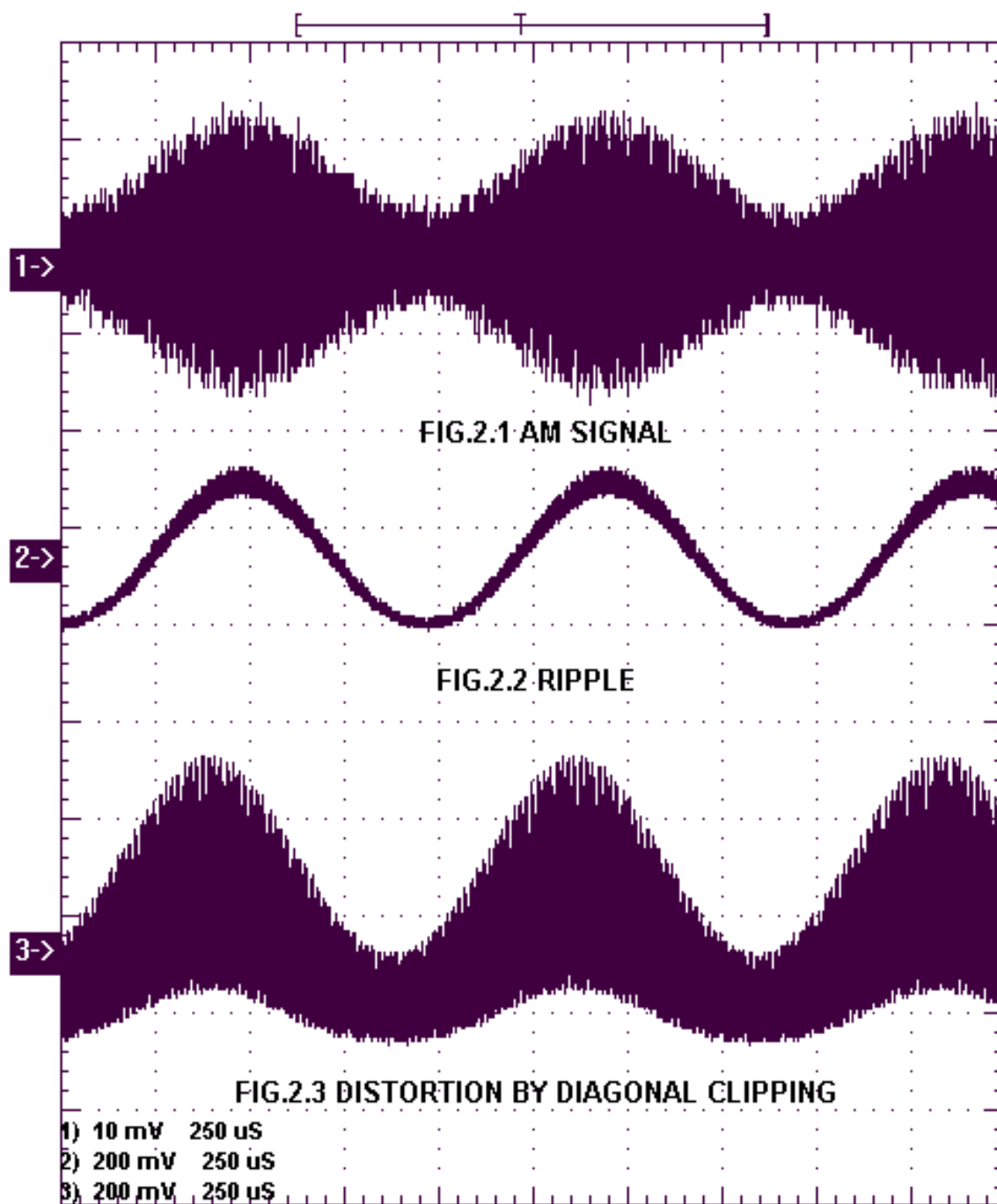
SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500 KHz position, 400 KHz to 500 KHz variation disabled and output gives fixed frequency.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 3 of SF 1 in ACT-02 Switch fault section to ON position. This will open R30 from gain adjust pins of balance modulator 2. Balance modulator2 maximum output reduced to 100 mV from 2V.



EXPERIMENT NO.2



EXPERIMENT NO.2

NAME:-

Study of Amplitude Demodulation

OBJECTIVE:-

- A. To study and observe the operation of Envelope Detector.
- B. To study and observe the effect of time constant RC.
- C. To study and observe the operation of synchronous AM Detector.

THEORY:-

Envelope Detector

The extraction of the modulating signal from an AM signal can be carried out using an envelope detector. Consider, in fact, the AM signal shown in fig. 2.1, and note that the modulating signal constitutes the envelop of the waveform reported. The most common envelope detector consists of a diode followed by RC filter. Its operation is analogous to the one of a half-wave rectifier, as the output voltage follows the maximum values of the carrier. As the amplitude of the carrier is variable, by properly choosing R and C, the output of the detector can be faithfully reproduced.

Distortions of the Detected Signal

The demodulated signal can have two kinds of distortions:

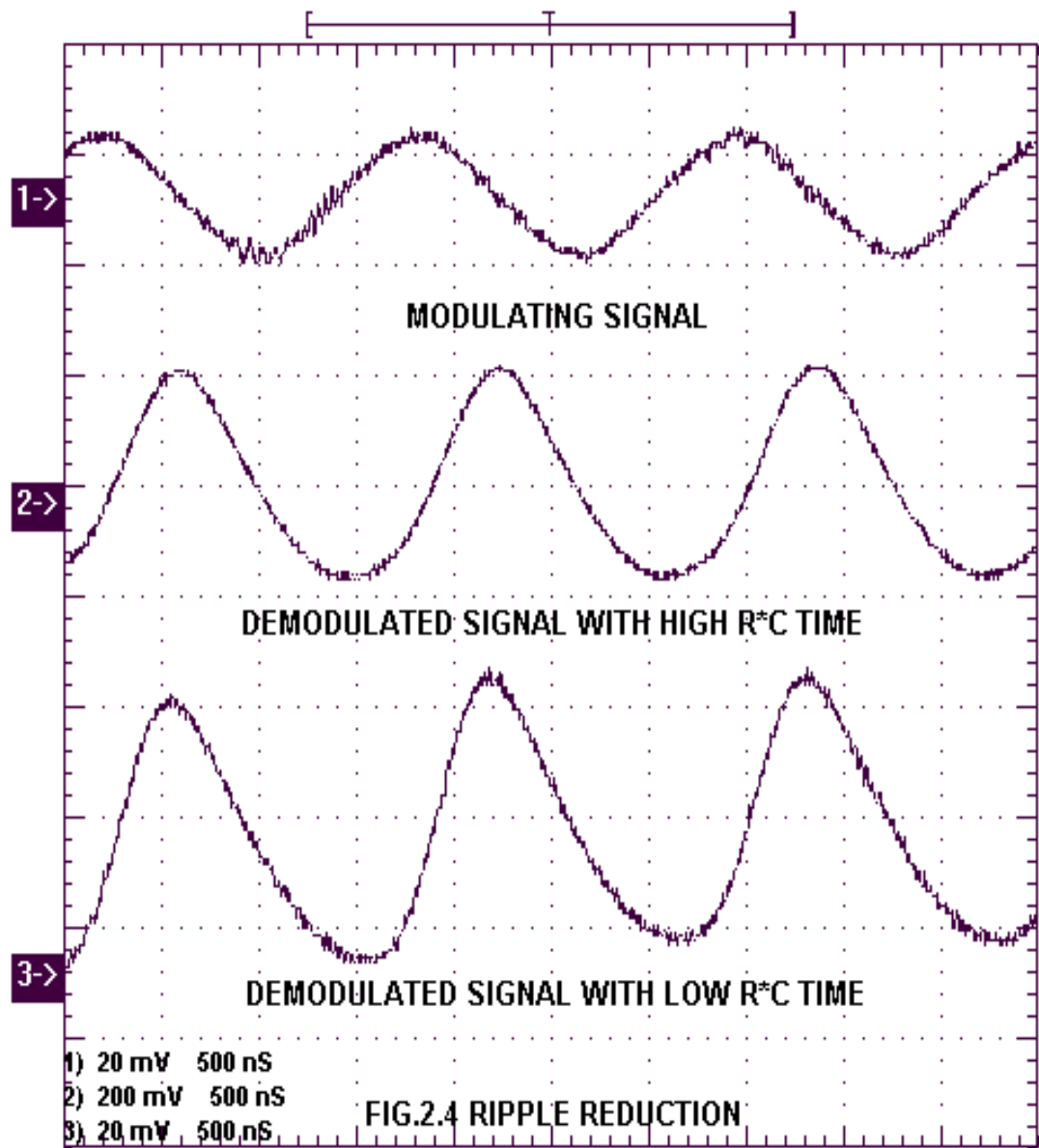
- If the time constant $R \cdot C$ is too small with respect to the period of the carrier, the envelope is approximated to a waveform affected by ripple, the more evident the lower is the value of $R \cdot C$
- If the time constant $R \cdot C$ is too high with respect to the period of the modulating signal, the detected signal always follows the behavior of the envelope, but sometimes it takes an exponential decreasing law (distortion by diagonal cutting)

Minimum Value of $R \cdot C$ (Ripple Minimization)

A first criterion for the choice of $R \cdot C$ has the purpose to reduce the amplitude of the ripple, which approximates the envelope to the minimum. To obtain this result the time constant $R \cdot C$ must be very big with respect to the period T of the carrier, so to avoid an excessive discharge of C between two consecutive peaks of the AM signal:

$$R \cdot C \gg T$$

Figure 2.4 shows two examples of detected signals, obtained with different values of $R \cdot C$ cannot be increased very high, because too high values will cause distortion, too.



Maximum Value of R•C (Distortion By Diagonal Cutting)

A further condition for the proper choice of the time constant R•C is detected referring to figure 2.4.

If the time constant is too high it can happen that, from a certain moment t₁, the output of the detector does not follow the envelope of the modulated signal anymore, but continues to diminish with the discharge law of C, which is different to the law according to which the envelope varies.

Then, the detector diode keeps, cut off until the instantaneous t₂, in which the outputs reach the envelope. In the interval (t₂-t₁), in fact, the output is constantly higher than the maximum values of the modulated signal, and the diode is inversely biased.

The distortion presented by the detected signal is known as Distortion by diagonal cutting.

The maximum value of R•C is calculated supposing that the capacitor C discharges on the resistor R to a higher or equal rhythm to the one in which the envelope of the modulated signal drops. The detected condition is the following:

$$R \cdot C \leq \frac{1}{f_{\max}} \cdot \frac{V (1-m^2)}{m}$$

with m = modulation index

$f_{\max}$ = maximum frequency of the modulating signal.

Note that for m=1 the condition cannot be satisfied, and so the detected signal will be certainly distorted.

Detection Efficiency

The efficiency η of the diode envelope detector is defined as the ratio between the amplitude of the output signal of the detector and the amplitude of the envelope of the input AM signal. Supposing that at the carrier frequency, the reactance of the capacitor $[1/(2\pi \cdot F \cdot C)]$ is much smaller than the resistance R (hypothesis checked if $RC \gg 1/F$ to reduce the ripple), the efficiency η practically depends on the ratio between R and the differential resistance r_d of the diode

Last Considerations for the Choice of R•C in the Envelope Detector

- To obtain a high detection efficiency the resistance R must be much higher than the differential resistance r_d of the diode ($R \gg r_d$)
- To minimize the ripple in the detected signal the product R•C must be much higher than the period T of the carrier ($R \cdot C \gg T$)

To avoid distortion by diagonal cutting the product R•C must be smaller or equal to a value, which depends on the modulating frequency, and on the modulation index. In the case of the modulating signal with variable amplitude and frequency, the maximum values of the frequency as well as of the modulation index must be considered.

Synchronous AM Detector

In the synchronous AM detector, also known as Coherent AM Detector, the signal modulated in amplitude is mixed to a coherent carrier signal and the result then, passes across a low pass filter, which supplies the desired demodulated signal. The modulated signal v_M can be expressed by the following relation:

$$v_M(t) = V_m(t) \cdot \sin(2\pi F \cdot t)$$

Where: $V_m(t) = A \cdot [1 + m \cdot \sin(2\pi F \cdot t)]$

A is the amplitude of the envelope of the AM signal, and F is the frequency of the carrier.

If the AM signal is multiplied with a un-modulated signal which has the same frequency and phase, the output of the multiplier is as composite signal $v_o(t)$ which can be expressed in this way:

$$v_o(t) = [A_C \cdot \sin(2\pi F \cdot t)] \cdot [V_m(t) \cdot \sin(2\pi F \cdot t)] = K_o \cdot V_m \cdot [1 + \cos(2\pi \cdot 2F \cdot t)]$$

Where K_o is considered as the gain of the multiplier circuit

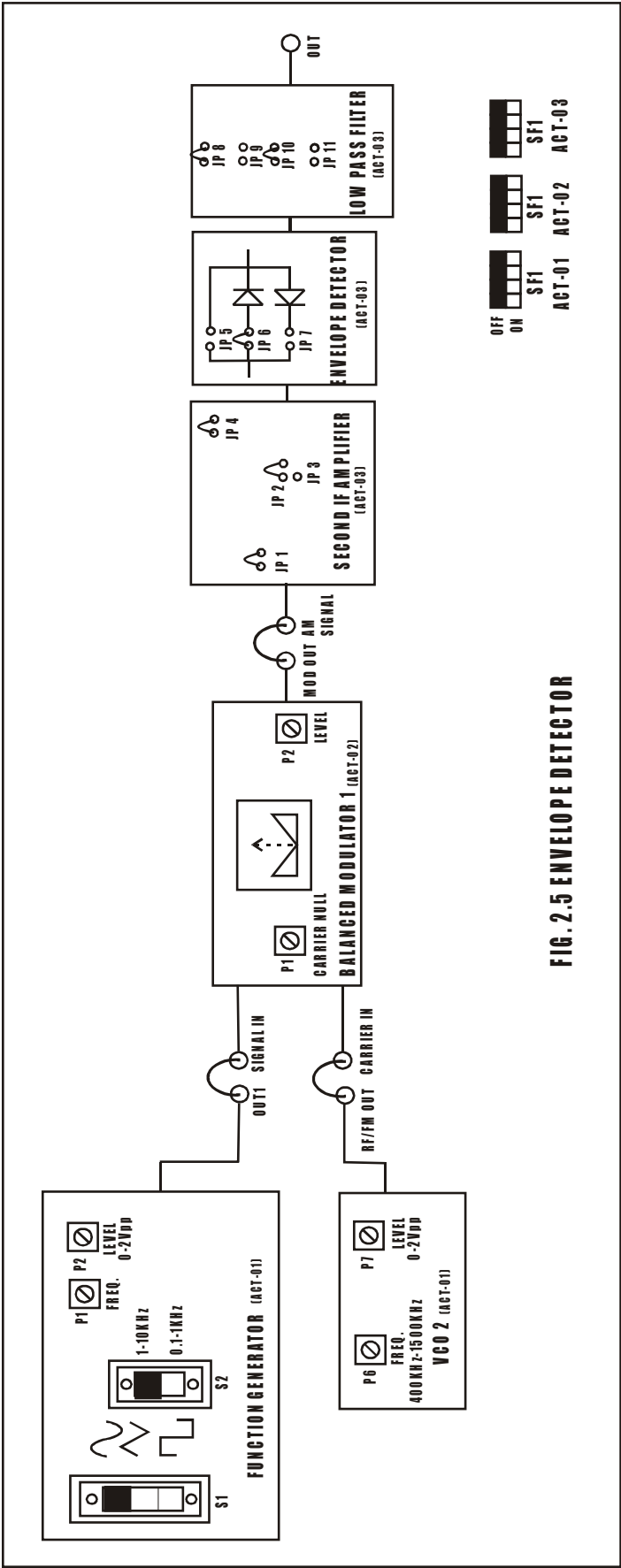
Make the signal $v_o(t)$ pass across a low pass filter, to remove the component with $2F \cdot t$ frequency, the resulting signal V_{out} is:

$$V_{out} = K_o \cdot V_m(t)$$

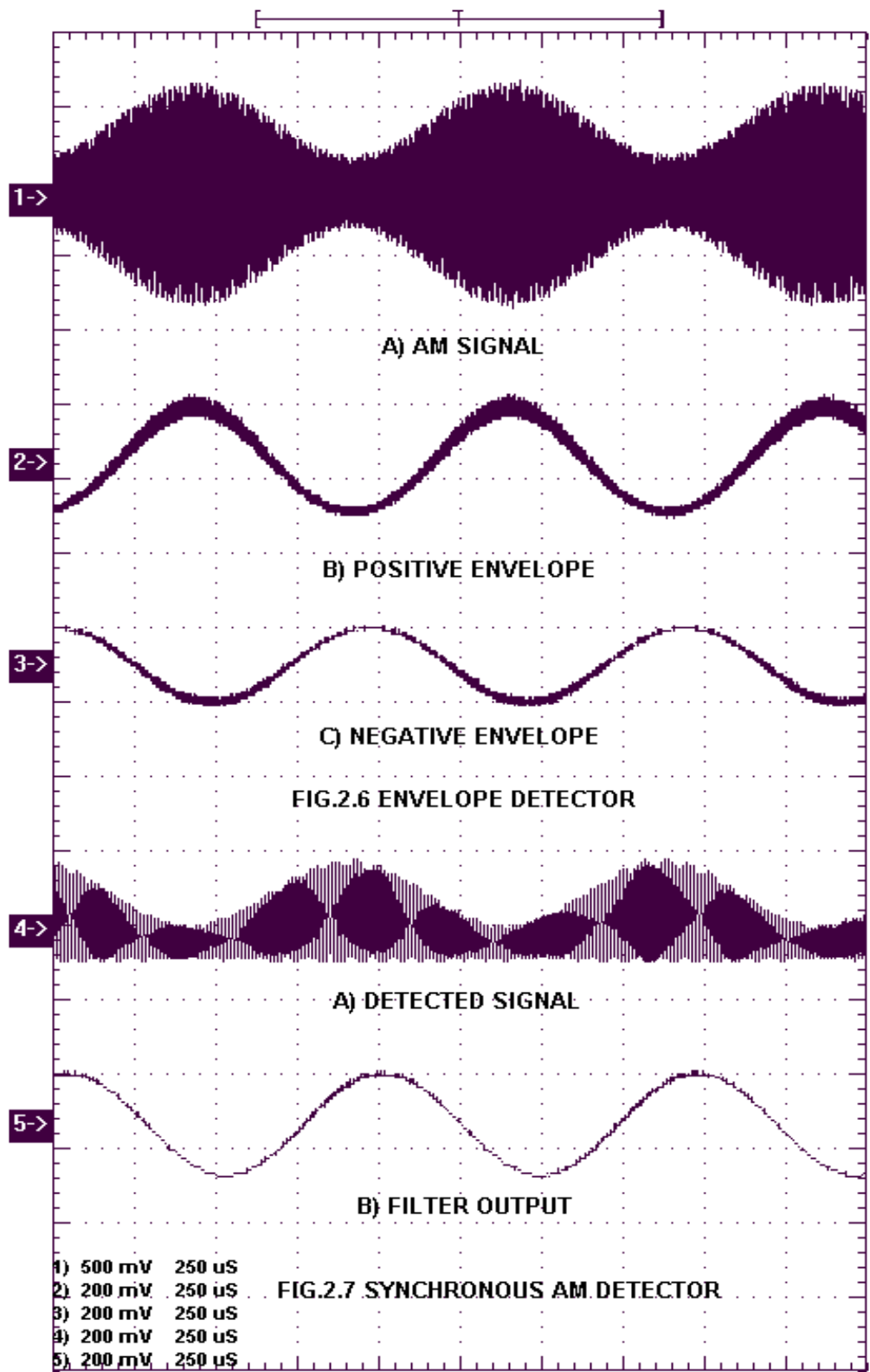
This corresponds to the AM detected signal

A PLL (Phase Locked Loop) system is generally used to regenerate a signal at the receiver, which is exactly synchronous to the carrier of the AM signal. In the applications where a higher quality is required, the use of the AM synchronous detector presents some important advantages with respect to the envelope detector:

- Low distortion
Capacity to properly demodulate also the AM signal with high modulation depth, or with quick variations of the modulation signal (as for example in pulse modulations)
- Capacity to provide gain instead of the attenuation introduced by the diode.



•



A. To study and observe the operation of Envelope Detector

EQUIPMENT:-

- Modules ACT-01, ACT-02, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01; ACT-02 and ACT-03 and switch it ON.
- Refer to the fig.2.5 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Using pot **P6** keep VCO2 frequency at **450KHz**.
 - Using pot **P7** keep VCO2 amplitude at **1Vpp**.
- In ACT-02
 - Keep BALANCED MODULATOR1 **CARRIER NULL** rotated completely clockwise or slightly lower than clockwise using pot **P1**.
 - Keeps BALANCED MODULATOR1 **LEVEL** at **20mVpp** using pot **P2**.
- In ACT-03
 - Keep Jumpers JP1, JP2, JP4, JP6, JP8, and JP10 Shorted.
 - Keep Jumpers JP3, JP5, JP7, JP9, and JP11 Open.
- Connect output of function generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Observe the modulated signal envelope, which corresponds to the waveform of the modulating signal at **MOD OUT** post of BALANCED MODULATOR1 of ACT-02.
- Connect the output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of IF Amplifier (ACT-03) **AM SIGNAL** post.
- Connect the oscilloscope before and after the detector diode (posts **IF OUT** and **AGC 2** of ACT-03) and detect the amplitude modulated signal and the demodulated signal (fig.2.6).
- Check that the detected signal follows the behaviour of the AM signal envelope.
- Move the jumper from JP6 to JP7, and check that the negative envelope is detected.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 2 of SF 1 in ACT-03 Switch fault section to ON position. This will open C7 from 2nd IF section low pass filter, filter output gets distorted.
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600 KHz.

B. To study and observe the Effect of time constant RC

EQUIPMENT:-

- Modules ACT-01, ACT-02, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

NOTE: keep all the switches in switch fault section in off position

PROCEDURE:-

- Perform the operation as done in Procedure (A)
- The values of R and C connected after the detector diode are respectively $R7=22K\Omega$ and $C7=10nF$. Keep the oscilloscope connected to post **IF OUT** and post **OUT** on ACT-03 increase the frequency of the modulating signal, taking it to 10KHz. Note the diagonal cutting distortion on the detected signal.
- Diminish the modulation depth (reduce the amplitude of the modulating signal) and check that the detected signal can properly follow the envelope of the AM signal.

-
- Take the modulation depth to about the 50% and change C7 (10nF) with C6 (1nF). By replacing jumper JP10 to JP9. The diagonal cutting distortion reduces, but the ripple added to the detected signal increases. The oscilloscope gives waveforms similar to the ones of fig. 2.2.

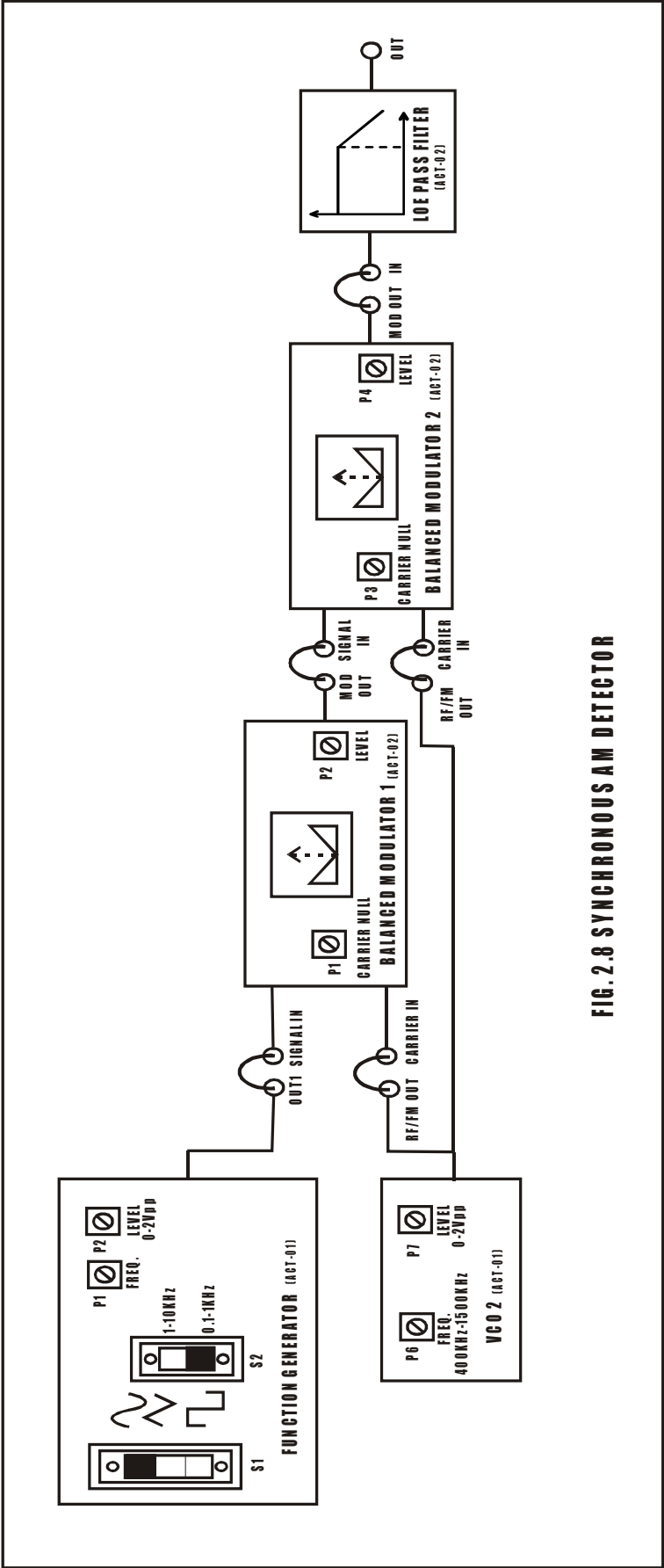


FIG. 2.8 SYNCHRONOUS AM DETECTOR

C. To study and observe the operation of synchronous AM Detector

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.2.8 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Using pot **P6** keep VCO2 frequency at **450 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **1Vpp**.
- In ACT-02
 - Keep BALANCED MODULATOR1 **CARRIER NULL** rotated completely clockwise or slightly lower than clockwise using pot **P1**.
 - Keep BALANCED MODULATOR1 **LEVEL** at maximum using pot **P2**.
 - Keep BALANCED MODULATOR2 **CARRIER NULL** in center position, so that the circuit operates as frequency converter (balanced modulator with suppressed carrier) using pot **P3**.
 - Keep BALANCED MODULATOR2 **LEVEL** at maximum using pot **P4**.
- Connect output of function generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of Balance modulator 2 (ACT-02) **SIGNAL IN** post.
- Connect the carrier of Balance modulator 1 (ACT-02) **CARRIER IN** post to the input of Balance modulator 2 (ACT-02) **CARRIER IN** post.
- Connect output of Balance modulator 2 (ACT-02) **MOD OUT** post to the input of low pass filter (ACT-02) **IN** post.
- Connect the oscilloscope to the input of the demodulator post **SIGNAL IN** of Balanced modulator 2 and to the output post **MOD OUT**. The waveform detected at the output (fig.2.7) is the result of the product between AM signal and the carrier that in the actual application is locally regenerated in the receiver. Note that this waveform consists of a frequency double the carrier, to which the waveform of the modulating signal is added.

- The low pass filter eliminates the high frequency component, supplying a clean detected signal. As the filter has a cutting frequency of 3.4 KHz, all superior frequencies are attenuated.
- Increase the modulation depth of the AM signal, and check that the detected signal does always faithfully follow the envelope. There is no distortion by diagonal cutting.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs. Upper shape gets distorted.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 3 of SF 1 in ACT-02 Switch fault section to ON position. This will open R30 from gain adjust pins of balance modulator 2. Balance modulator2 maximum output reduced to 100 mV from 2V.
- Put switch 4 of SF 1 in ACT-02 Switch fault section to ON position. This will open filter capacitor C24 from low pass filter, filter output gets distorted.

EXPERIMENT NO.2A

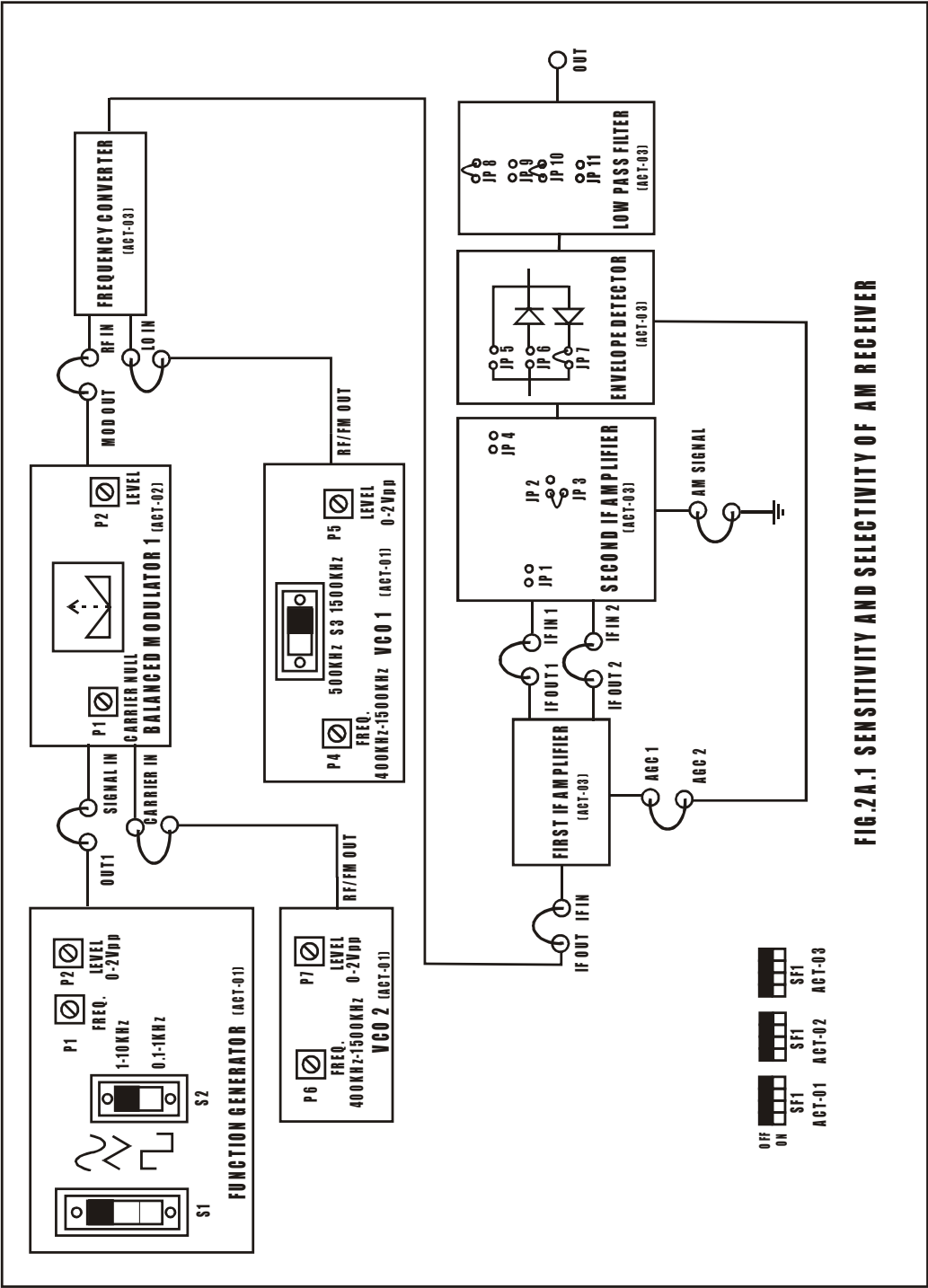


FIG.2A.1 SENSITIVITY AND SELECTIVITY OF AM RECEIVER

EXPERIMENT NO: 2A

NAME:-

Study of Sensitivity, Selectivity, of an AM Receiver

OBJECTIVE:-

- A. Study of sensitivity of an AM receiver.
- B. Study of selectivity of an AM receiver.

THEORY:-

2A.1 Sensitivity

The sensitivity of a receiver is its ability to amplify weak signals. Sensitivity of a receiver is the minimum voltage required at the input of the receiver to produce the original signal. It is often defined in terms of the voltage that must be applied to the receiver input terminals to give a standard output power, measured at the output terminal. For AM broadcast receivers, several of the relevant quantities have been standardized.

The most important factors determining the sensitivity of a super heterodyne receiver are the gain of the IF amplifiers and that of the RF amplifier.

2A.2 selectivity

A parallel tuned circuit has its greatest impedance at resonance and decreases at higher and lower frequencies. If the tuned circuit is included in the circuit design of an amplifier, it results in an amplifier, which offers more gain at the frequency of resonance and reduced amplification above and below this frequency. This is called selectivity. We can see the effects of using an amplifier with selectivity.

The radio receiver is tuned to a frequency of 820 KHz and, at this frequency; the amplifier provides a gain of five. Assuming the incoming signal has an amplitude of 10mV, its output at this frequency would be $5 \times 10\text{mV} = 50\text{mV}$. The stations being received at 810KHz each have a gain of one. With the same amplitude of 10mV, this would result in outputs of $1 \times 10\text{mV} = 10\text{mV}$. The stations at 800KHz and 840KHz are offered a gain of only 0.1 (approx.). This means that the output signal strength would be only $0.1 \times 10\text{mV} = 1\text{mV}$.

The overall effect of the selectivity is that whereas the incoming signals each have the same amplitude, the outputs vary between 1mV and 50mV so we can select, or 'tune', the amplifier to pick out the desired station.

The greatest amplification occurs at the resonance frequency of the tuned circuit. This is sometimes called the center frequency.

In common with nearly all radio receivers, adjusts the capacitor value by means of the local oscillator control to select various signals.

A. Study of Sensitivity of an AM Receiver

EQUIPMENTS:-

- Modules ACT-01, ACT-02 & ACT-03.
- Power supply +/-12 V.
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency counter.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01; ACT-02 and ACT-03 and switch it ON.
- Refer to the fig.2A.1 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **855 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1Vpp**.
 - Using pot **P6** keep VCO2 frequency at **400 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **200mVpp**.
- In ACT-02
 - Keep BALANCED MODULATOR1 **CARRIER NULL** rotated completely clockwise or slightly lower than clockwise using pot **P1**.
 - Keep BALANCED MODULATOR1 **LEVEL** at **50mVpp** using pot **P2**.
- In ACT-03
 - Keep Jumpers JP3, JP7, JP8, and JP10 Shorted.
 - Keep Jumpers JP1, JP2, JP4, JP5, JP6, JP9, and JP11 Open.
- Connect output of function generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect the output of Balance modulator 1 (ACT-02) **OUT** post to the input of Frequency Converter (ACT-03) **RF IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **LO IN** post.
- Connect the output of Frequency Converter (ACT-03) **IF OUT** post to the input of 1st IF (ACT-03) **IF IN** post.
- Connect the output of 1st IF (ACT-03) **IF OUT 1** post to the input of 2nd IF (ACT-03) **IF IN 1** post and connect **AM SIGNAL** post to **GND** post.
- Connect post **IF OUT 2** of 1st IF (ACT-03) to post **IF IN 2** of 2nd IF (ACT-03).

- Connect post **AGC 1** of 1st IF (ACT-03) to post **AGC 2** of 2nd IF (ACT-03). This establishes Automatic Gain Control.
- Observe the modulated signal envelope, which corresponds to the waveform of the modulating signal at **OUT** post of the balanced modulator of ACT-02.
- Connect the oscilloscope to the **IN** and **OUT** post of envelope DETECTOR and detect the AM signal.
- Check that the detected signal follows the behavior of the AM signal envelope vary the amplitude of the balanced modulator output, and check the corresponding variations at the demodulated signal.
- Adjust the input to **RF IN** post in such a way that you should get minimum detected output of about 0.3Volt at the output of Envelope detector.
- You can take the readings as per the table mentioned below for various carrier frequencies (VCO2) and corresponding Local Oscillator frequency (VCO1) settings.

VCO2(KHz)	400	500	600	700	800	900	1000
VCO1(KHz)	850	950	1050	1150	1250	1350	1450
Input to RF (mv)	60	25	50	45	43	46	46

Plot the graph of INPUT TO RF ON Y-AXIS AND VCO1 FREQUENCY ON X-AXIS (FIG.2A.2)

B1. Study of Selectivity of an AM Receiver

EQUIPMENTS:-

- Modules ACT-01, ACT-02 & ACT-03.
- Power supply +/-12 V.
- 20MHz Oscilloscope.
- Connecting Links.
- Frequency counter.

NOTE: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01; ACT-02 and ACT-03 and switch it ON.
- Refer to the fig.2A.1 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** in **1500 KHz** position.

- Using pot **P4** keep VCO1 frequency at **1390 KHz**.
- Using pot **P5** keep VCO1 amplitude at **0.5Vpp**.
- Using pot **P6** keep VCO2 frequency at **900 KHz**.
- Using pot **P7** keep VCO2 amplitude at **200mVpp**.
- In ACT-02
 - Keep BALANCED MODULATOR1 **CARRIER NULL** rotated completely clockwise or slightly lower than clockwise using pot **P1**.
 - Keeps BALANCED MODULATOR1 **LEVEL** at **50mVpp** using pot **P2**.
- In ACT-03
 - Keep Jumpers JP3, JP7, JP8, and JP10 Shorted.
 - Keep Jumpers JP1, JP2, JP4, JP5, JP6, JP9, and JP11 Open.
- Connect output of function generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect the output of Balance modulator 1 (ACT-02) **OUT** post to the input of Frequency Converter (ACT-03) **RF IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **LO IN** post.
- Connect the output of Frequency Converter (ACT-03) **IF OUT** post to the input of 1st IF (ACT-03) **IF IN** post.
- Connect the output of 1st IF (ACT-03) **IF OUT 1** post to the input of 2nd IF (ACT-03) **IF IN 1** post and connect **AM SIGNAL** post to **GND** post.
- Connect post **IF OUT 2** of 1st IF (ACT-03) to post **IF IN 2** of 2nd IF (ACT-03).
- Connect post **AGC 1** of 1st IF (ACT-03) to post **AGC 2** of 2nd IF (ACT-03). This establishes Automatic Gain Control.
- Observe the modulated signal envelope, which corresponds to the wave form of the modulating signal at **OUT** post of the balanced modulator of ACT-02.
- Connect the oscilloscope to the **IN** and **OUT** post of envelope DETECTOR and detect the AM signal.
- Check that the detected signal follows the behavior of the AM signal envelope vary the amplitude of the Balanced modulator output, and check the corresponding variations at the demodulated signal.
- Vary the Carrier frequency (VCO2) in terms of 10KHz on both sides from 870KHz to 940 KHz.
- Take readings of Amplitude levels at Envelope Detector output for various Carrier frequencies.

Table Similar to the One given below can be observed at the users end.

Carreir freq. Vco2(khz)	Envelope detector output (mv)
870	5
880	24
890	200
900	500
910	300
920	160
930	100
940	22

(Table given for reference only actual values may differ from above)

Plot the graph accordingly as per the readings taken. For the carrier frequency and o/p Amplitude (FIG. 2A.3).

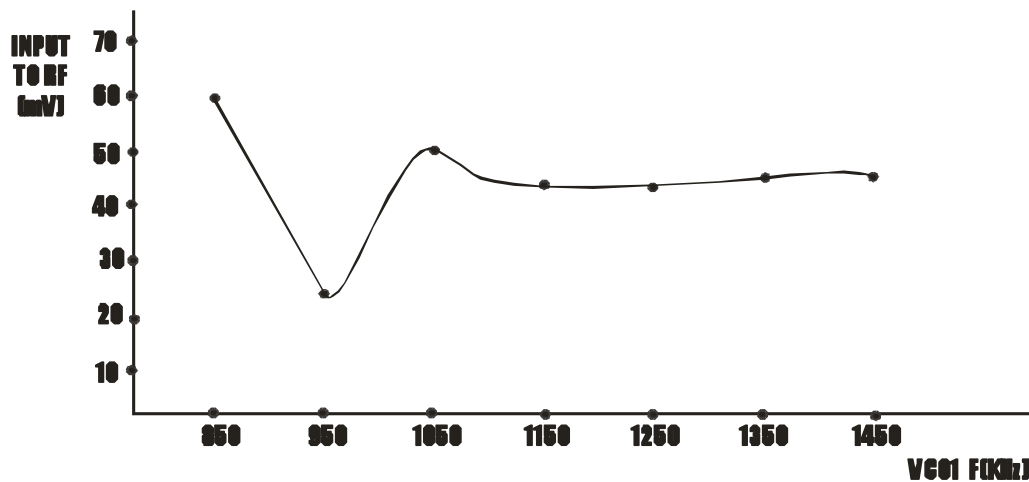


FIG.2A.2 SENSITIVITY OF AN AM RECEIVER

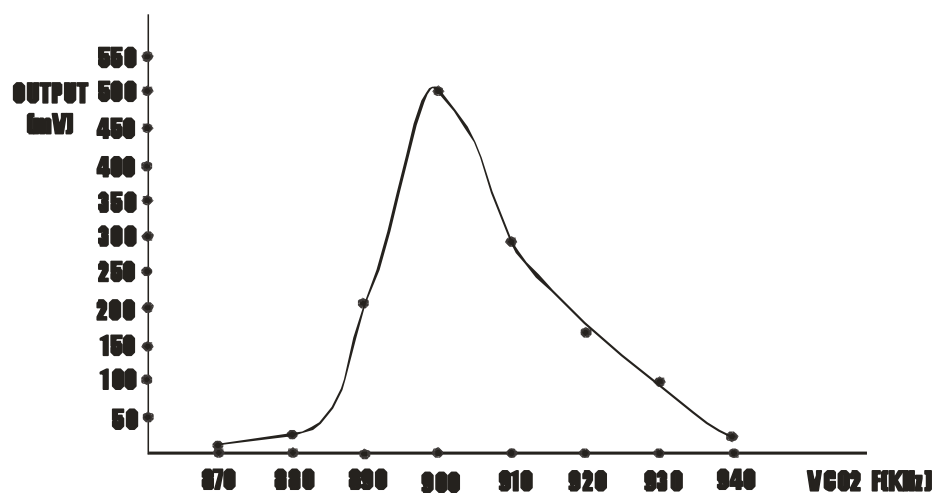
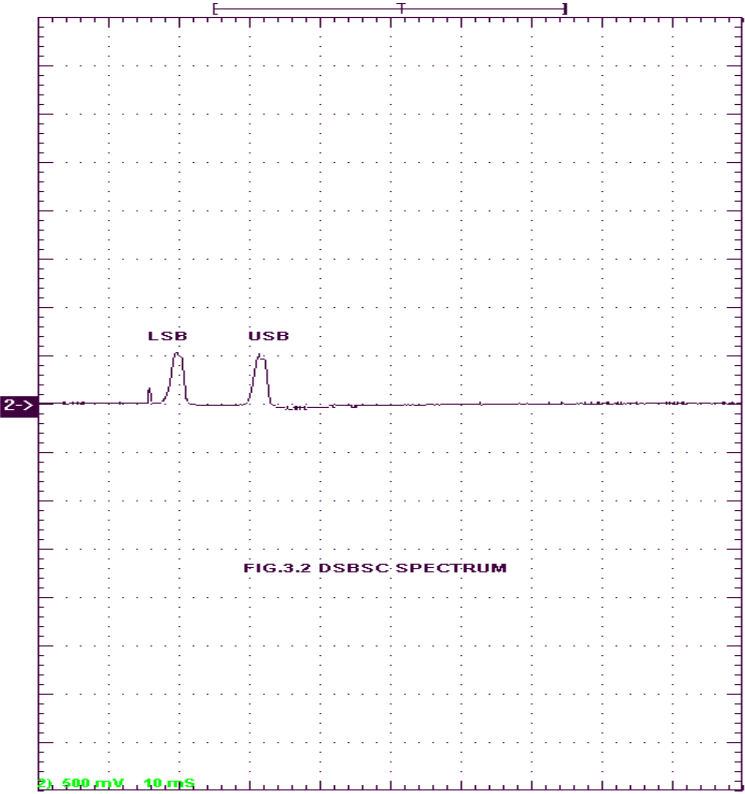
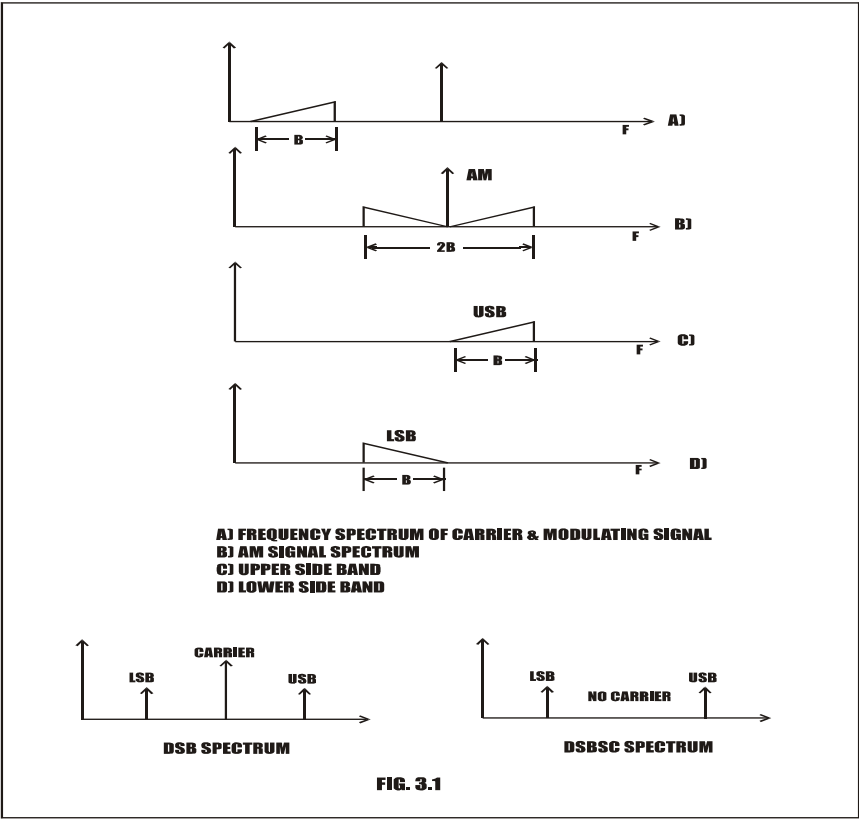


FIG.2A.3 SELECTIVITY OF AN AM RECEIVER

EXPERIMENT NO.3



EXPERIMENT NO.3

NAME:-

Single Side Band Modulation

OBJECTIVE:-

- A1. To observe the frequency response of Ceramic filter.
- A2. To plot the frequency response curve of Ceramic filter.
- B. To study the operation of Amplitude Modulator with suppressed carrier.
- C. To study Single Side Band Generation.
- D. To Study and observe the Spectrum of the Signal generated by the Balance Modulator.
- E. To Study the Demodulation of the SSB Signal.
- F. To Study the operation of the Ring Modulator.

THEORY:-

The Amplitude modulation theory, as stated in chapter 1, shows that the modulation process produces a carrier and two side bands.

See $v_m(t)$ is a modulation signal:

$$v_m(t) = B \sin(2\pi f \cdot t)$$

and $v_c(t)$ the carrier:

$$v_c(t) = A \sin(2\pi F \cdot t)$$

the modulated signal $v_M(t)$ consists of three sine components:

$$A \sin(2\pi F \cdot t) \quad \text{carrier}$$

$$\frac{A}{2} \cos[(2\pi(F-f) \cdot t)] \quad \text{lower side band}$$

$$\frac{A}{2} \cos[(2\pi(F+f) \cdot t)] \quad \text{upper side band.}$$

If the modulating signal does not consist of a single sine wave, but of a generic signal with frequency spectrum ranging between two frequencies, with the amplitude modulation this spectrum is moved over or under the carrier (fig. 3.1).

It is evident that the carrier takes no information, as it keeps constant in amplitude as well as in frequency independently from the modulating signal.

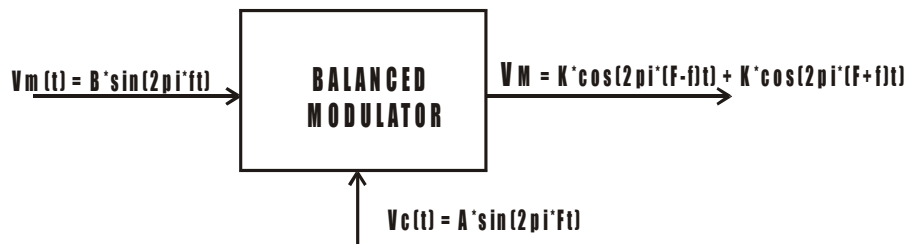


FIG.3A. BALANCED MODULATOR

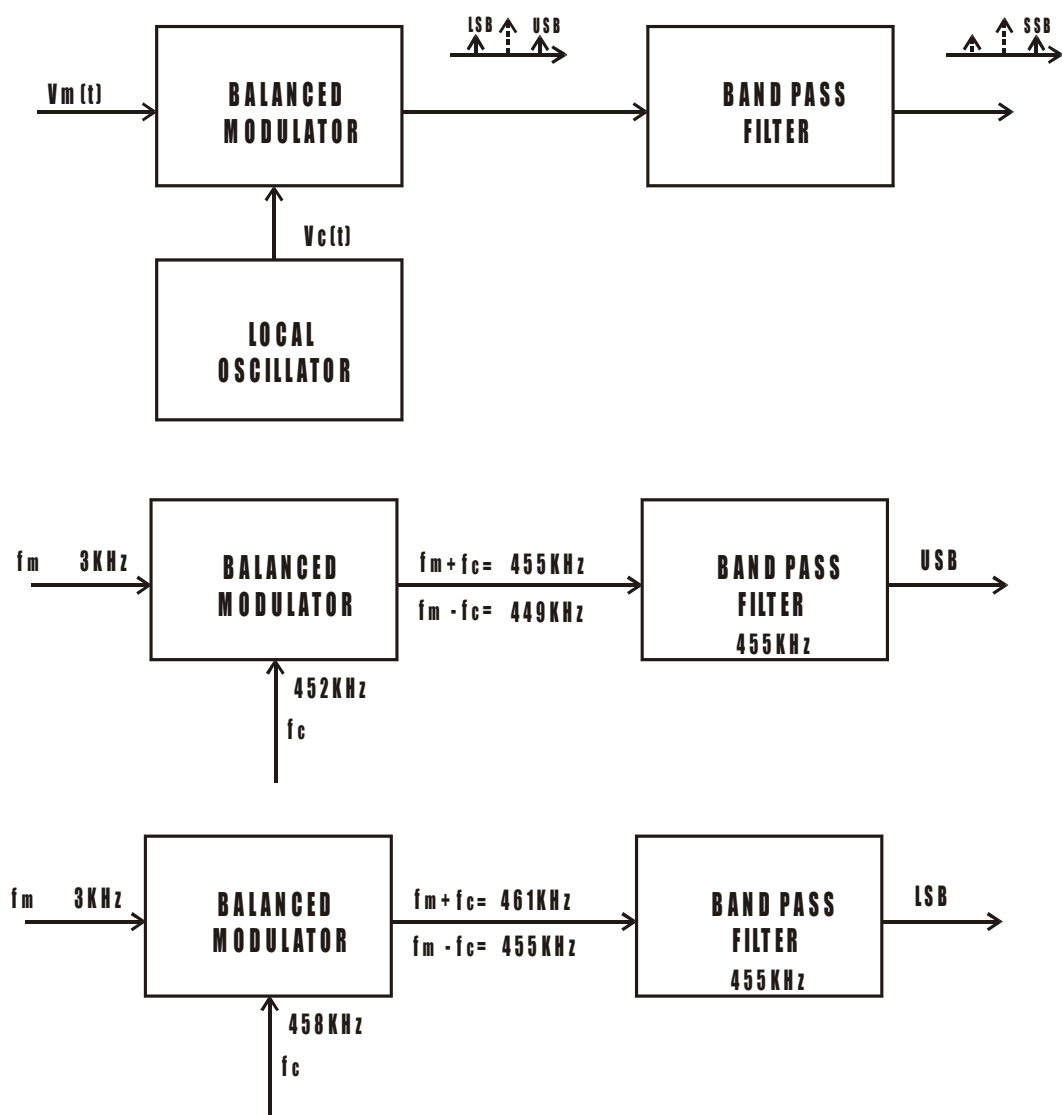


FIG.3B. SSB GENERATION WITH FILTERING METHOD

It is also evident that the two side bands are exactly one image of the other; the amplitudes of both vary in the same way according to $m \cdot A/2$, and so also the frequencies are different from the frequency of the carrier of the same quantity f . You can see that all the information can be transmitted using one single side band: the carrier is superfluous and the other side band is redundant.

SSB against AM

In respect to the amplitude modulation the single side band presents the following advantages:

- The band of the modulated signal is reduced to half. This means that, in the same range of frequencies, there can be a double number of communication channels.
- All power emitted by the transmitter is associated to the information to be transmitted, differently from the AM in which most power is associated to the carrier.

The most evident advantages of the SSB as compared to AM are mentioned below:

- Higher circuit complexity in the modulator as well as in the demodulator.
- Need to locally generate the carrier in the receiver to detect the modulated signal correctly.

Applications of the SSB

- Radio transmissions
- Multichannel telephone transmission with the technique of Frequency Division Multiplexing (FDM)
- High-speed data transmission (modem V35/V36/V37).

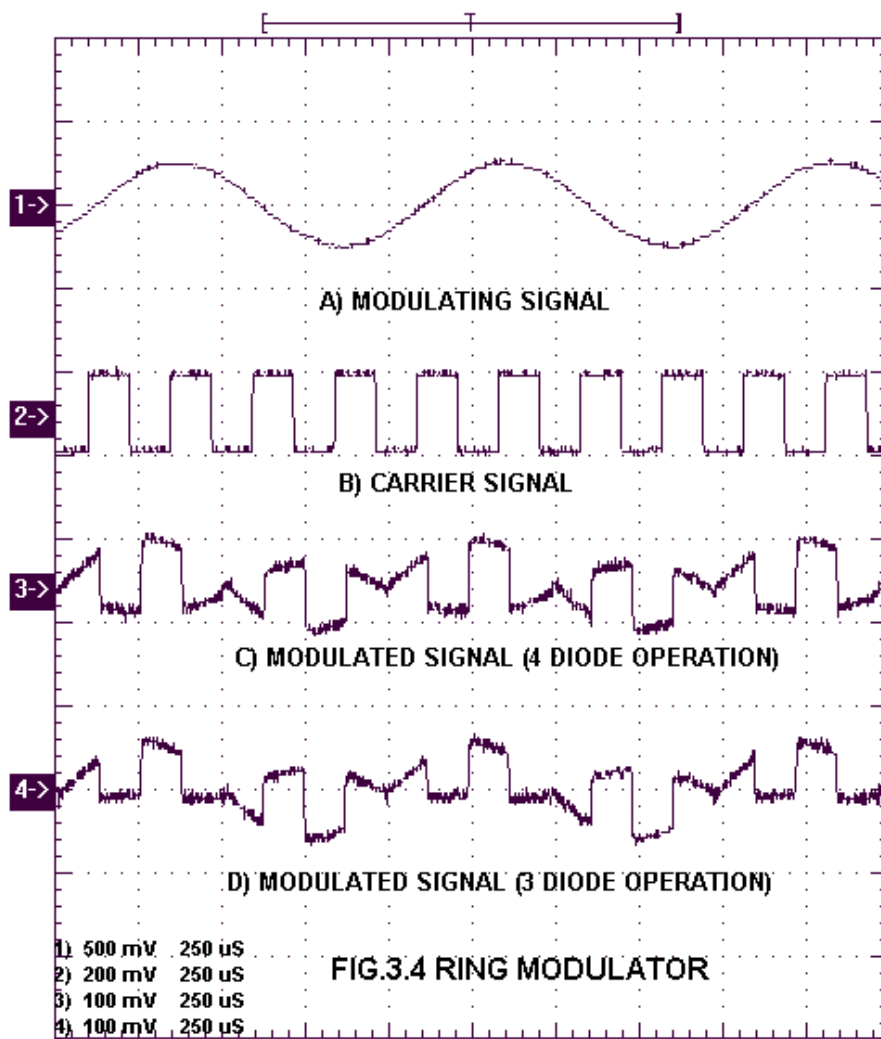
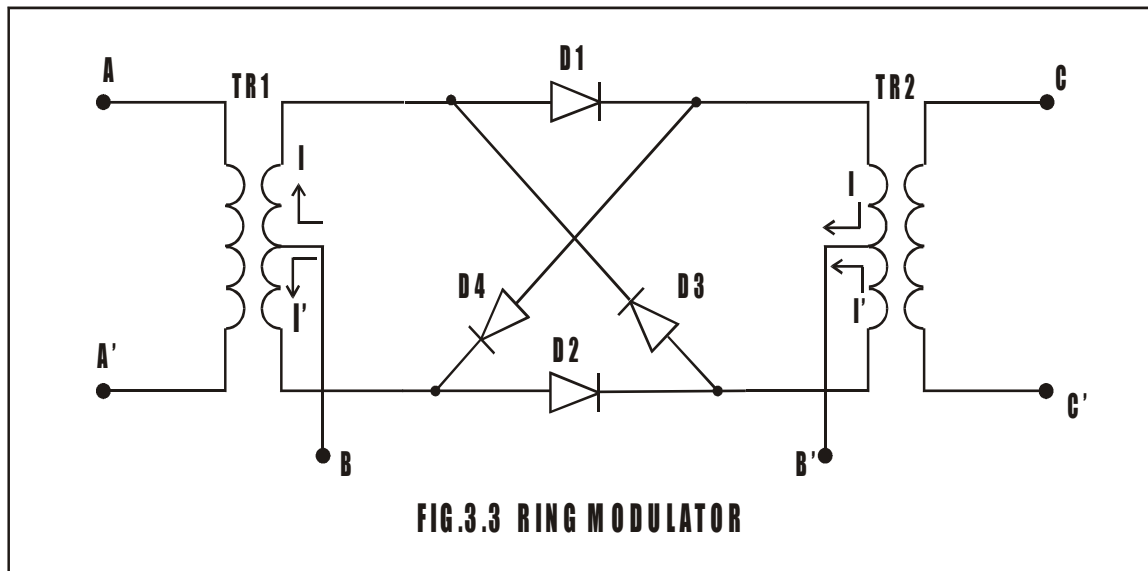
SSB Generation by Filtering

The purpose of the methods used for the generation of an SSB signal to suppress the carrier and one side band.

The most used method is the filtering, which is actuated in two consequent phases (fig.3B)

- An amplitude modulation with suppressed carrier is generated, also known as Double Side band (DSB)
- A next band pass filter extracts one of the two side bands.

The most common modulators are the balanced modulators and the ring modulators, described hereafter. The filters used for the separation of one side band must present a high steepness of the attenuation curve between the pass band and the attenuated band. Quartz, ceramic, active or passive L-C filters are used as function of the operating frequencies and the applications.



To obtain the lower or the upper side band, the frequency of the carrier generator is varied, keeping the filter as it is.

Balanced Modulator

The “balanced modulator” is a circuit, which can generate an amplitude modulation signal with suppressed carrier, consisting of single side bands. To obtain this result, it is sufficient to multiply the carrier signal and the modulating one between them. If $v_m(t)$ is the modulating signal:

$$V_m(t) = B \sin(2\pi f \cdot t)$$

And $v_c(t)$ is the carrier:

$$V_c(t) = A \sin(2\pi F \cdot t)$$

The modulated signal $v_M(t)$ obtained by their multiplication is constituted by two sine components (fig. 3A)

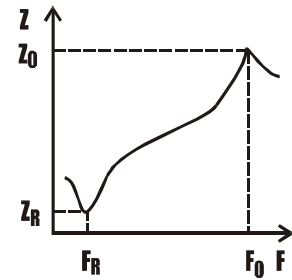
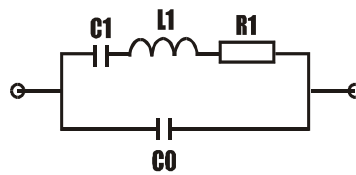
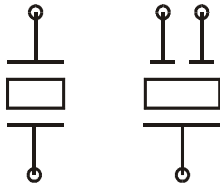
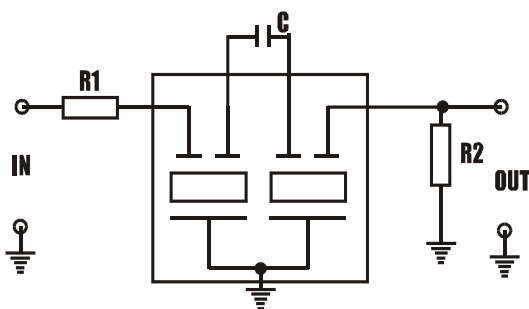
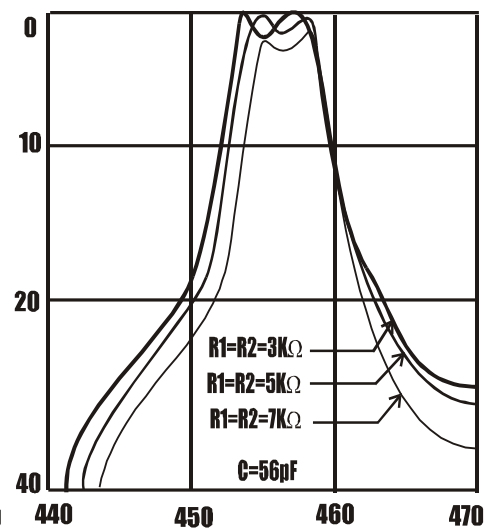
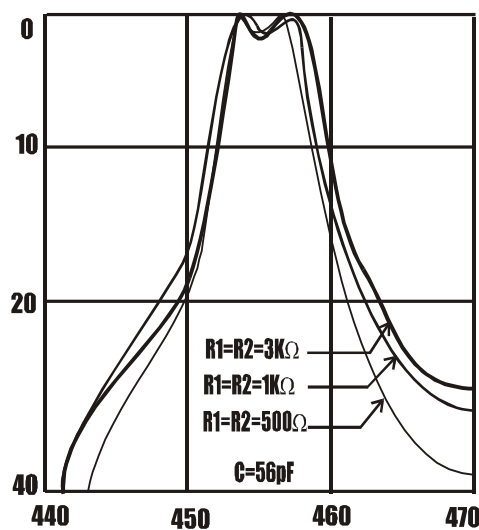
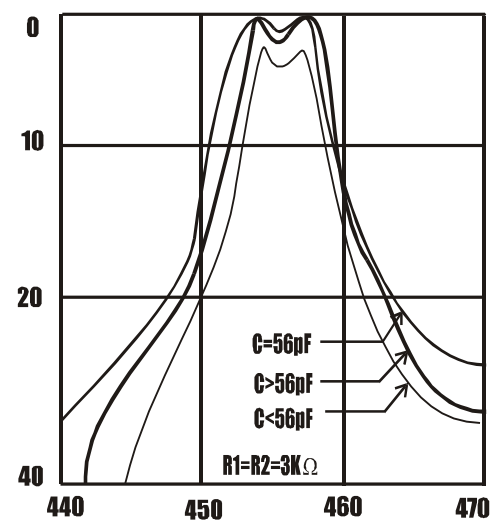
$K \cos[(2\pi(F-f)) \cdot t]$	lower side band
$K \cos[(2\pi(F+f)) \cdot t]$	upper side band

An integrated circuit commonly used as signal multiplier, and then as balanced modulator. This has a quadruple differential amplifier, driven by a further differential amplifier. The output signal consists of a constant (gain), which multiplies the product of two input signals.

Ring Modulator

The ring modulator is another example, which can generate an amplitude modulation with suppressed carrier. The electrical diagram is shown in fig. 3.3. The following are the operations:

- During the positive half-wave of v_c , supposing post B positive in respect to post B' the diodes D1 and D2 are in conduction and across the output there is a signal v_m .
- During the negative half wave of v_c , with the post B' now positive in respect to post B, diodes D3 and D4 are in conduction and the signal v_m is reported across the output.
- If the diodes have the same electrical characteristics and if the central posts of TR1 and Tr2 are correct, the two currents I and I' caused by carrier v_c are equal, and so there contribution across the output is null; in this way the carrier is null.
- The waveforms of the signals related to the ring modulator are reported in fig. 3.4. Removing the two diodes D3 and D4 you obtain a different form of ring modulator.

CERAMIC FILTER**A) TWO & THREE TERMINAL FILTER****B) EQUIVALENT ELECTRICAL CIRCUIT****C) IMPEDANCE****D) CERAMIC FILTER WITH COUPLED CELLS****FIG.3C. FREQUENCY RESPONSE OF CERAMIC FILTER**

Ceramic Filter

A ceramic filter is a band pass filter using a piezoelectric ceramic material as Electro-mechanic transducer and mechanic resonator. The symbols of the cells with 2 and 3 terminals are reported in fig.3C.A. Fig.3C.B&C reports the equivalent circuit of the filter with 2 terminals and its impedance at frequency variations. In many cases the filter consists of 2 cells coupled using a small capacity. Important parameters of the ceramic filter are the input impedance, the output impedance and the coupling capacity between two sections. Fig. 3.C. Shows ideal frequency response of the ceramic filter with 455 KHz (the same used for the exercises fig.3.5) at variation of the coupling capacity and input and output impedance matching.

The ceramic filter mounted on ACT-02 has the following main characteristics:

- Central frequency : 455 KHz
- Band width at –3dB : 4.5 KHz $\pm$ 1KHz
- Input and Output impedance : 3 K Ω
- Coupling Capacity : 56 pF.

Response Curve of the Filter Detected with Wobulated Generator.

The response curve (or frequency response) of a generic quadruple is determined applying a variable frequency signal across the input and detecting the amplitude of the output signal. The attenuation of the quadruple, measured at different frequencies, is given by:

$$A = V_o / V_i$$

And, in decibel: $A_{dB} = 20 \bullet \text{Log} (V_o / V_i)$.

The attenuation representation as function of the frequency determines the response curve of the quadruple.

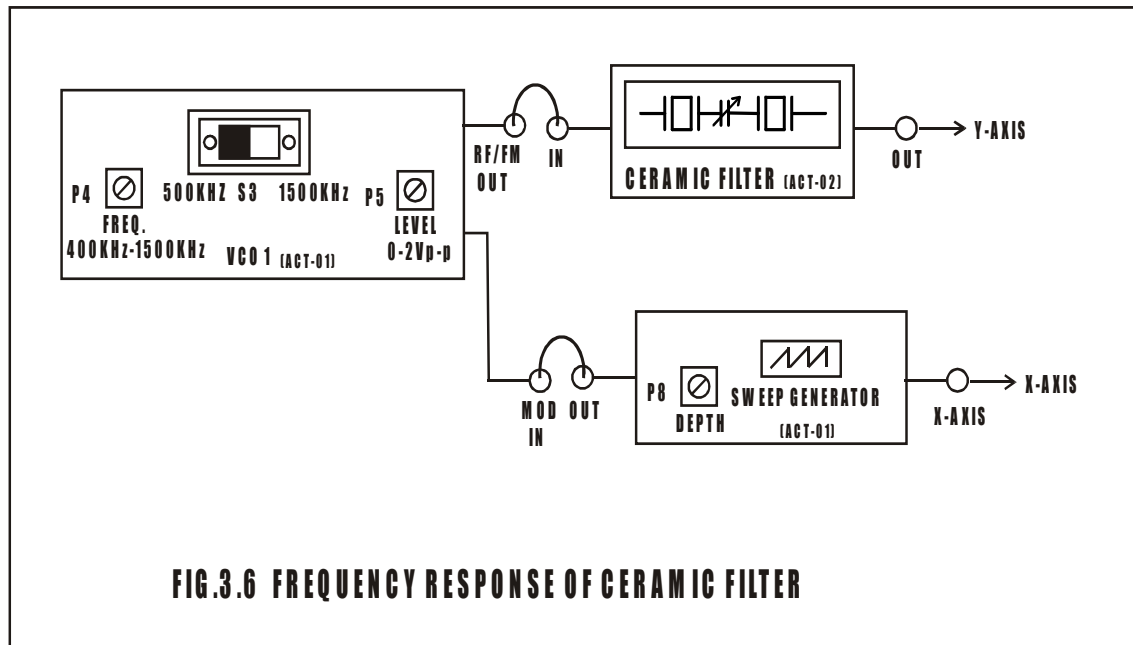
SSB signal demodulation

The SSB signal demodulation requires the presence of the carrier, which must be locally generated in the receiver.

To obtain the modulating signal from the modulated signal, multiply the modulating signal and the locally generated carrier. Filter out the result to extract the modulating signal. The circuit carrying out the multiplication of the two signals can be the same used to generate the modulation with suppressed carrier in transmission. When used as demodulator the circuit is commonly called product detector.

Let's now examine the demodulator operation. If for example:

$$V_M(t) = K \bullet \cos [(2\pi (F+f) \bullet t]$$



The signal (upper side band in the example) is:

$$V_c(t) = A \cdot \sin(2\pi F \cdot t)$$

The carrier locally generated in the receiver. The product $M(t) \cdot v_c(t)$

$$K' \cdot \sin(2\pi f \cdot t)$$

$$K' \cdot \sin(2\pi (2F+f) \cdot t)$$

Then, supposing to have the upper side band and applying the carrier again, across the output there are the carrier modulator and a really higher frequency, which can be removed with a low pass filter. If the carrier inserted in reception again does not have the same frequency of the carrier suppressed in transmission, the frequency of the demodulated signal is translated by the difference between the two carriers, altering in this way the reception. To reduce these differences to the minimum, the transmission carrier as well as the reception one is usually generated by quartz oscillators.

A1. To Observe the Frequency Response of Ceramic Filter

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply
- Oscilloscope
- Connecting Links
- Frequency Meter

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.3.6 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **450 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1.5Vpp**.
 - Using pot **P8** keep SWEEP **DEPTH** almost to the minimum (counterclockwise).
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Ceramic filter (ACT-02) **IN** post.
- Connect the output of Ceramic filter (ACT-02) **OUT** post to the Y-axis of Oscilloscope.
- Connect the modulation in (ACT-01) **MOD IN** post of VCO1 to the post **OUT** of SWEEP circuit (ACT-01).
- Connect the **X-axis** post of sweep detector of ACT-01 to the **X-axis** of Oscilloscope.
- Keep Oscilloscope in X- Y mode.
- Adjust the central frequency of the VC01 and the amplitude of the Sweep (DEPTH) to obtain a curve on the oscilloscope which is similar to the ones of fig.3.5
- Keep volt/div of CRO, X-axis= 200mV, Y-axis= 500mV.
- Vary the coupling capacity CV1 and check the 3 following conditions: the response curve becomes narrower (CV1<56pF); the response is flat (CV1=56pF); the response has 2 resonance peaks (CV1>56pF).

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500 KHz position, 400 KHz to 500 KHz variation disabled and output gives fixed frequency.

A2. To Plot the Frequency Response Curve of Ceramic Filter

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.
- Frequency Meter.

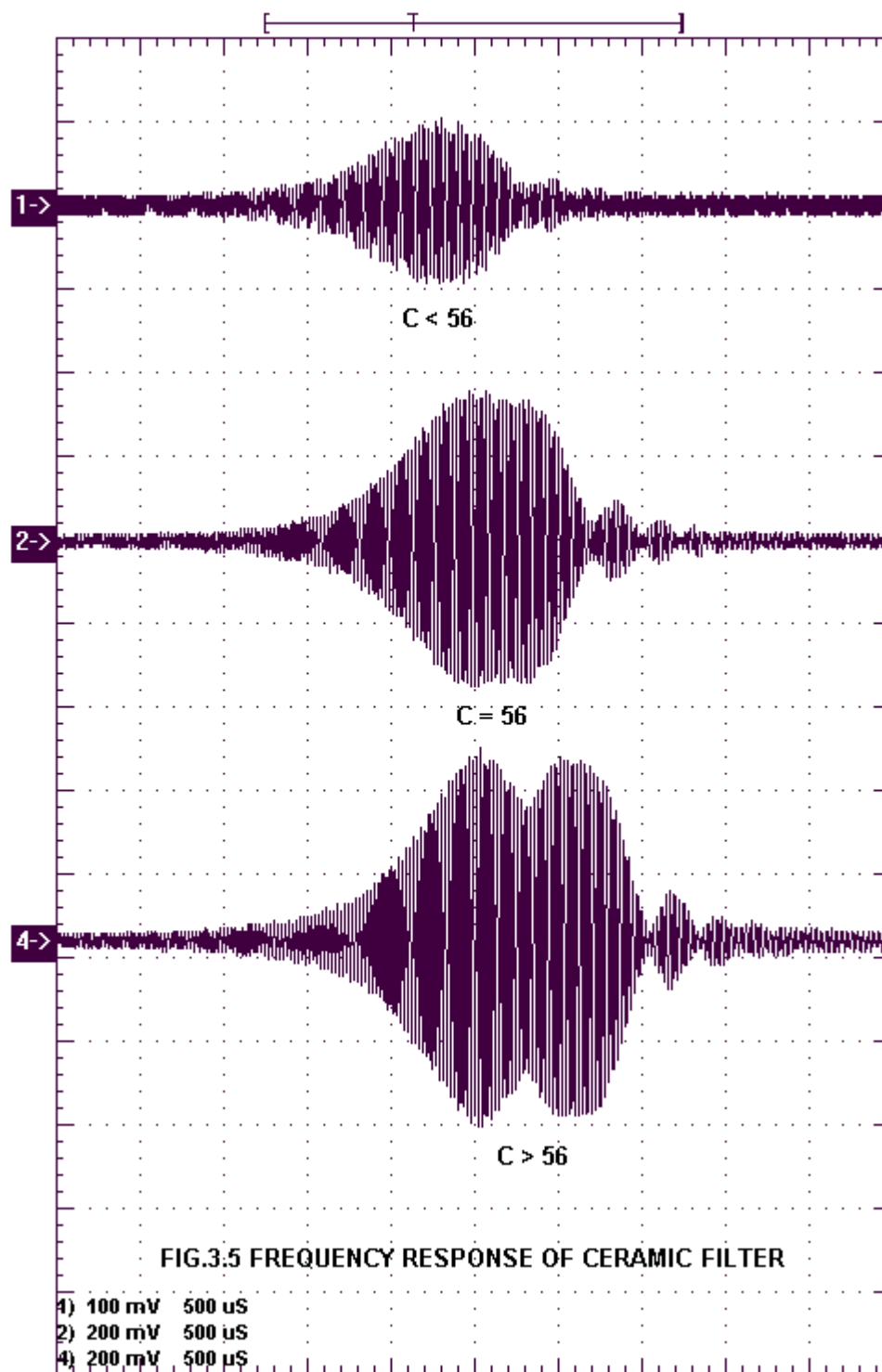
Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Do the procedure same as (A1).
- Apply a signal with frequency of 455 KHz, corresponding to the central frequency of the filter to the input of Ceramic filter (ACT-02) **IN** post.
- If V_o and V_i are the peak-to-peak voltages measured across the output and the input of the filter. The attenuation A of the filter at 455 KHz is given by $A = V_o / V_i$ and, in decibel: $A_{dB} = 20 \cdot \log (V_o / V_i)$
- Vary the frequency from 445 KHz to 465 KHz in steps of 1 KHz and measure the corresponding input and output voltages i.e. V_i & V_o respectively. Calculate A_{dB} in correspondence to each frequency and report all in a table.
- With the data in the table, plot a graph setting A_{dB} on the Y-axis (vertical) and the frequency on the X-axis (horizontal): you obtain the frequency response curve of the filter, similar to the one fig.3.5.
- Evaluate the pass band B of the filter:

$$B = f_2 - f_1$$

With frequencies f_2 and f_1 in correspondence to which the attenuation A_{dB} increases of 3dB in respect to its minimum value (fig.3.5) you will find a bandwidth of about 3 – 4KHz



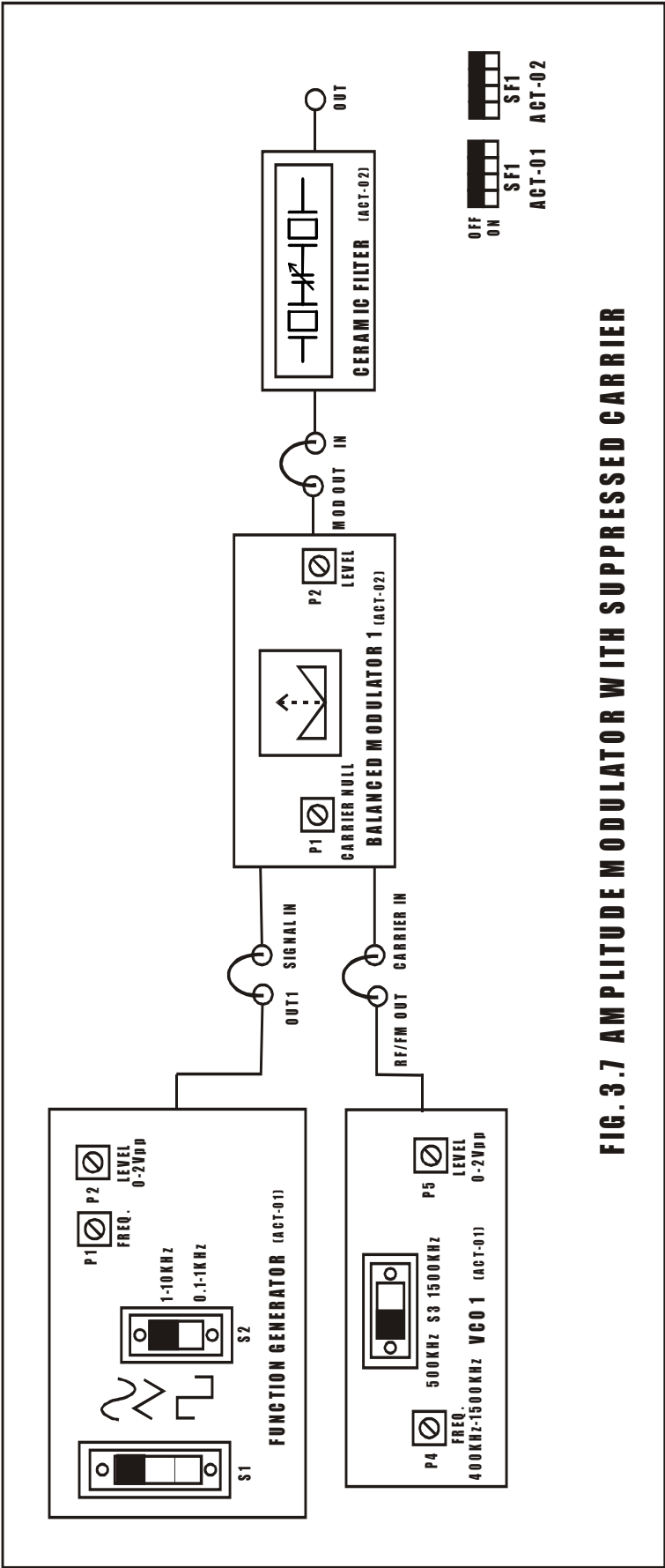


FIG. 3.7 AMPLITUDE MODULATOR WITH SUPPRESSED CARRIER

B. To Study the operation of Amplitude Modulator with Suppressed Carrier

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.
- Frequency Meter.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.3.7 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **450 KHz**.
 - Using pot **P5** keep VCO2 amplitude at **1.5Vpp**.
- In ACT-02
 - Keeps BALANCED MODULATOR1 **CARRIER NULL** in center position using pot **P1**.
 - Keep BALANCED MODULATOR1 **LEVEL** at maximum using pot **P2**.
- Connect the sine wave output (ACT-01) **OUT1** post to the input of Balance modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect the oscilloscope to the inputs of the modulator and detect the modulating signal and the carrier signal (fig.3.8 a/b)
- Move the probe from **SIGNAL IN** post to **MOD OUT** post (modulator output), where the modulated signal is detected (fig.3.8.c). Reset the level of the modulating signal, and adjust the **CARRIER NULL** to obtain the minimum of the output carrier of the modulator. Take the amplitude of the modulating signal about 0.5Vpp. Note that the waveform of the modulating signal does not correspond to the envelope of the modulated signal, as it occurs, instead, in case of signal AM.
- Vary the amplitude of the modulating signal and check the corresponding variation of the modulated signal amplitude. Note that it is different from the Amplitude modulation where the modulated signal is never null, the modulated signal annuls when the modulating signal is null

- Vary frequency and amplitude of the modulating signal, and check the corresponding variations of the modulated signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500 KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.

C. To study single side band generation

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.
- Frequency Meter.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.3.7 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **3 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **1Vpp**.
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **452 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1.5Vpp**.

- In ACT-02
 - Keeps BALANCED MODULATOR1 **CARRIER NULL** in center position using pot **P1**
 - Keep BALANCED MODULATOR1 **LEVEL** at maximum using pot **P2**.
 - Connect the sine wave output (ACT-01) **OUT1** post to the input of Balance modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect the output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of Ceramic filter (ACT-02) **IN** post.
- With the oscilloscope, monitor the output signal of the ceramic filter post **OUT**, and check that it is a sine wave. You can conclude that the filter extracts only one of the two components generated by the balanced modulator.
- Measure the frequency f_c of the carrier, f_m of the modulating signal and f_{ssb} of the SSB signal across the output of the ceramic filter post **OUT**.
- Check that: $f_{ssb} = f_c + f_m$
This means that the band extracted by the ceramic filter corresponds to the Upper Side Band
- Repeat the last measurements setting the frequency of the carrier to 458 KHz. You obtain:
 $f_{ssb} = f_c - f_m$
This means that the band extracted by the ceramic filter corresponds to the Lower Side Band
- Increase the frequency of the modulating signal (FUNCTION GENERATOR) and check that the SSB signal attenuates and tends to zero. Explain the reason.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.

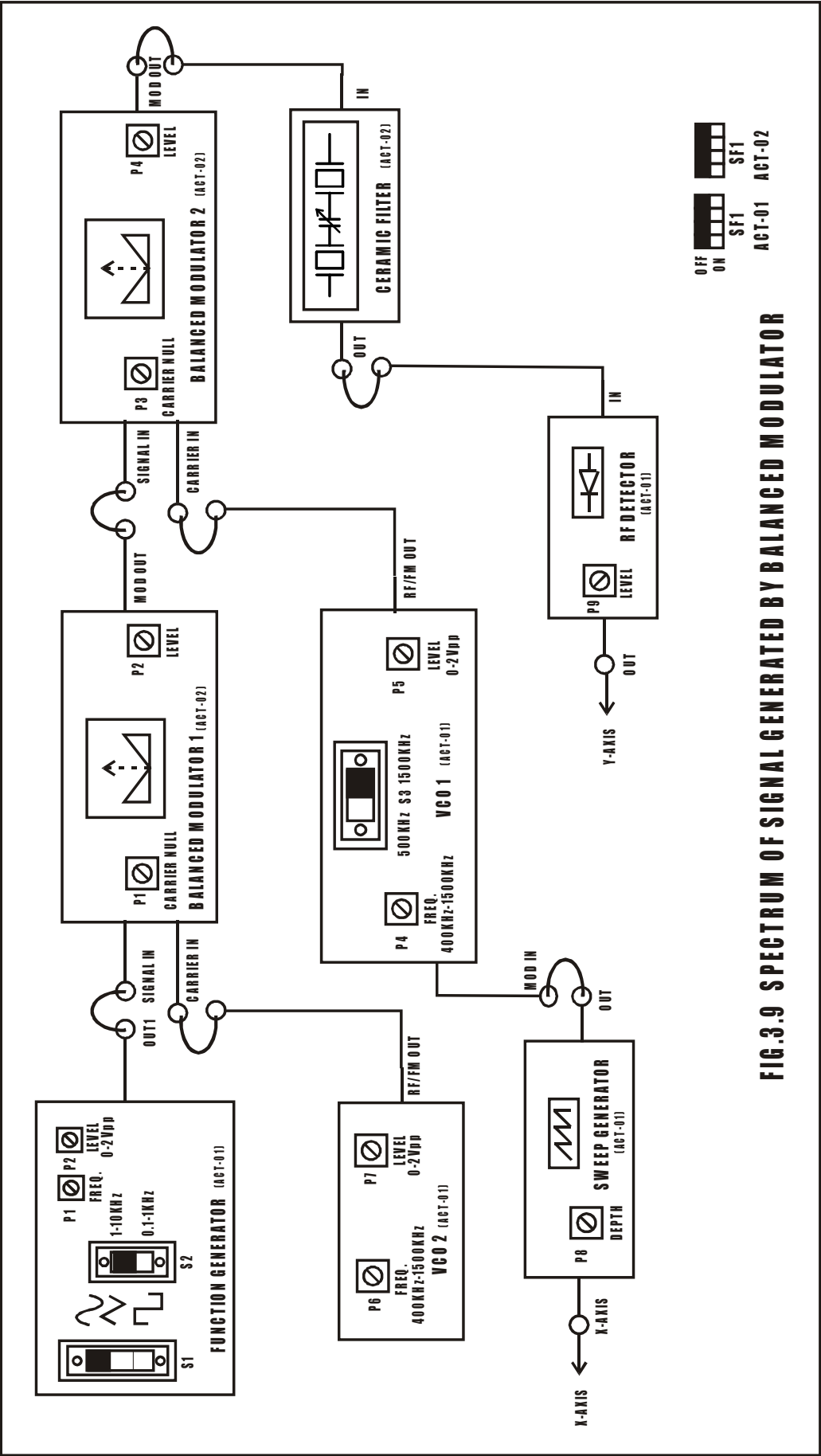


FIG.3.9 SPECTRUM OF SIGNAL GENERATED BY BALANCED MODULATOR

D. To Study and observe the Spectrum of the signal Generated by the Balance Modulator

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.
- Frequency Meter.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

Refer to the: “Appendix A” to explain the operating principle of the Spectrum Analyzer.

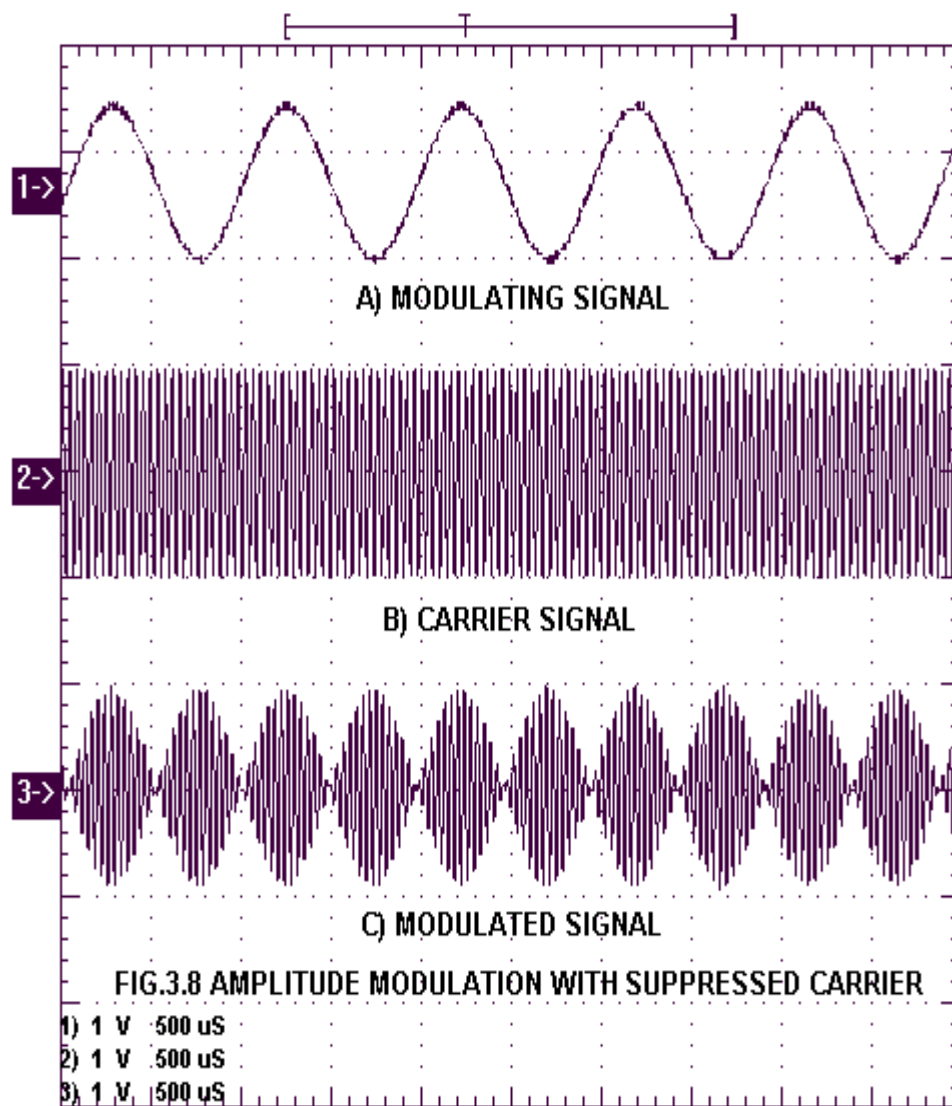
- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.3.9 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **10 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **900 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
 - Using pot **P6** keep VCO2 frequency at **450 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **1Vpp**.
 - Using pot **P8** keep SWEEP **DEPTH** almost completely counter clockwise.
 - Using pot **P9** keep RF DETECTOR **LEVEL** completely clockwise.
- In ACT-02
 - Keeps BALANCED MODULATOR1 **CARRIER NULL** in center position using pot **P1**.
 - Keep BALANCED MODULATOR1 **LEVEL** at maximum using pot **P2**.
 - Keep BALANCED MODULATOR2 **CARRIER NULL** in center position, so that the circuit operates as frequency converter (balanced modulator with suppressed carrier) pot **P3**.
 - Keep BALANCED MODULATOR2 **LEVEL** at maximum using pot **P4**.
- Connect output of generator (ACT-01) **OUT1** post to the input of Balance Modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of Balance modulator 2 (ACT-02) **SIGNAL IN** post.

- Connect output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 2 (ACT-02) **CARRIER IN** post.
- Connect output of Balance modulator 2 (ACT-02) **MOD OUT** post to the input of Ceramic filter (ACT-02) **IN** post.
- Connect output of Ceramic filter (ACT-02) **OUT** post to the input of RF detector (ACT-01) post **IN**.
- Connect modulation in of (ACT-01) **MOD IN** post of VCO1 to the out of SWEEP circuit of (ACT-01) **OUT** post.
- Set the oscilloscopes in X-Y mode.
- Connect the SWEEP generator **X-axis** post of ACT-01) to the X-axis, and the detected signal post **OUT** of RF detector ACT-01) to the Y-axis.
- Vary the frequency of the carrier (VCO2) until the oscilloscope shows a representation similar to the one of fig.3.2. To obtain the best wave form adjust: the deviation of the SWEEP generator (DEPTH) fully clockwise; vary VCO1 carrier to obtain at center, the pass band of the ceramic filter (variable capacity) and vary the CARRIER NULL of the balanced modulator
- The waveform represents the spectrum of the signal generated by the balanced modulator, constituted by the 2 side bands.
- Vary frequency, amplitude of the modulating signal and examine how the spectrum varies.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 3 of SF 1 in ACT-02 Switch fault section to ON position. This will open R30 from gain adjust pins of balance modulator 2. Balance modulator2 maximum output reduced to 100 mV from 2V.



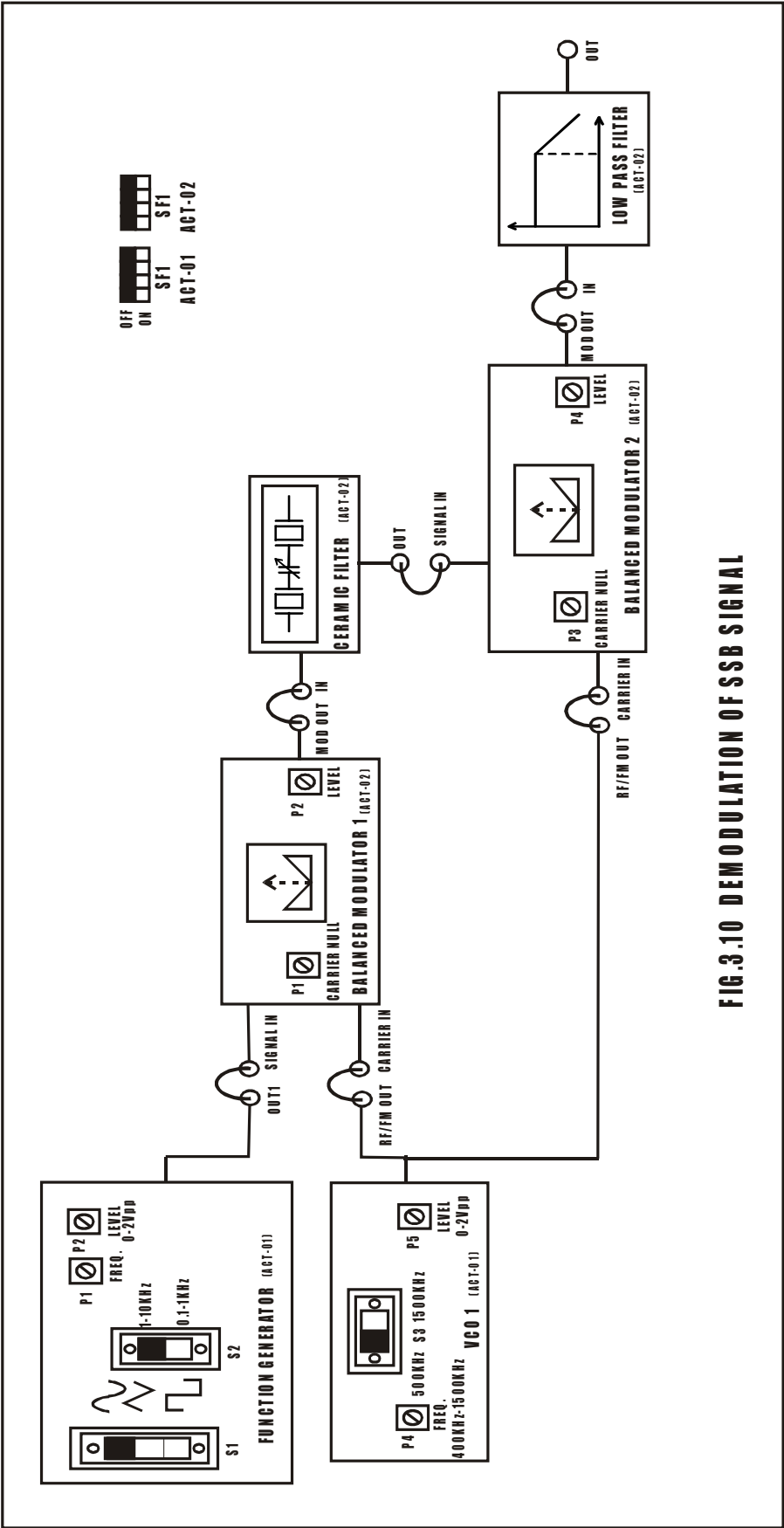


FIG.3.10 DEMODULATION OFSSB SIGNAL

E. To Study the Demodulation of the SSB Signal

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.
- Frequency Meter.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

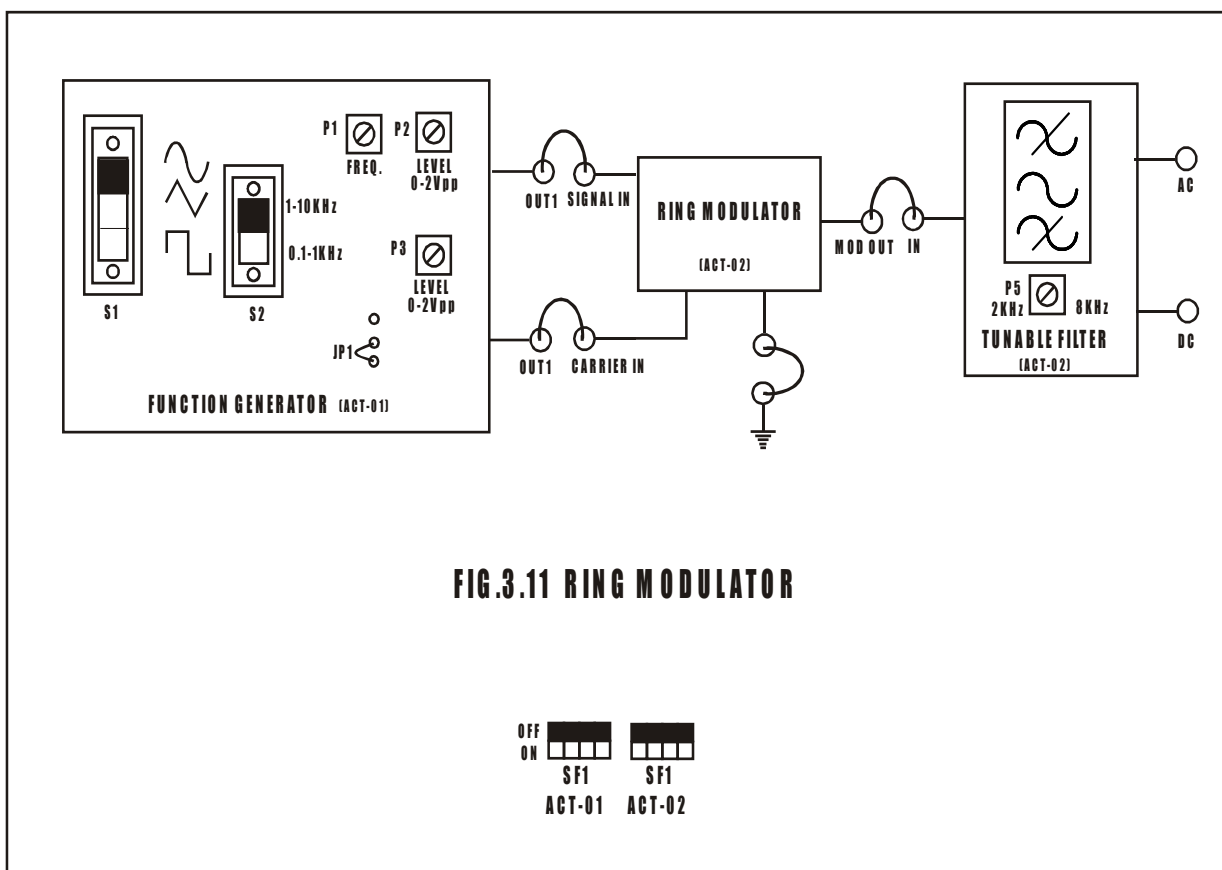
- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.3.10 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **3 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **1Vpp**.
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **452 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1.5Vpp**.
- In ACT-02
 - Keeps BALANCED MODULATOR1 **CARRIER NULL** in center position using pot **P1**
 - Keep BALANCED MODULATOR1 **LEVEL** at maximum using pot **P2**.
- Connect the sine wave output (ACT-01) **OUT1** post to the input of Balance modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect the output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of Ceramic filter (ACT-02) **IN** post.
- Connect the output of SSB modulator post **OUT** of ceramic filter to the input **SIGNAL IN** post of the balance modulator 2. In this case balance modulator is used as product detector.
- Across the input **CARRIER IN** of the balance modulator 2 apply the same **CARRIER IN** supplied to the balance modulator 1.
- Connect the output of the balance modulator II post **MOD OUT** to the low pass filter post **IN**
- Observe the signals across the following posts:
 - Post **OUT** of ceramic filter (ACT-02) i.e. output of the SSB modulator: it is a sine wave which corresponds to the Upper Side Band, at the base of the frequency set for the carrier.

- Post **MOD OUT** of Balance Modulator 2 (ACT-02) i.e. output of the product detector: there is a sine wave with frequency equal to the one of the Modulating signal, to which a component with much higher frequency is laid over.
- Post **OUT** (ACT-02) i.e. output of the low pass filter: the high frequency component has been removed and the demodulated signal remains, equal to the transmitted modulating signal.
- Increase the frequency of the modulating signal and check that the detected Signal attenuates and tends to a null.
- Disconnect VCO1 from the balanced modulator 2 POST **CARRIER IN** and Connect VCO2. In this way, you supply the product detector with a different Carrier form the one used in the modulator.
- Vary the frequency of VCO2, to find out a frequency which is more equal to The one used by the modulator (generated by VCO1). Check that it is really Difficult to obtain the starting modulating signal across the output of the filter This is due to the fact that it is very difficult to set the two VCO's to the same Frequency

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 3 of SF 1 in ACT-02 Switch fault section to ON position. This will open R30 from gain adjust pins of balance modulator 2. Balance modulator2 maximum output reduced to 100 mV from 2V.
- Put switch 4 of SF 1 in ACT-02 Switch fault section to ON position. This will open filter capacitor C24 from low pass filter, filter output gets distorted.



F. To study the operation of the Ring Modulator

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Connecting Links.
- Frequency Meter.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

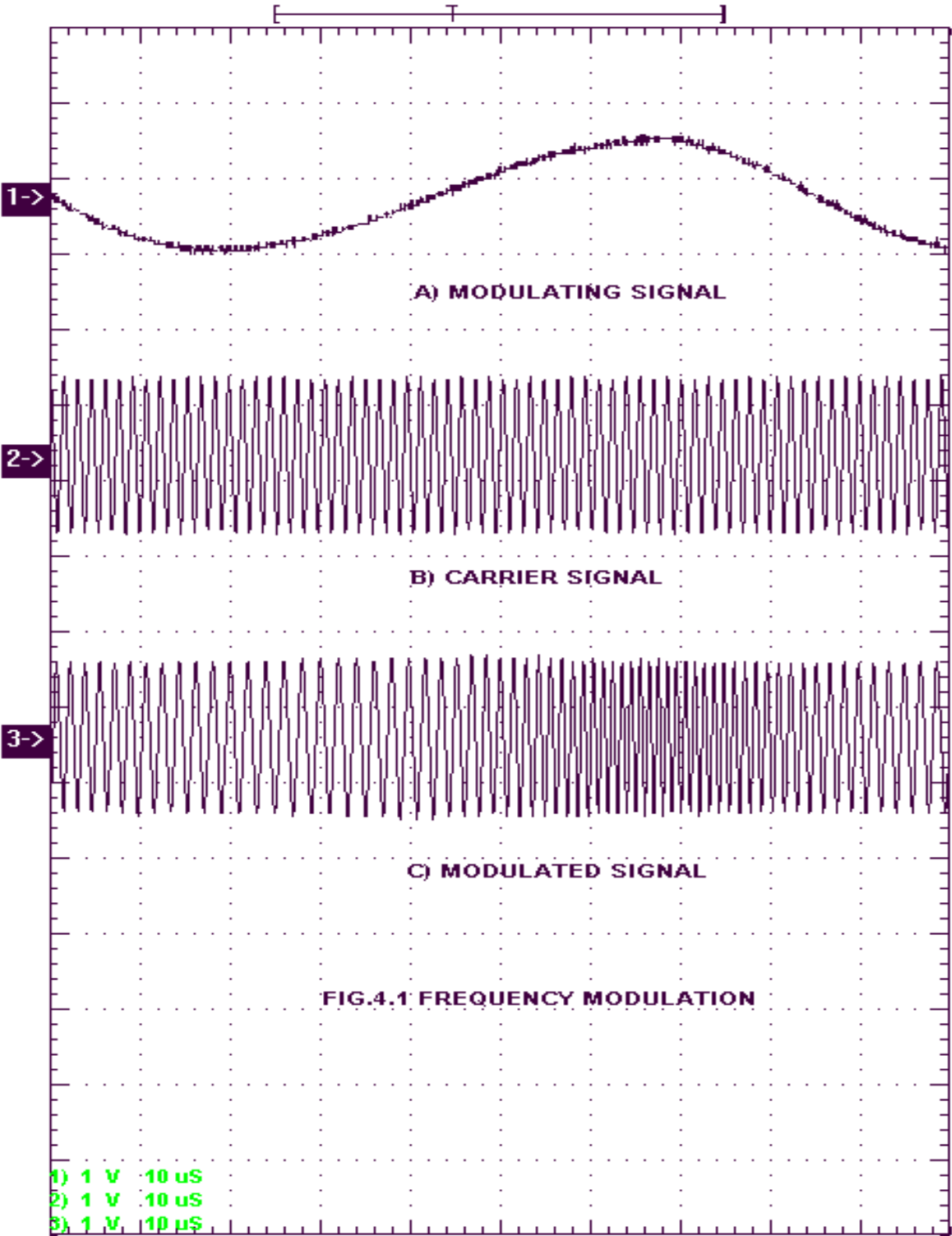
- Connect power supply with proper polarity to the kits ACT-01 and ACT-02 and switch it ON.
- Refer to the fig.3.11 & Carry out the following connections and settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep jumper **JP1** in **4F** position.
 - Keep Amplitude of **OUT2** at 200mV.
- Connect the output of Function Generator (ACT-01) **OUT1** post to the input of Ring modulator (ACT-02) post **SIGNAL IN**.
- Connect the output of Function Generator (ACT-01) **OUT2** post to the **CARRIER IN** post (ACT-02) of Ring modulator. Connect **GND** to CARRIER IN post. Connect the modulation out of Ring modulator (ACT-02) post **MOD OUT** to the input of tunable filter (ACT-02) **IN** post.
- Connect the oscilloscope to the inputs of the modulator and detect the modulating signal and the carrier signal as shown in fig. 3.4 a/b.
- Move the probe from input to output of the modulator, where the modulated signal is detected as shown in fig.3.4c
- Connect the oscilloscope and frequency meter to post **OUT** output of the band pass filter with variable tuning. Vary the central frequency of the filter, and check that two sine waves are extracted with frequency of about 4-1 KHz (lower side band) and 4+1 KHz (upper side band)

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 1 of SF 1 in ACT-02 Switch fault section to ON position. This will open D4 from ring modulator; Ring modulator operates in three-diode configuration.

EXPERIMENT NO.4



EXPERIMENT NO.4

NAME:-

TO Study Frequency Modulation

OBJECTIVE:-

- A1 To plot the modulation characteristic of Frequency modulator.
- A2 To calculate the modulation sensitivity of Frequency modulator.
- A3 To calculate non-linearity of Frequency modulator.
- B To observe and measure frequency deviation and modulation index of FM.
- C To Study marker insertion to evaluate the Frequency deviation F.
- D To observe the Spectrum of the FM Signal.

THEORY:-

Frequency Modulation

It is a type of modulation in which the frequency of the high frequency (Carrier) is varied in accordance with the instantaneous value of the modulating signal.

Main Aspects

Consider a sine wave signal $v_m(t)$ with pulse w (fig.4.1):

$$v_m(t) = B \cdot \sin(\omega t)$$

And another sine wave $v_c(t)$ with upper Ω pulse:

$$v_c(t) = A \cdot \sin(\Omega t).$$

The signal $v_m(t)$ is called modulating signal, the signal $v_c(t)$ is called carrier signal.

Vary the frequency of the carrier $v_c(t)$ in a way proportional to the amplitude of the modulating signal $v_m(t)$. You obtain a $v_m(t)$ frequency modulated diagonal, which can be expressed by the relation:

$$v_m(t) = A \cdot \sin[\theta(t)]$$

With $\theta(t)$ instantaneous angle function of $v_m(t)$

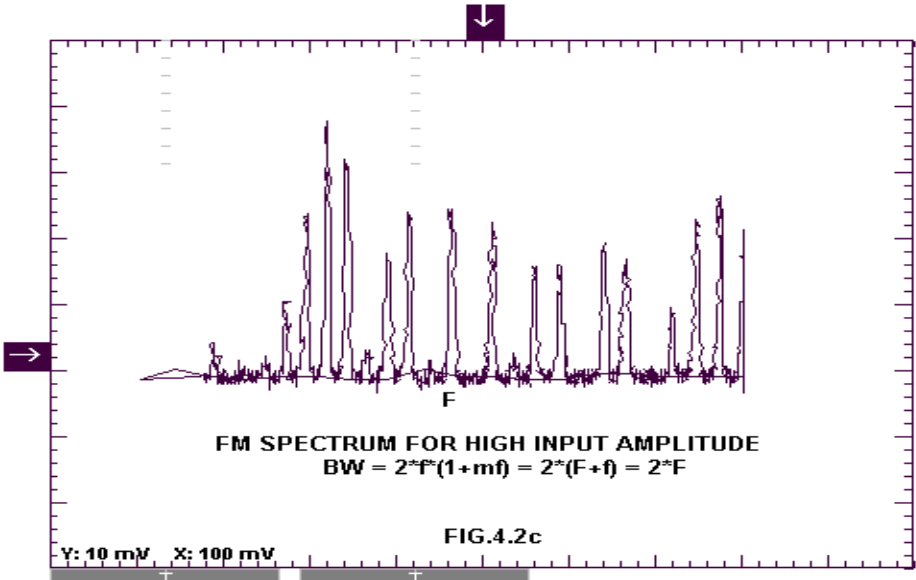
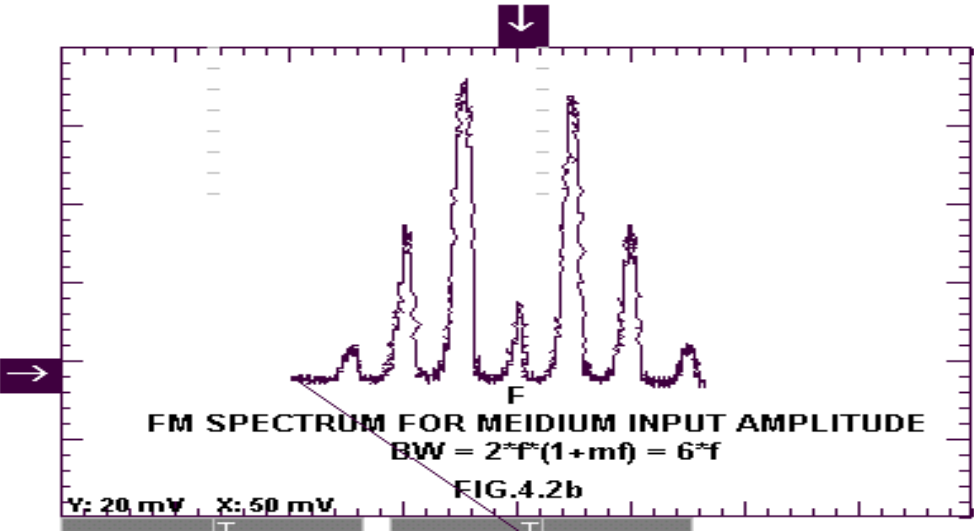
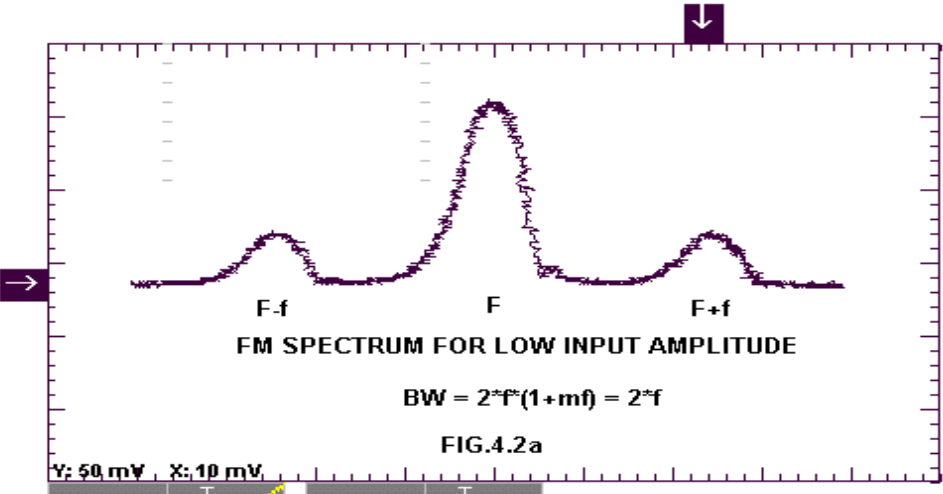
Mathematical Expression of the Frequency Modulated Signal

The instantaneous pulse $\Omega(t)$ of the FM signal by definition:

$$\Omega(t) = \Omega + K \cdot v_m(t)$$

With $h\Omega$ =carrier pulse

K = modulation sensitivity



The instantaneous angle $\Omega(t)$ to be used as subject of the sine to obtain the mathematical operation of the FM signal, is detected by integrating $\Omega(t)$:

$$\theta(t) = \int \Omega(t) dt$$

In the case of modulating sine wave signal [$v_m(t) = B \cdot \sin(w \cdot t)$], $\theta(t)$ it results:

$$\theta(t) = \Omega(t) - (K \cdot B/w) \cdot \cos(wt).$$

The expression of the frequency modulated signal $v_m(t)$ becomes:

$$v_m(t) = A \cdot \sin[\Omega(t) - (K \cdot B/w) \cdot \cos(wt)].$$

Frequency Deviation ΔF and Modulation Index M_f

The instantaneous frequency $F(t)$ of the carrier modulated by a sine wave, results:

$$F(t) = \frac{\Omega(t)}{2\pi} = \frac{\Omega}{2\pi} + K \cdot B \cdot \sin(wt)$$

And oscillates between a minimum F_{min} and a maximum value F_{max}

$$F_{min} = \frac{\Omega}{2\pi} - \frac{K \cdot B}{2\pi} \quad F_{max} = \frac{\Omega}{2\pi} + \frac{K \cdot B}{2\pi}$$

The frequency deviation ΔF represents the maximum shift between the modulated signal frequency, over and under the frequency of the carrier:

$$\Delta F = \frac{F_{max} - F_{min}}{2}$$

We define as modulation index m_f the ratio between ΔF and the modulating frequency f :

$$m_f = \frac{\Delta F}{f}$$

Theoretical Spectrum of the Modulating Signal

Using the modulation index m_f , the modulated signal is expressed by the relation:

$$v_m(t) = A \cdot \sin[\Omega(t) + \cos(wt)]$$

This, with trigonometric relations, can be transformed into

$$v_m(t) = A \cdot \{\sin(\Omega t) \cdot \sin[m_f \cdot \cos(wt)] - \cos(\Omega t) \cdot \cos[m_f \cdot \sin(wt)]\}$$

Using the Bessel function, the last relation becomes:

$$v_m(t) = A \cdot \{J_0 \cdot \sin(\Omega t) - J_1 \cdot [\sin(\Omega - w)t - \sin(\Omega + w)t] + \\ + J_2 \cdot [\sin(\Omega - 2w)t + \sin(\Omega + 2w)t] + \\ - J_3 \cdot [\sin(\Omega - 3w)t + \sin(\Omega + 3w)t] + \dots\}$$

With $J_3(m_f)$ Bessel function of the first kind, with order n and subject m_f

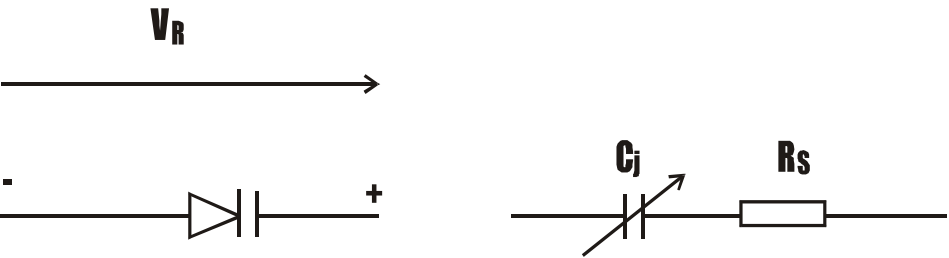


FIG.4.3 SYMBOL & EQUIVALENT CIRCUIT OF VARICAP DIODE

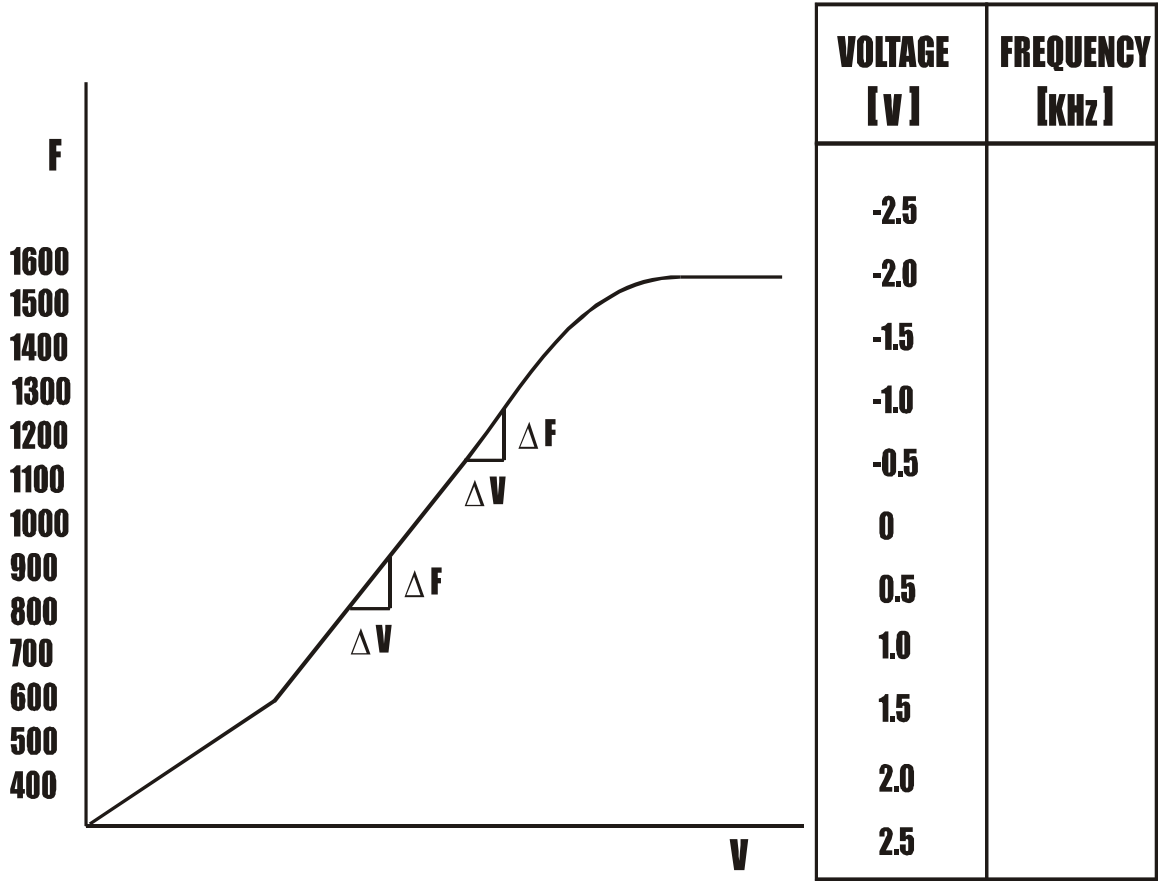


FIG.4.4 MODULATION CHARACTERISTIC CURVE

Actual Spectrum of the Modulated Signal

The last consideration give that the spectrum of a signal modulated in frequency by a sine wave with frequency f is composed by an infinite set of harmonics, symmetrical in respect to the carrier, (at distance f , $2f$, $3f$...from it) and with amplitude proportional to the Bessel functions. Theoretically, an FM signal produces an infinite spectrum. But in practice, the Bessel functions of order superior to $(1 + m_f) = 2(\Delta F/f)$

With m_f very small ($\ll 1$) the last relation is simplified as:

$$B_w = 2 \cdot f$$

While for high values of m_f (> 10) it becomes:

$$B_w = 2 \cdot \Delta F.$$

The figure 4.2 shows spectrum of FM.

Frequency Modulation Generation

The circuits used to generate a frequency modulation must vary the frequency of a high frequency signal (carrier) as function of the amplitude of a low frequency signal (modulating signal). In practice there are two main methods used to generate the FM:

- Direct method: an oscilloscope is used in which the reactance of one of the elements of the resonant circuit depends on the modulating voltage. The most common device with variable reactance is the Varactor or Varicap, which is a particular diode whose capacity varies as function of the reverse bias voltage. The frequency of the carrier is established with AFC circuits (Automatic Frequency Control) or PLL (Phase Locked Loop)
- Indirect method: the FM is obtained in this case by a Phase Modulation, after the modulating signal has been integrated. In the phase modulator the carrier can be generated by a quartz oscillator, and so its frequency stabilization is easier.

In the circuit used for the exercise, the frequency modulation is generated by Hartley oscillator whose frequency is determined by fixed inductance and the capacity (variable) supplied by Varicap diodes.

The Varicap Diode

The Varicap (or Varactor) is a diode whose terminals act like a variable capacitor, whose capacitance is depend on the applied reverse voltage. The symbol and the equivalent circuit of the Varicap diode are shown in fig.4.3, where:

- C_j = junction capacity
- R_s = series resistance (it drops as the applied reverse voltage increases)

The junction capacity C_j depends on the reverse voltage V_R applied to the diode, according to the relation:

$$C_j = \frac{C_0}{(1 + \frac{V_R}{V_D})^\eta}$$

Where: V_R = reverse voltage applied to Varicap

C_0 = junction capacity for $V_R=0$

V_D = junction potential (0.6 V in the silicon diodes)

η = it depends on the manufacturing process; it ranges between 0.3 and 0.6 approx. The factor of merit Q of the diode is expressed by: $Q = 1/(w \cdot C_j \cdot R_s)$.

A1. To plot the Modulation Characteristics of Frequency Modulator

EQUIPMENT:-

- Modules ACT-01
- Power supply
- Oscilloscope
- Voltmeter
- Frequency Meter
- Connecting Links

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

The characteristic modulation curve is given by the output frequency of the modulator as function of the input modulating voltage (fig.4.4). It is possible to plot the curve of fig.4.4 post by post, using a potentiometer amplitude variation of the modulating signal is done and the corresponding output frequency of the modulator is measured.

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-02 and switch it on.
- In ACT-01
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **400 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
- Connect the oscilloscope and frequency meter to the output of the modulator **RF/FM OUT**.
- Connect the voltmeter to the cursor of the frequency regulation potentiometer post **VOLTS**.
- Vary the voltage in steps of 0.5 Volt and fill a table with the voltage values and the corresponding frequencies.
- Plot a graph with the measured voltage and frequency values. You obtain a curve similar to the one of fig.4.4
- From the analysis of the curve you can note that some segments do not have linear variation, while considering the whole characteristics you find a high non-linearity.

A2. To Calculate the Modulation Sensitivity of Frequency Modulator

EQUIPMENT:-

- Modules ACT-01
- Power supply
- Oscilloscope
- Voltmeter
- Frequency meter
- Connecting Links

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Perform the procedure as done in A1.
- Consider the modulator operation in the segment of curve within 700 to 1300 KHz, with central frequency of 1000 KHz. From the analysis of the curve of fig.4.4 it is possible to calculate the modulation sensitivity of the modulator.
- The modulation sensitivity S is defined as:

$$S = \frac{dF(v)}{dv}$$

Where F(v) is the instantaneous frequency function of the modulating voltage v. The last relation can be approximated writing the incremental ratio:

$$S = \frac{\Delta F}{\Delta v}$$

With reference to the curve of fig.4.4, in correspondence to the central frequency (1000 KHz) you obtain:

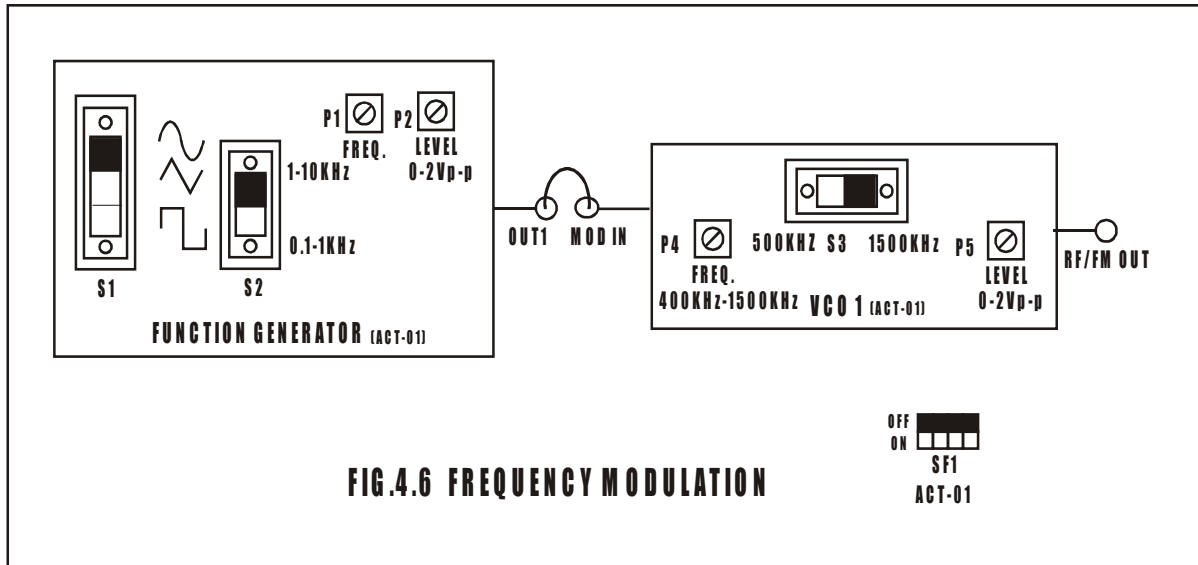
$$\Delta F = 50 \text{ KHz} \quad v \approx 125 \text{ mV} \quad \text{from which: } S_0 = 50/125 = 0.4 \text{ KHz/mV}$$

A3. To calculate Non-Linearity calculate Frequency Modulator

EQUIPMENT:-

- Modules ACT-01.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position



PROCEDURE:-

- Perform the procedure as done in A1 & A2.
- The non linearity N.L. of the modulator is defined as percentage relative shift of the sensitivity S from the S_0 value corresponding to the central frequency:

$$N.L. = \frac{S - S_0}{S_0} \times 100$$

As the sensitivity, also the linearity can be plotted point by point by the curve of fig.4.4. Considering for example the point around 1300KHz, the calculation is carried out as follows:
 $\Delta F' = 50\text{KHz} \quad V' \approx 175 \text{ mV}$
 From which:
 $S' = 50/175 \approx 0.29 \text{ KHz/mV}$
 $N.L. = \frac{0.4 - 0.29}{0.4} \times 100 = 27.5\%$
- Move the switch **S3** of VCO1 to 500KHz, vary the modulating voltage with steps of 50mV and plot the modulation curve from 400 to 500KHz about. You obtain a similar curve to the one of fig.4.4. Analyze the modulation sensibility and the non-linearity of the considered curve segment.

B. To Observe and measure Frequency Deviation and Modulation Index of FM

EQUIPMENT:-

- Modules ACT-01.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

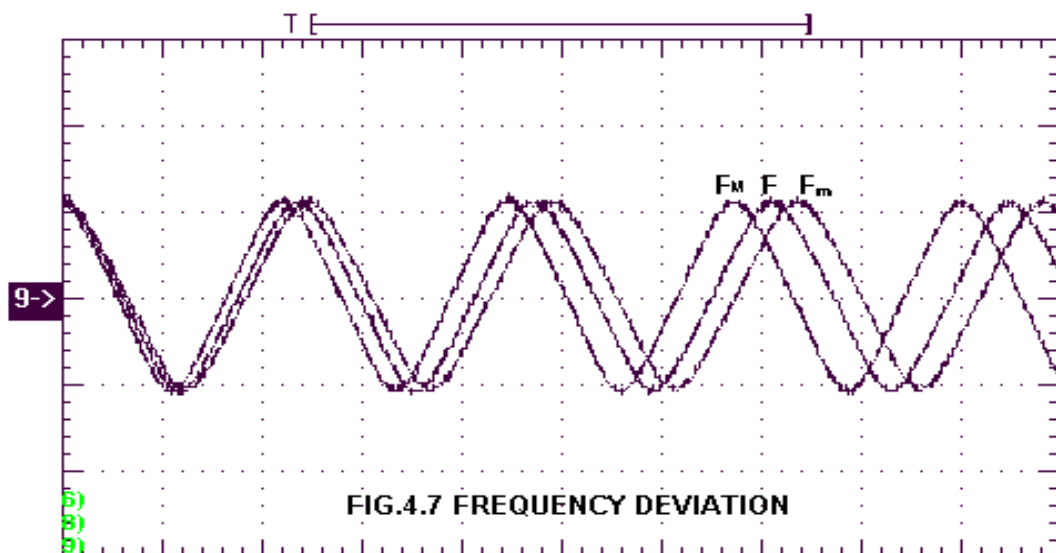
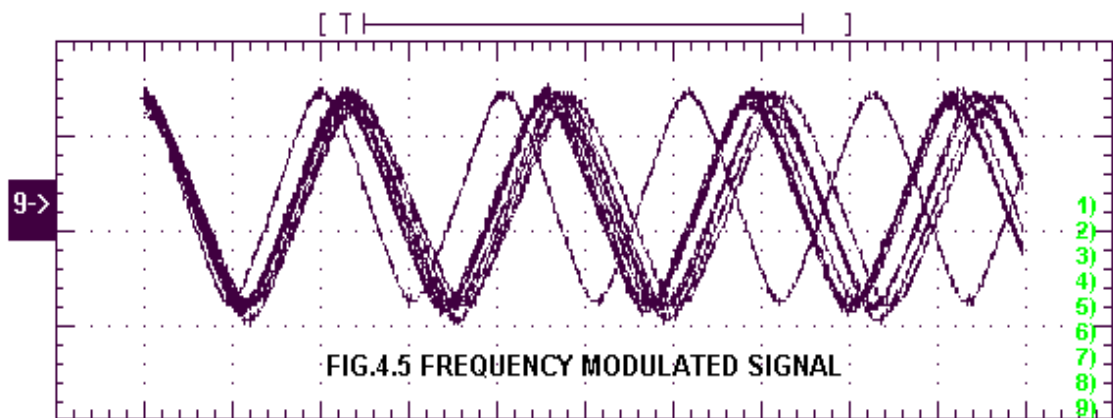
- Connect the power supply with proper polarity to the kits ACT-01 & ACT-02 and switch it on.
- Refer to the fig.4.6 & Carry out the following connections & Switch settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.2Vpp**.
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency on the center.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
- Connect the output of function generator (ACT-01) **OUT1** post to the modulation input of VCO1 (ACT-01) **MOD IN** post.
- Connect the oscilloscope to the output of the modulator post **RF/FM OUT** of VCO1. You obtain a waveform similar to the one of fig.4.5.
- The frequency deviation ΔF can be calculated as follows (refer to fig.4.7):
 - From the oscilloscope evaluate F_M and F_m , detecting the periods of the respective sine waves
 - The frequency deviation ΔF is defined as: $\Delta F = (F_M - F_m)/2$

You can note that if the modulator operates in a linear zone so F_M and F_m are over and under the central frequency F of the same quantity ΔF , otherwise this does not occur.
- The value of the modulation index m_f is calculated by the relation $m_f = \Delta F/f$, where f is the frequency of the modulating signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs. Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency



C. To study Marker Insertion to Evaluate the Frequency Deviation

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-02 and switch it on.
- Refer to the fig.4.8 & Carry out the following connections and switch settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in lower position (**0.1-1KHz**).
 - Using pot **P1** keep sine wave signal frequency at **0.1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **1Vpp**.
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency on the center.
 - Using pot **P5** keep VCO1 amplitude at **1Vpp**.
 - Using pot **P6** keep VCO2 frequency on the center.
 - Using pot **P7** keep VCO2 amplitude at **2Vpp**.
- Connect the output of function generator (ACT-01) **OUT1** post to the modulation input of VCO1 (ACT-01) **MOD IN** post.
- Connect the output of VCO2 (ACT-01) **RF/FM OUT** post to the **MARKER IN** post of VCO1 of ACT-01.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of RF detector ACT-01 **IN** post.
- Connect the output of RF detector (ACT-01) **OUT** post to the Y-axis of Oscilloscope.
- Connect the **MOD IN** post of VCO1 of ACT-01 to the X-axis of Oscilloscope.
- Set the oscilloscope in X-Y mode. You obtain a curve similar to the one of fig.4.9

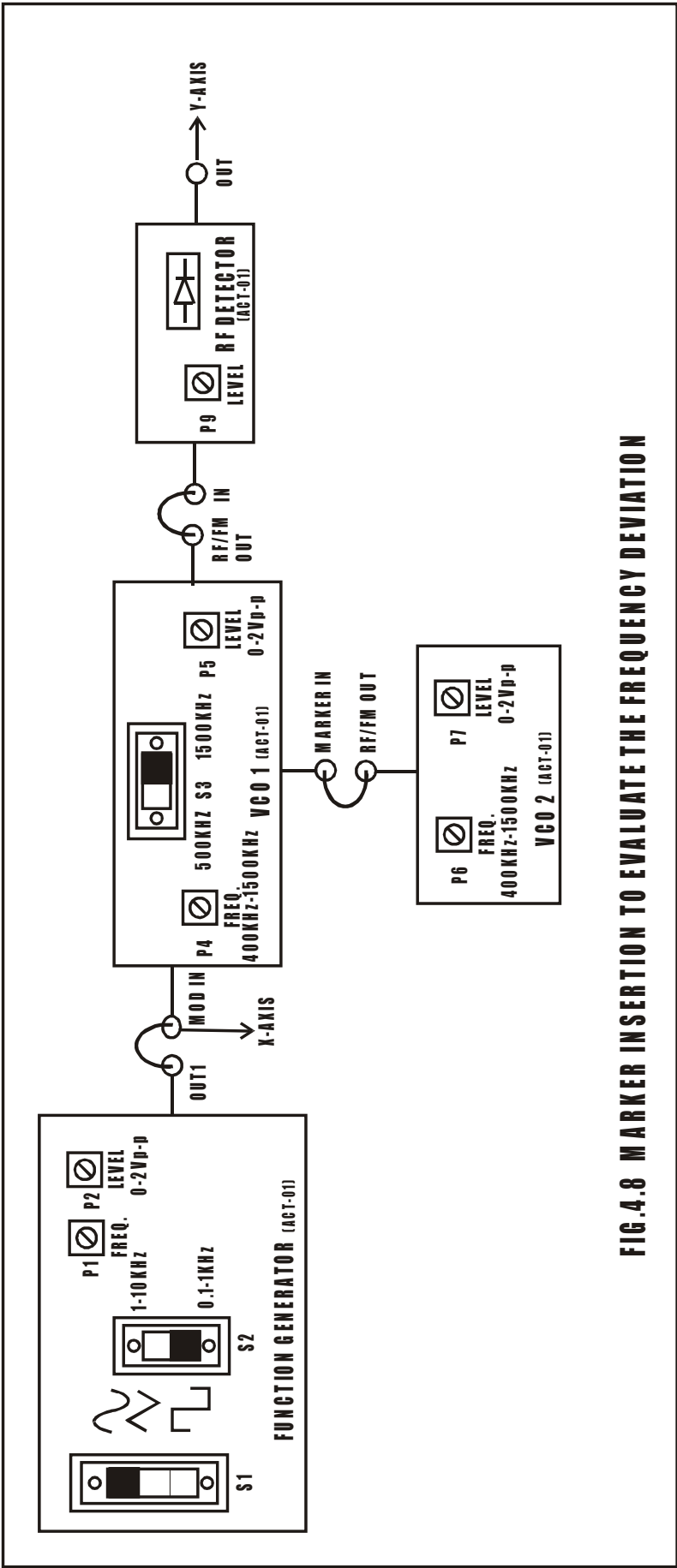
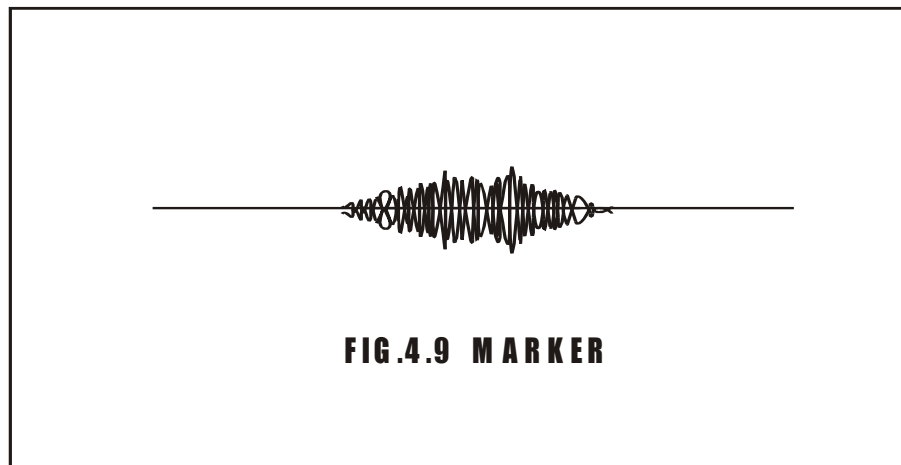
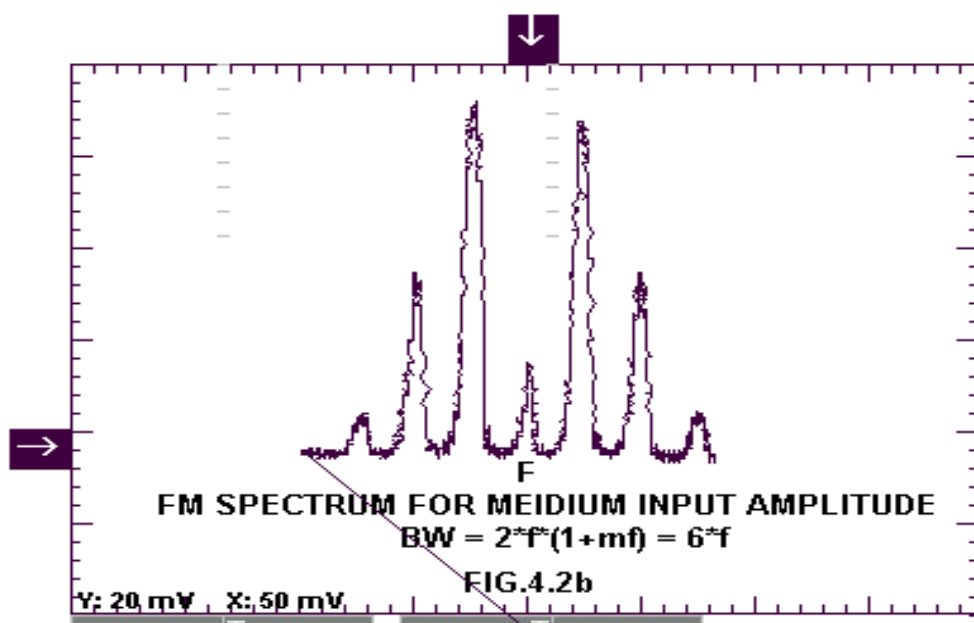


FIG.4.8 MARKER INSERTION TO EVALUATE THE FREQUENCY DEVIATION



- The measurement operation can be explained in the following way:
 - In the output stage of the Frequency Modulator (VCO1) the FM signal (with instantaneous frequency f) and the sine wave signal supplied by the VCO2 (with frequency f_2) are added.
 - The beating caused by the signal produces different spectral components, among which two the sum and the difference of input frequencies $[(f+f_2)$ and $(f-f_2)]$. Only the low frequency component frequencies $(f-f_2)$ are detected by RF DETECTOR, while the high frequency component $(f+f_2)$ is removed by the output low pass filter of the RF DETECTOR.



- As f is variable, when f gets near f_2 the beating can be displayed by the Oscilloscope as small “peaks”
- Varying the frequency of VCO2 the peaks move from one extreme to the other of the curve.
- With a frequency meter, measure the frequencies of VCO2 corresponding to the two extremes of the curves. The difference between the two extremes of the curve supplies the frequency deviation (peak-to-peak) of the Frequency Modulator.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.

D. To observe the Spectrum of the FM Signal

EQUIPMENT:-

- Modules ACT-01, ACT-02.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

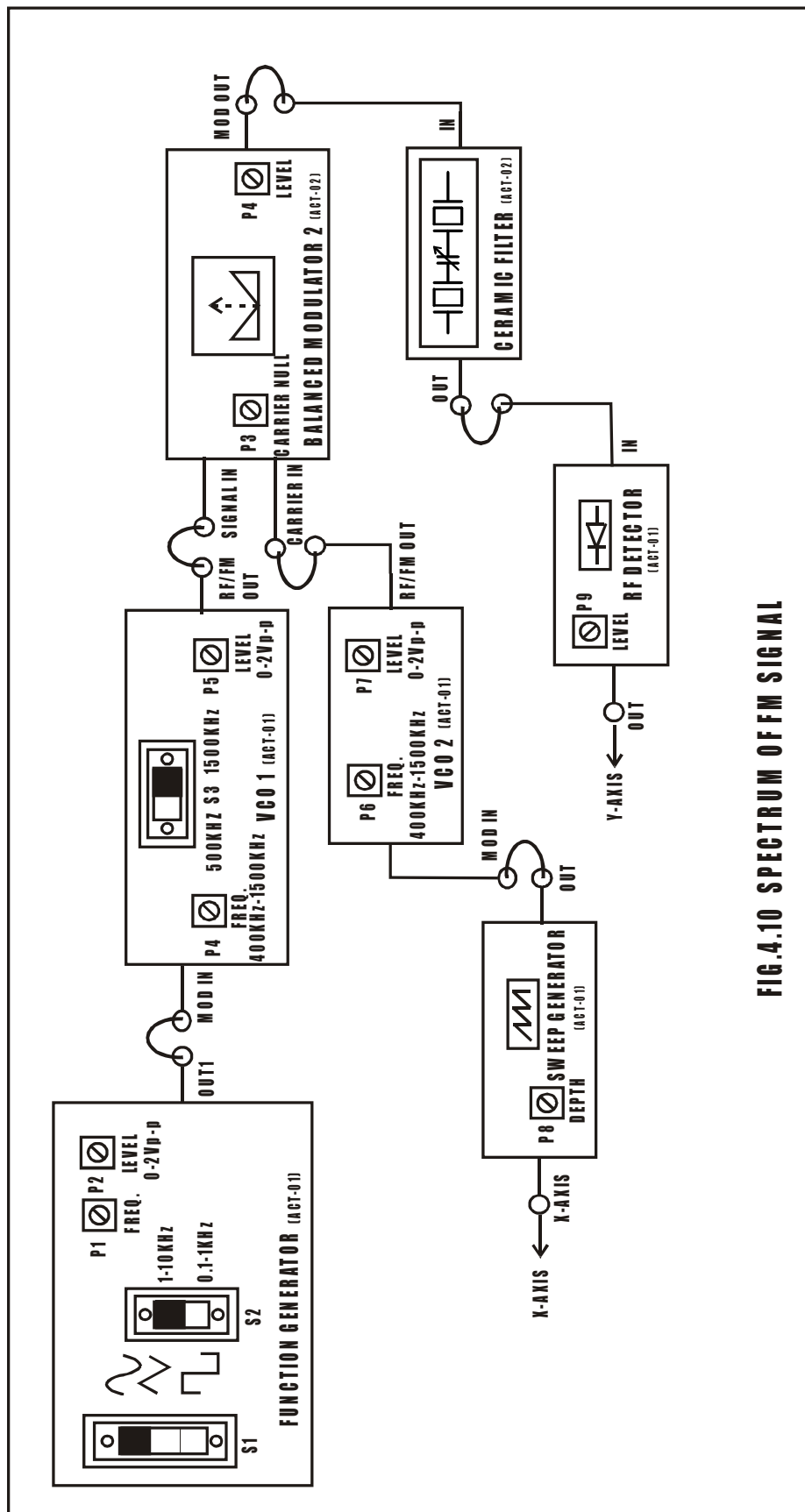


FIG.4.10 SPECTRUM OF FM SIGNAL

PROCEDURE:-

Refer to the “Appendix A’ to explain the operating principle of the spectrum analyzer.

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-02 and switch it on.
- Refer to the fig.4.10 & Carry out the following connections and switch settings.
- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **10 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **100mV**.
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency on the center.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
 - Using pot **P6** keep VCO2 frequency on the center.
 - Using pot **P7** keep VCO2 amplitude at **1Vpp**.
 - Using pot **P8** keep SWEEP **DEPTH** almost completely counter clockwise.
 - Using pot **P9** keep RF DETECTOR **LEVEL** completely clockwise.
- In ACT-02
 - BALANCED MODULATOR 2: CARRIER NULL in central position, so that the circuit operates as frequency converter (balanced modulator with suppressed carrier) using pot **P3**.
 - BALANCED MODULATOR 2: OUT LEVEL clockwise using pot **P4**.
 - Trimmer **CV1** of the CERAMIC FILTER completely clockwise.
- Connect the output of function generator (ACT-01) **OUT1** post to the **MOD IN** post of VCO1 (ACT-01).
- Connect **RF/FM OUT** post of VCO1 (ACT-01) to the **SIGNAL IN** post of Balance modulator 1 (ACT-02).
- Connect output of VCO2 (ACT-01) **RF/FM OUT** post to the Carrier input of Balance modulator 1 (ACT-02) **CARRIER IN** post.
- Connect output of Balance modulator 1 (ACT-02) **MOD OUT** post to the input of Ceramic filter (ACT-02) **IN** post.
- Connect output of Ceramic filter (ACT-02) **OUT** post to the input of RF detector (ACT-01) **IN** post.
- Connect **MOD IN** post of VCO2 (ACT-01) to the **OUT** post of SWEEP circuit (ACT-01).
- Connect **OUT** post of RF detector (ACT-01) to the Y-axis of Oscilloscope.
- Connect **X-AXIS** post of SWEEP circuit (ACT-01) to the X-axis of Oscilloscope.
- Set the oscilloscope in X-Y Mode.
- Keep volt/div of CRO, X-axis= 0.2, Y-axis= 0.2.
- Vary the frequency of VCO2 until the oscilloscope gives a similar representation to the one of the fig.4.2. To obtain the best wave form adjust: the shift of the SWEEP generator (DEPTH); the band pass of the ceramic filter (variable capacity); the CARRIER NULL of the balanced modulator1.vary modulating signal & RF level for clear signal.

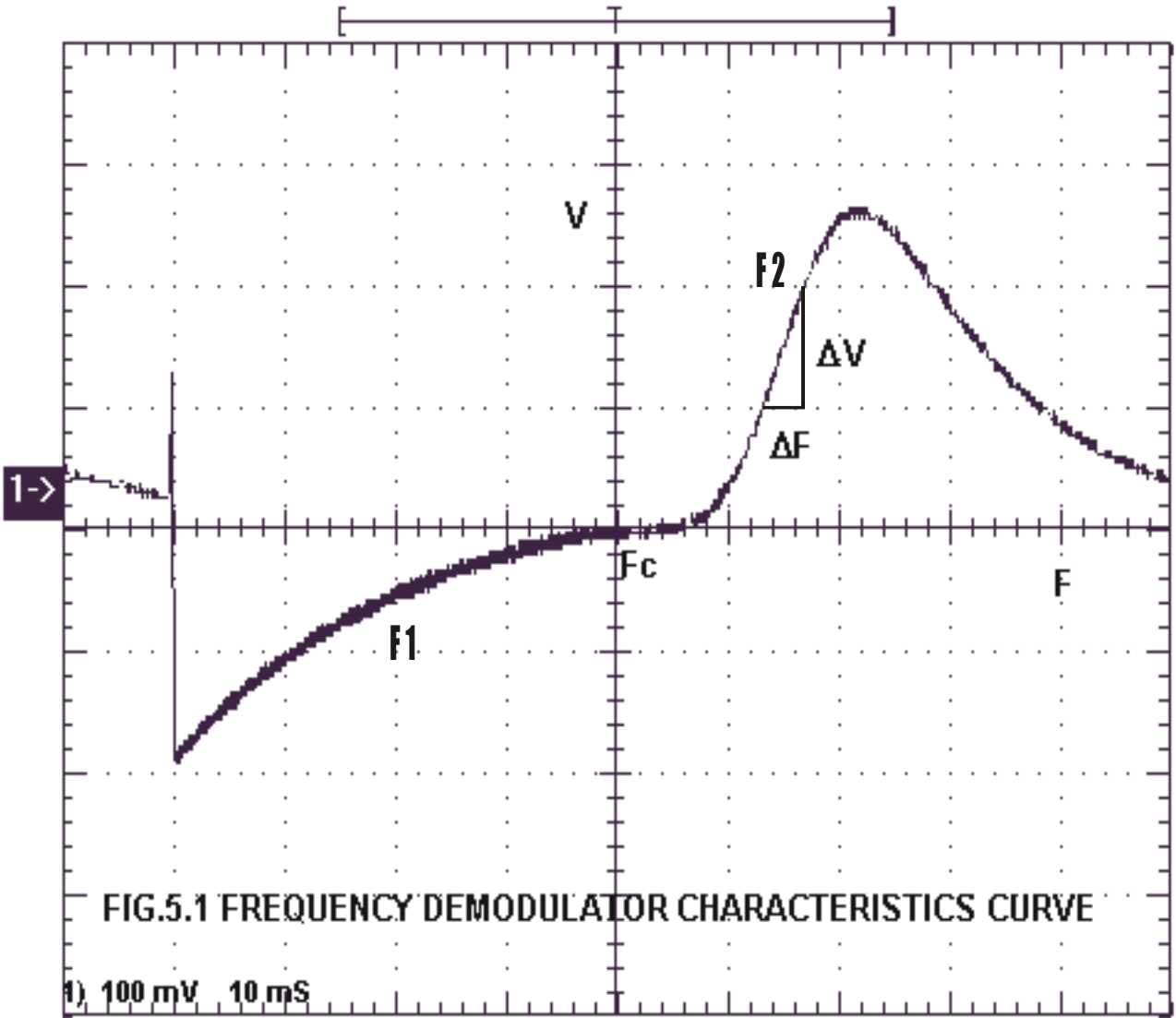
- The waveform represents the spectrum of the FM signal. Check that the distribution of the harmonics is symmetrical in respect to the one of the carriers.
- Vary the amplitude and frequency of the modulating signal, and examine how the spectrum varies. Check that when the amplitude increases also the number of spectral lines increases (the increase of the amplitude of the modulating causes the frequency deviation to increase ΔF ; this causes the modulation index m_f to increase and consequently see fig. 4.2 the spectrum to enlarge). Check that the spectral line corresponding to the carrier varies in amplitude and annuls in correspondence to determinate frequency deviation values ΔF .

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 3 of SF 1 in ACT-02 Switch fault section to ON position. This will open R30 from gain adjust pins of balance modulator 2. Balance modulator2 maximum output reduced to 100 mV from 2V.

EXPERIMENT NO.5



EXPERIMENT NO.5

NAME:-

Study of Frequency Demodulation

OBJECTIVE:-

- A1 To plot the Demodulation characteristic of the Frequency Demodulator (Foster Seeley Demodulator).
- A2 To calculate the Demodulation Sensitivity and Non Linearity of Frequency Demodulator.
- A3 To observe the Demodulation characteristic curve on oscilloscope.
- A4 To observe the Waveforms of the Demodulated Signal.
- B To Study the Ratio Demodulator.
- C To Study the Quadrature Detector.
- D To Study the Limiter Circuit.

THEORY:-

To demodulate a frequency modulated signal a circuit is necessary, which supplies the output with voltage proportional to the frequency deviation of the input modulated signal.

The ideal characteristic of the demodulator is a straight line. To obtain characteristics as the one of fig. 5.1, which presents a linear behavior only for a certain frequency range (demodulator usage range the fig.5A reports:

- a) The instantaneous frequency f of the modulated signal, oscillating between F_1 and F_2 (F_c frequency of the carrier)
- b) The voltage/frequency characteristic curve of the demodulator.
- c) The detected signal.

Sensitivity and Demodulation Non-Linearity

The Sensitivity and Non Linearity are characteristic parameters of the frequency demodulator. Both parameters can be detected by the characteristic demodulation curve, shown in fig. 5B.

Sensitivity S is defined by:

$$S = \frac{DV(f)}{df} \approx \frac{\Delta V}{\Delta f}$$

Where $V(f)$ is the instantaneous output voltage, function of the instantaneous input frequency f . If S_c and S_1 are the sensitivities calculated respectively corresponding to the central frequency on post 1, the Non Linearity N.L. in post 1 is defined by:

$$N.L. = \frac{S_c - S_1}{S_c} \cdot 100$$

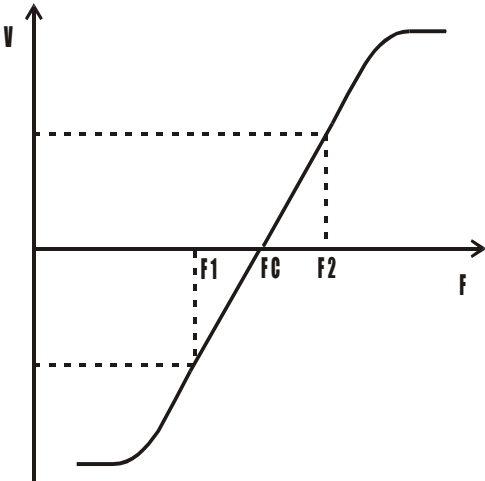


FIG.5A

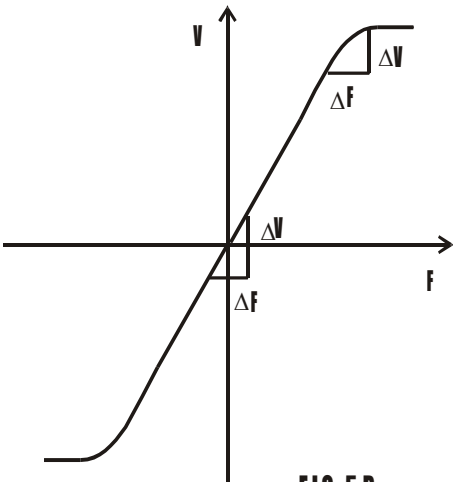
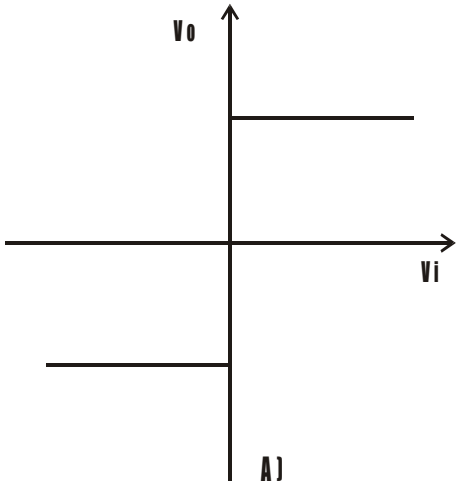
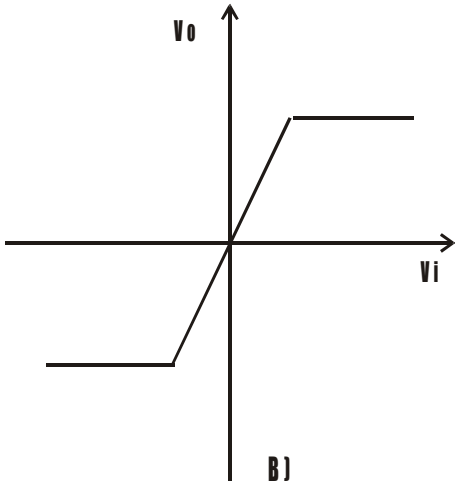


FIG.5B



A)



B)

FIG.5C LIMITER CHARACTERISTICS A) IDEAL B) ACTUAL

Frequency Demodulator Circuits

For the detection of the frequency-modulated signals different circuit solutions have been used, some are out of use and others are used at the moment. Among the first ones, we mention:

- Travis Detector: it is based on amplitude variation, as function of frequency, introduced by a resonant circuit. The amplitude variation is detected via the diode.
- Foster-Seeley Detector: it is based on the phase variation as function of frequency, introduced by a resonant circuit. The originative modulated signal and the shifted one are properly added, and the resulting signal is detected with the diode.

Among the modulators used till now, we mention

- Ratio Detector: it has a behavior analogous to the Foster-Seeley one, but it is unaffected by the modulated signal amplitude.
- Quadrature Detector: is used in integrated circuits. The direct FM signal and the same signal shifted of 90° are multiplied between them: the resulting signal is proportional to the frequency deviation of the input FM signal.
- PLL detector: it constitutes one of the applications of the Phase Locked Loop and is, as compare to the last circuits, less sensitive to noise.

Amplitude Limiter

The frequency demodulators are generally sensitive to the amplitude variation of the input FM signal. The output of the demodulator depends not only on the frequency variation of the input signal, but also on its eventual amplitude variation (caused for example by the noise or by disturbances of different types). To minimize this inconvenience, use a limiter circuit before it, which removes or reduces the unwished amplitude variations.

The characteristics curves of the ideal limiter fig.5C.A where the amplitude of the output signal is constant for any input signal amplitude and in actual limiter fig.5C.B the output amplitude keeps constant only if the input signal gets over the minimum value.

Foster-Seeley Detector

Fig. 5.2 shows the typical circuit of a Foster-Seeley Detector.

The FM signal is inductively coupled to the resonant circuit L2-C2, tuned to the central frequency of the modulated signal. The same signal is also taken, with C1, to the main socket of L2. The diodes D1 and D2, with the respective low pass filters C-R, form two envelope detectors.

F_0 is the frequency at which the circuit L2-C2 is tuned. The operation of the circuit is analyzed in three situations:

- 1) Instantaneous frequency f of the input FM signal equal to F_0 .
- 2) Instantaneous input frequency higher than F_0 .
- 3) Instantaneous input frequency lower than F_0 .

- $\underline{f = F_0}$

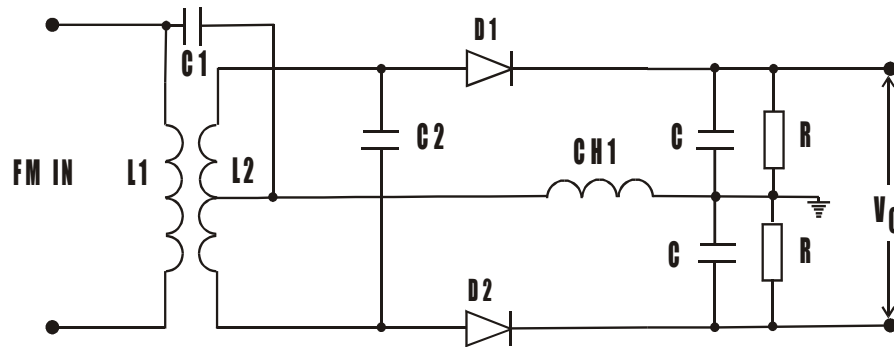


FIG.5.2 FOSTER-SEELEY DETECTOR

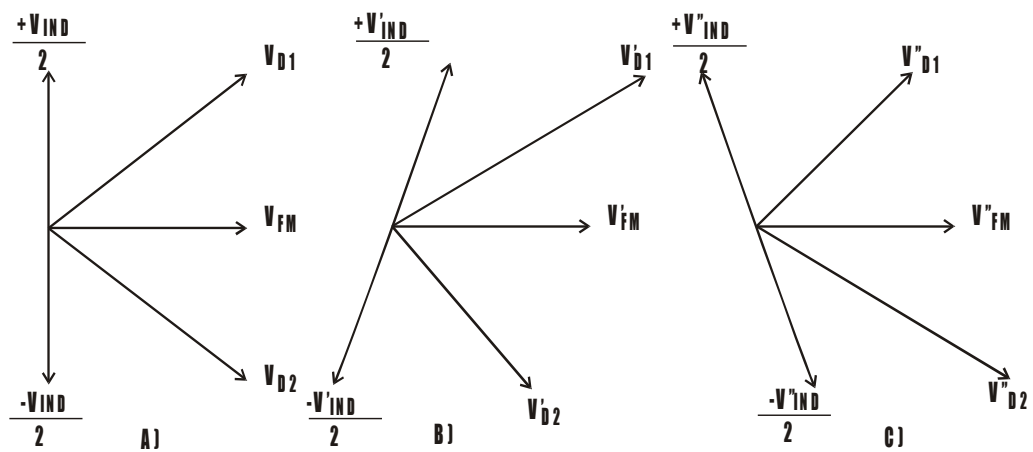


FIG.5.3 PHASE DIAGRAM

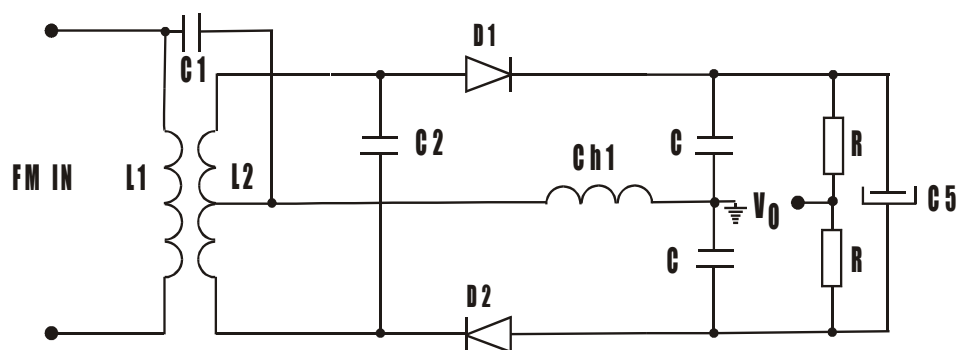


FIG.5.4 RATIO DETECTOR

In the two secondary of L2 we add two voltages (fig. 5.3a). One is induced with L1 by the input signal V_{FM} ; the other is the input signal coupled directly via C1. Being at the resonance frequency, the induced voltage v_{ind} will be shifted by 90° with respect to the voltage V_{FM} . The voltage coupled directly through C1 can be considered in phase with the input V_{FM} , if the reactance of C1 is small at the signal frequency,

The voltages reaching D1 and D2 are the vectorial sum of V_{FM} and $\pm v_{ind}/2$, and have the same amplitude but opposite sign. The output v_o , which is the sum of the two detected signals, in this case will be null.

- $f > F_0$

When the instantaneous frequency of the input FM signal is superior to F_0 , the resonant circuit L2-C2 has an inductive behaviour and the vectorial diagram fig. 5.3b is obtained. The voltages of the diodes have different amplitudes in this case, and the resulting output voltage will be positive.

- $f < F_0$

When instantaneous frequency of the input FM signal is lower than F_0 , the resonant circuit L2-C2 has a capacitive behaviour and the vectorial diagram of fig. 5.3c is obtained. The voltages on the diodes have still different amplitudes, but the resulting voltage will be negative.

The main disadvantage of the Foster-Seeley demodulator is that it detects amplitude variations of the input signal, as the voltage amplitude v_{D1} and v_{D2} on the diodes depends also on amplitude of the input signal. This inconvenience is minimized in the ratio demodulator.

Ratio Detector

The operation of the circuit, as concerns the coupling of the FM signal to the two detection circuits and the vectorial diagrams, is similar to what seen for the Foster-Seeley Detector. The capacitor C5, with higher value, is used to highly reduce the amplitude voltage fluctuations v_{ab} , due to amplitude variations of the input signal. In this way the output voltage v_o is not affected by unwished amplitude variations. We can write, in fact:

$$V_o = \frac{V_{ae} + V_{eb}}{V_{ae} + V_{eb}} - V_{eb} = \frac{V_{ae} - V_{eb}}{(V_{ae} / V_{eb}) - 1}$$

$$V_o = \frac{V_{ab}}{2} \cdot \frac{(V_{ae} / V_{eb}) - 1}{(V_{ae} / V_{eb}) + 1} = \frac{2}{2} \cdot \frac{2}{(V_{ae} / V_{eb}) + 1}$$

As v_{aeb} is practically constant, the output v_o depends only on the ratio v_{ae}/v_{eb} , which varies only by the effect of the variations of the input signal frequency, and does not cause amplitude variations.

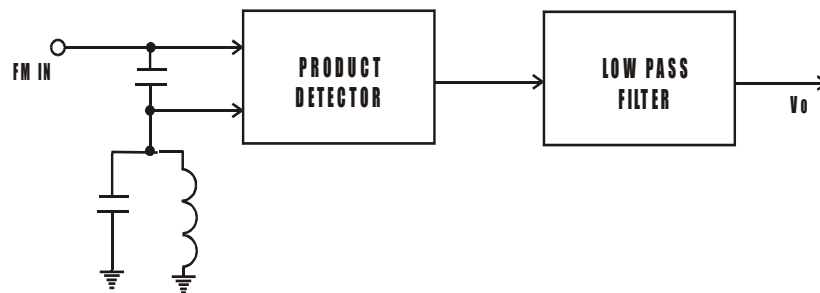


FIG.5D QUADRATURE DETECTOR

Quadrature Detector

Fig.5D shows the functional diagram of the quadrature detector. A product detector multiplies between them the direct FM signal and the same shifted signal by a resonant circuit LC.

At the frequency of resonance, corresponding to the central frequency of the FM signal, the shift is 90° . At the variation of the input signal the shift introduced by the circuit LC will vary.

The multiplication of the direct FM signal and the shifted FM signal produces many components, among which the low frequency component proportional to the information. The low pass filter separates this signal.

A1. To plot the Demodulation Characteristic of the Frequency Demodulator (Foster Seeley Demodulator)

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

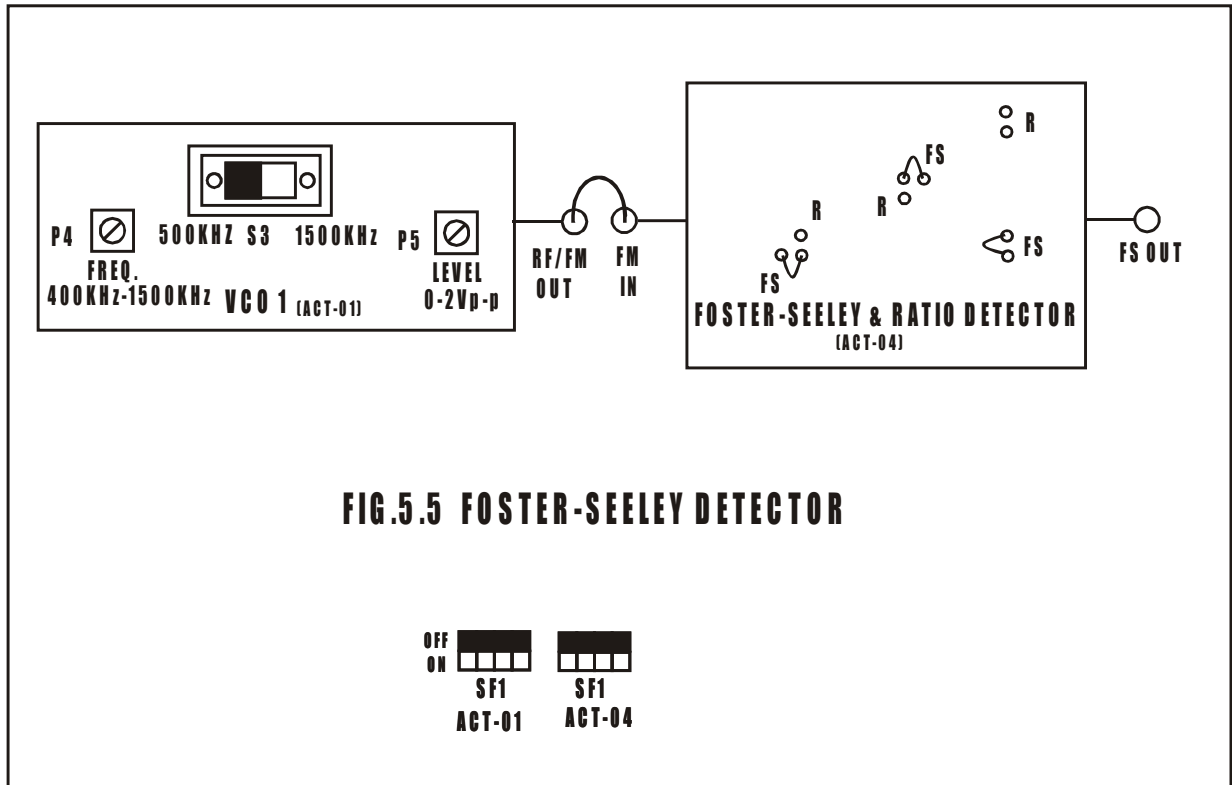
Note: keep all the switches in switch fault section in off position

PROCEDURE:-

The demodulation characteristic curve is given by the output voltage of the demodulator as function of the input instantaneous frequency (fig. 5.1) It is possible to plot the curve of fig. 5.1 by varying the input frequency to the Detector and measuring the corresponding output voltages.

Calibration of the Demodulator

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.5.5 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S3** on 500 KHz position.
 - Keep VCO1 **LEVEL** about 1 Vpp using pot **P5**.
- In ACT-04:
 - Set the frequency demodulator in Foster-Seeley (jumpers in FS position).
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Foster Seeley Demodulator (ACT-04) **FM IN** post
- Connect the frequency-meter to the input of the demodulator.
- Connect the voltmeter (or the DC oscilloscope) to the output of the demodulator (between post **FS OUT** and ground)
- Set the input frequency to 450 KHz, and check that the output voltage is 0 volt. In the contrary, adjust the central frequency of the Detector (adjust the variable capacitor on the rear side of the module).
- Vary the input frequency from 400 to 500 KHz, with steps of 5KHz, and report the frequencies and the corresponding output voltages on a table. The output voltage must vary from about -100mV to about $+100\text{mV}$.
- Fill a graph with the measured values. You obtain a curve similar to the one of fig. 5.1.



SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

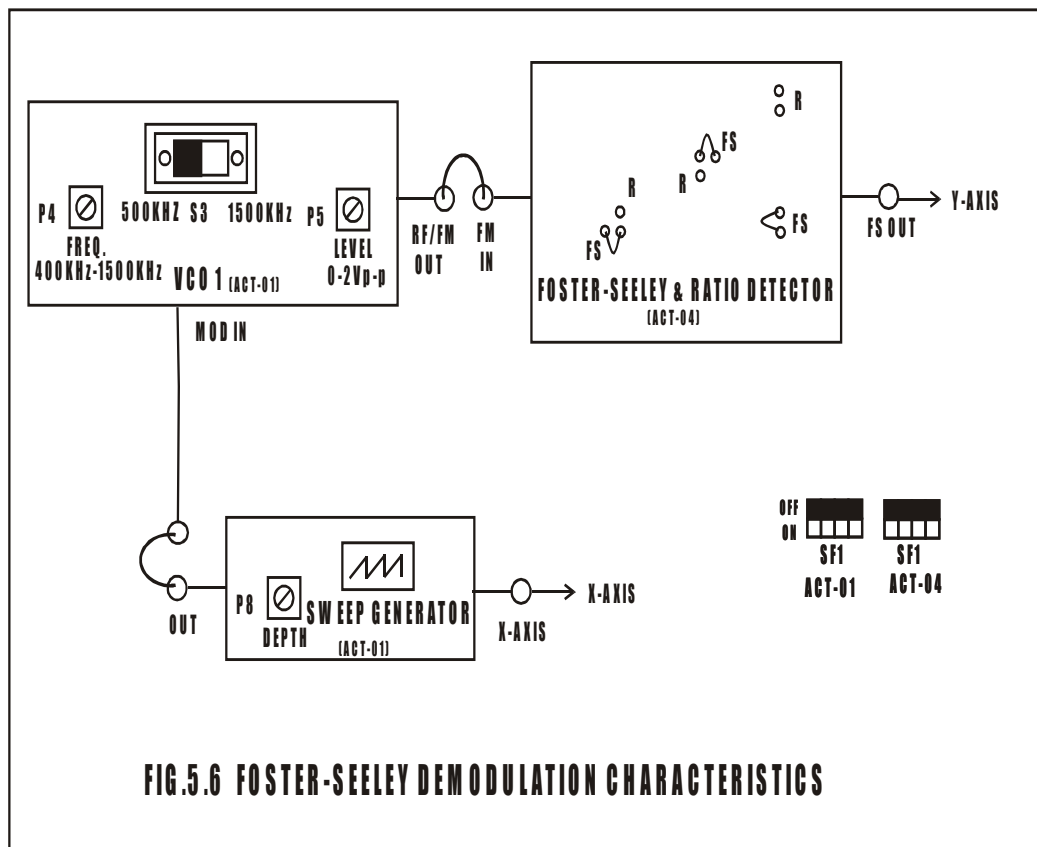
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.

A2. To Calculate the Demodulation Sensitivity and Non-Linearity of Frequency Demodulator

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position



PROCEDURE:-

- Perform the procedure as done in A1.
- Refer the curve plotted above from the analysis of the curve you can notice that the central segment has a linear behavior, while if you consider the whole characteristic there is a high non-linearity.
- Consider that you make the modulator work on the segment of the curve included between 445 and 455 KHz, with central frequency 450 KHz. From the definitions and the analysis of the curve, it is possible to calculate the demodulation sensitivity S and Non-linearity of the demodulator. You will find a value S equal to about 100mV/10KHz.

A3. To Observe the Demodulation Characteristic Curve on CRO

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.5.6 & Carry out the following connections switch settings.
- In ACT-01:
 - Keep switch **S3** on 500 KHz position.
 - Keeps VCO1 **FREQ.** about 450 KHz using pot **P4**.
 - Keep VCO1 **LEVEL** about 1 Vpp using pot **P5**.
 - **SWEEP DEPTH** almost to the minimum using pot **P8**.
- In ACT-04:
 - Frequency demodulator in Foster-Seeley mode (jumpers in FS position)
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Foster Seeley Demodulator (ACT-04) **FM IN** post.
- Connect the output of Foster Seeley Demodulator (ACT-04) post **FS OUT** to the Y-axis of Oscilloscope.
- Connect **MOD IN** post of VCO1 to the **OUT** post of SWEEP Circuit.
- Connect **X AXIS** post of SWEEP Circuit to the X-axis of Oscilloscope.
- CRO volt/div, X-axis= 200mV, Y-axis= 200mV.
- Keep Oscilloscope in X-Y mode.
- Adjust the central frequency of the VCO1 and the amplitude of the Sweep (DEPTH) to obtain the characteristic curve of the demodulator as shown in fig. 5.1

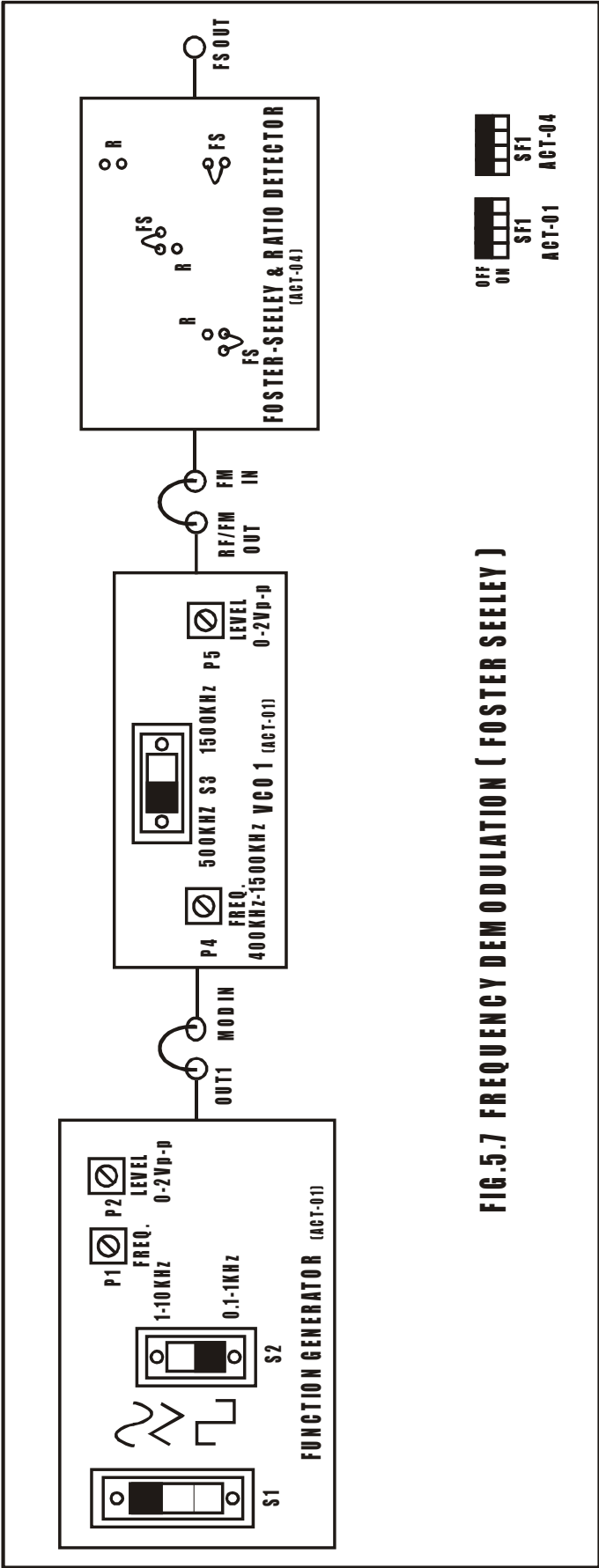


FIG.5.7 FREQUENCY DEMODULATION (FOSTER SEELEY)

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.

A4. To observe the waveforms of the Demodulated Signal

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply +/-12 Vdc
- Oscilloscope
- Volt meter
- Frequency meter
- Connecting Links

Note: keep all the switches in switch fault section in off position

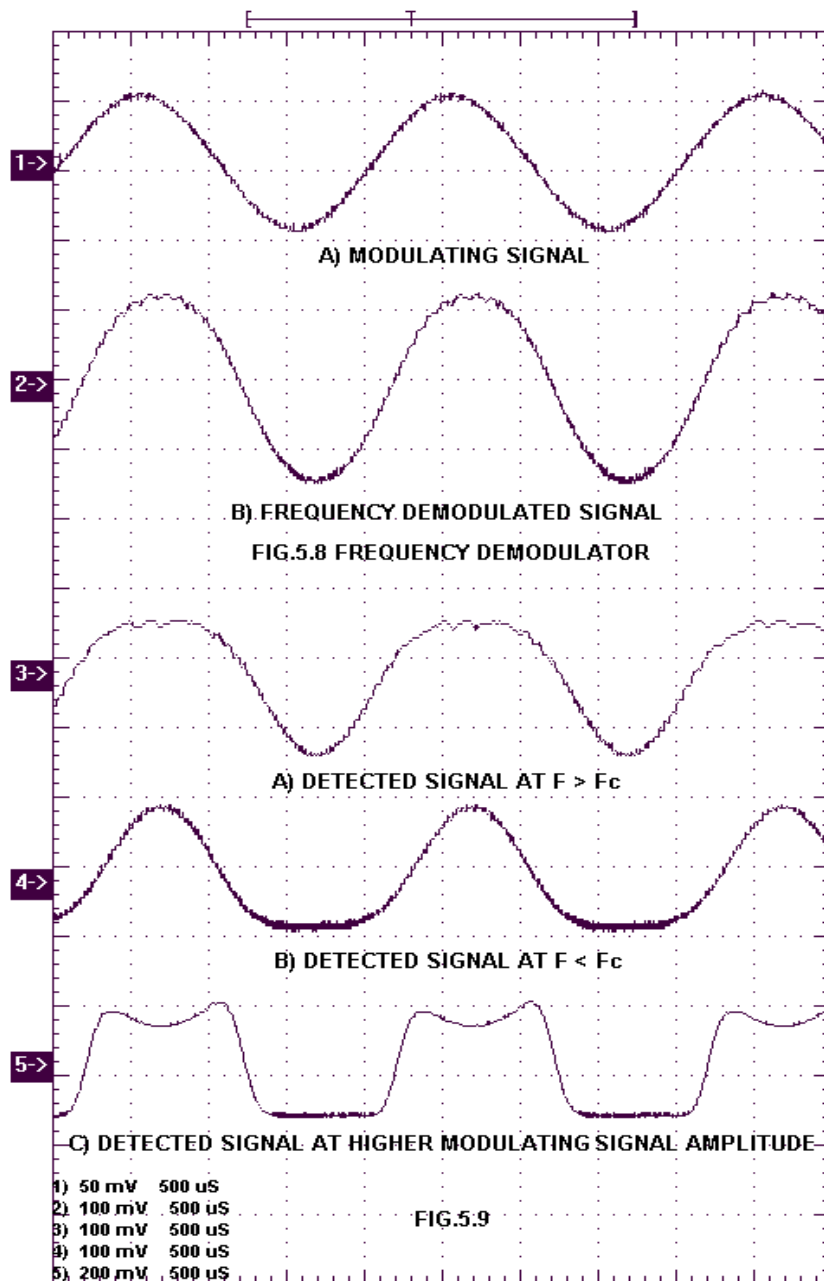
PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.5.7 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in lower position (**0.1-1KHz**).
 - Using pot **P1** keep sine wave signal frequency at **500Hz**.
 - Using pot **P2** keep sine wave signal amplitude at **100mVpp**.
 - Keep switch **S3** on 500 KHz position.
 - Keep VCO1 **FREQ.** about 450 KHz using pot **P4**.
 - Keep VCO1 **LEVEL** about 1 Vpp using pot **P5**.
- In ACT-04:
 - Frequency demodulator in Foster-Seeley mode (jumpers in FS position)
- Connect the output of Function Generator (ACT-01) **OUT1** post to the **MOD IN** post of VCO1 (ACT-01).
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Foster Seeley Demodulator (ACT-04) **FM IN** post.
- Connect the oscilloscope across post **MOD IN** of ACT-01 (modulating signal) and post **FS OUT** ACT-04 (detected signal).
- If the central frequency of the Detector and the carrier frequency of the FM signal coincide, you obtain two signals similar to the ones of fig.5.8. Note that the demodulated signal has null continuous component. Vary the amplitude of the FM signal and check that the amplitude of the detected signal varies, too.
- Increase the carrier frequency and note that a positive voltage is added to the detected signal. Still increasing the frequency, the detected signal presents a distortion similar to the one of fig.5.9a (in these condition you operate on a non linear zone of the Detector)
- Diminish the carrier frequency and note a symmetrical behavior in respect to the last case (fig. 5.9b)
- Reduce the carrier frequency to its proper value (450 KHz). Increase the amplitude of the modulating signal (LEVEL of the FUNCTION GENERATOR) to generate an FM signal with frequency deviation over the linear zone of the Detector. You obtain a much-distorted signal, similar to the one of fig. 5.9c.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency.
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.



B. To study the Ratio Demodulator

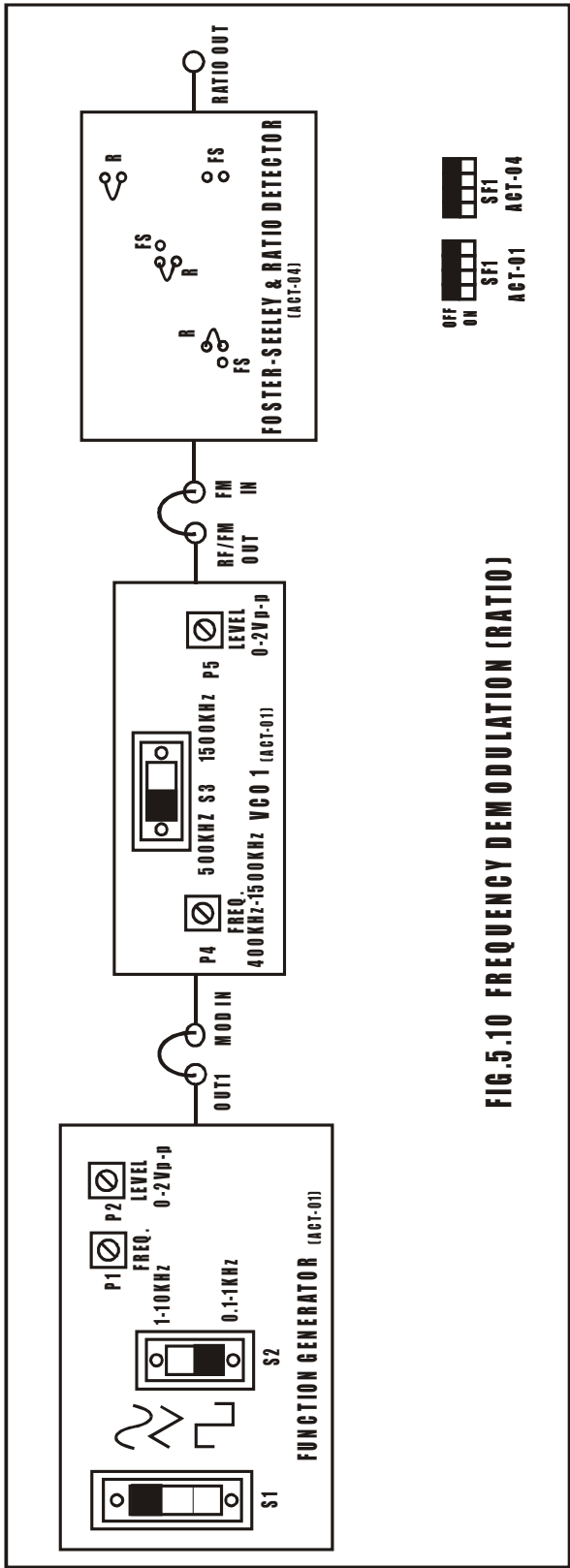
EQUIPMENT:-

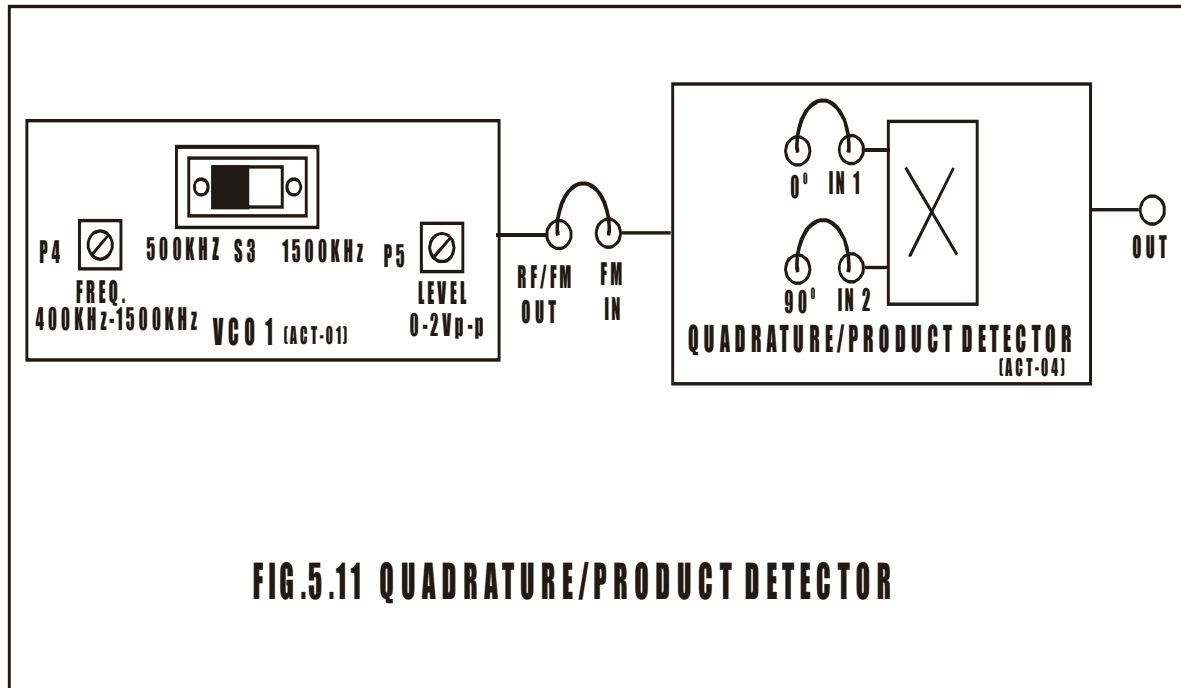
- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.5.10 & Carry out the following connections and switch settings.
- Perform the procedure as done in A1, A2, A3, and A4.
- Set the frequency demodulator in ratio mode (Jumpers in R position). Observe the detected signal at post **RATIO OUT** (ACT-04).
- Plot the Demodulation characteristic curve as done in A1.
- Measure Sensitivity and Non linearity for ratio demodulator as done in A2
- Observe the characteristic curve on CRO for ratio demodulator as done in A3.
- Observe the Demodulated wave forms for ratio demodulator as done in A4.
- When the wave forms of the detected signal are analyzed, it is possible to check the operation of the electrolytic capacitor (C5 of fig.5.4) in the following way:
 - Disconnect the capacitor from the circuit (remove the related jumper). Vary the amplitude of the FM signal and check that also the amplitude of the detected value varies.
 - Connect the capacitor and check that now amplitude variations of the detected signal are highly reduced.





SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.

C. To study the Quadrature Detector

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.5.11 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S3** on 500 KHz position.
 - Keep VCO1 **FREQ.** about 450 KHz using pot **P4**.
 - Keep VCO1 **LEVEL** about 0.5 Vpp using pot **P5**.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of product detector (ACT-04) **FM IN** post.
- Connect the ground post of VCO1 (ACT-01) to the ground post of product detector (ACT-04)
- Connect **0°** post to **IN1** post of the product detector (ACT-04).
- Connect **90°** post to **IN2** post of the product detector (ACT-04).
- Connect the oscilloscope (probes 10:1) to the two inputs of the product detector (posts **IN1** and **IN2**). Check that the two signals are shifted exactly of 90°.
- Carry out the various tests as done in Experiment A2, A3, and A4.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 3 of SF 1 in ACT-04 Switch fault section to ON position. This will open TR3 from Product detector, 90° output disturbed.

D. To study the Limiter Circuit

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Voltmeter.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

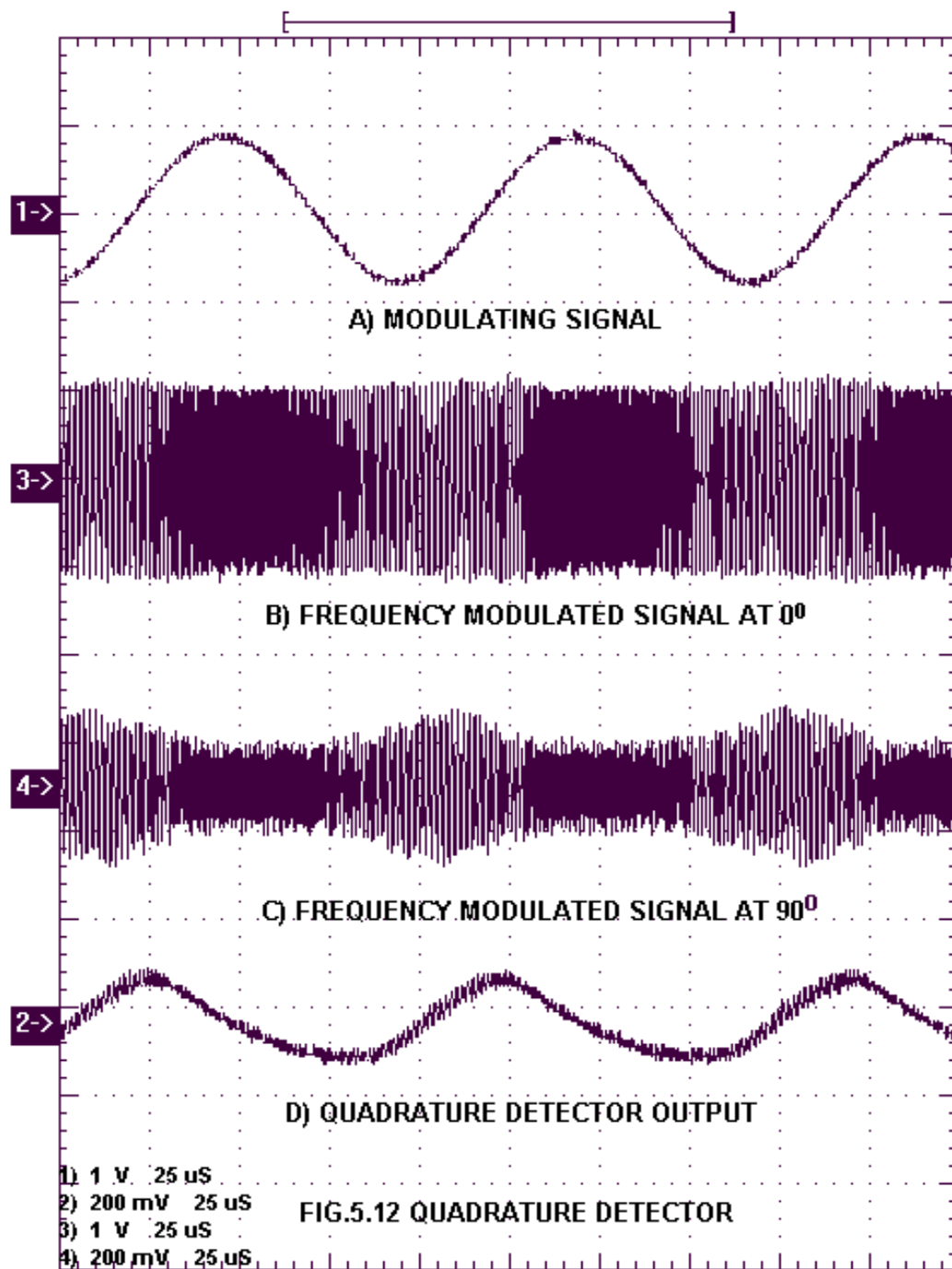
PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.5.13 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in lower position (**0.1-1KHz**).
 - Using pot **P1** keep sine wave signal frequency at **500Hz**.
 - Using pot **P2** keep sine wave signal amplitude at **100mVpp**.
 - Keep switch **S3** on 500KHz position.
 - Keep VCO1 **FREQ.** about 450 KHz using pot **P4**.
- Connect the output of Function Generator (ACT-01) **OUT1** post to the **MOD IN** post of VCO1 of ACT-01.
- Connect the output of VCO1 **RF/FM OUT** post (ACT-01) to the input of the limiter (ACT-04) **IN** post.
- Connect the oscilloscope to the input and output of the limiter (post **IN** and **OUT**). Gradually increase the input signal amplitude. Check that the amplitude of the output signal increases at start (to an input level of about 200mVpp), and then stabilizes around a value of about 1.5Vpp
- Connect the output of limiter (ACT-04) **OUT** post to the input of Foster Seeley Detector or Ratio Detector (ACT-04) **FM IN** post.
- Connect ground post of limiter to ground post of Foster Seeley Detector.
- Vary the amplitude of the FM signal and check that, for amplitude superior to about 200mVpp, the detected signal amplitude keeps constant.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.
- Put switch 2 of SF 1 in ACT-04 Switch fault section to ON position. This will open R38 from Limiter. Limiter action is disabled.



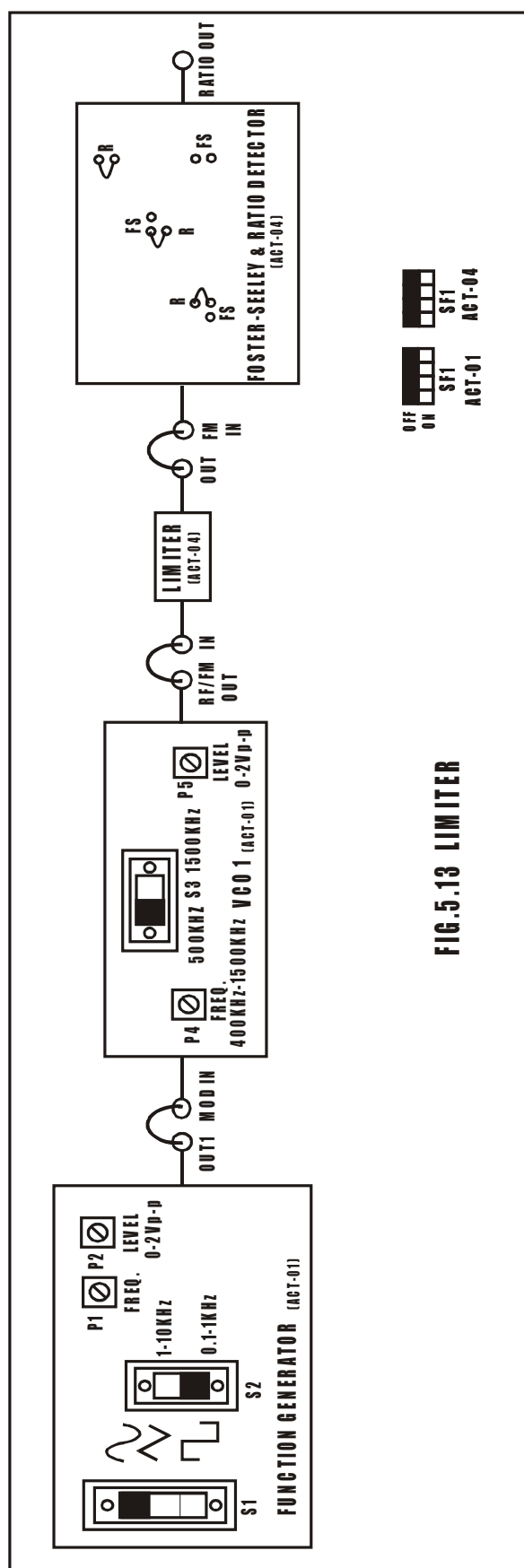
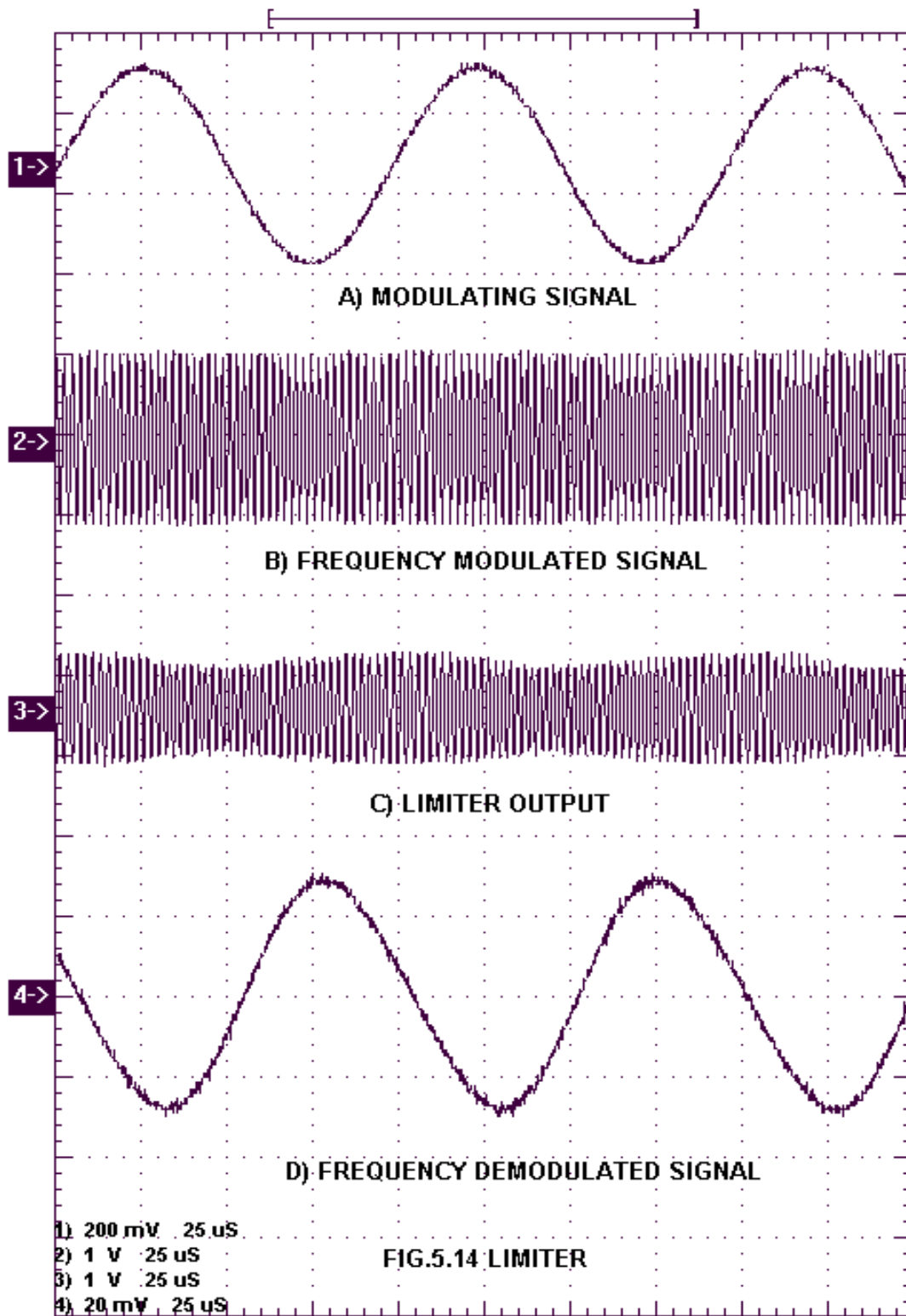
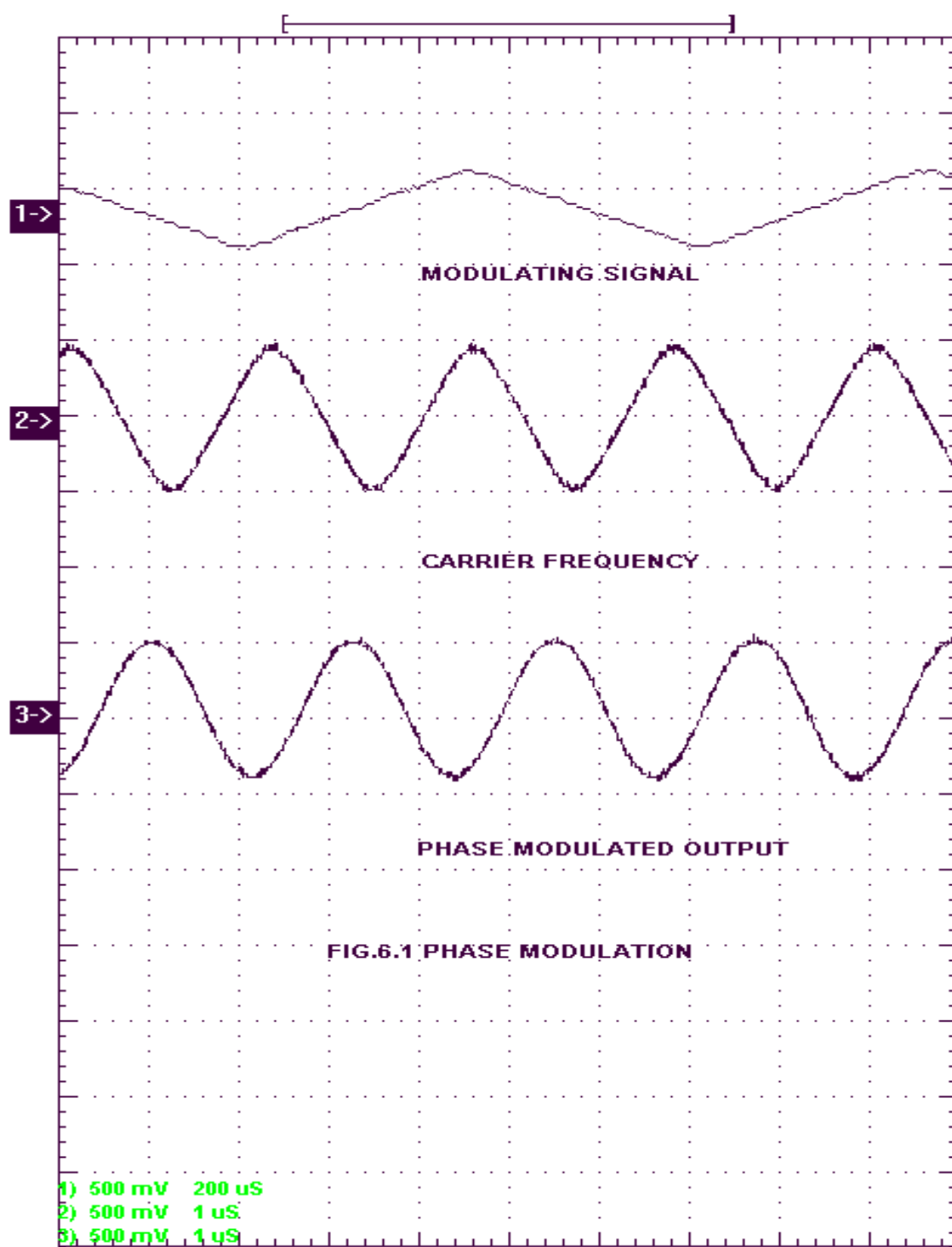


FIG. 5.13 LIMITER



EXPERIMENT NO.6



EXPERIMENT NO.6

NAME:-

Study of Phase Modulation

OBJECTIVE:-

- A To Study and observe the Waveforms of the Phase Modulated Signal.
- B To Study and observe the Waveforms of Phase Demodulated Signal.
- C To Study Frequency Demodulation via Phase modulator.

THEORY:-

Phase Modulation

It is a type of modulation in which the phase of high frequency sine wave (carrier) is varied in accordance with the instantaneous value of the modulating signal.

Main Aspects

Consider a generic low frequency signal $V_m(t)$ and a sine wave signal $V_c(t)$ with pulse Ω :

$$V_c(t) = A \sin [\Omega t + \theta(t)] = A \sin [\phi(t)]$$

The signal $V_m(t)$ is called modulating signal, the signal $V_c(t)$ is called carrier signal.

The phase modulation is produced by varying the angle θ , so that its instantaneous phase, with respect to the value θ_0 in the absence of modulation, is proportional to the amplitude of the modulating signal $V_m(t)$. So the following must be true:

$$\theta(t) = \theta_0 + K \cdot V_m(t)$$

Where K is the sensibility of modulation

Mathematical Expression Of The Phase Modulated Signal

For ease consider a null starting phase θ_0 , the instantaneous angle becomes:

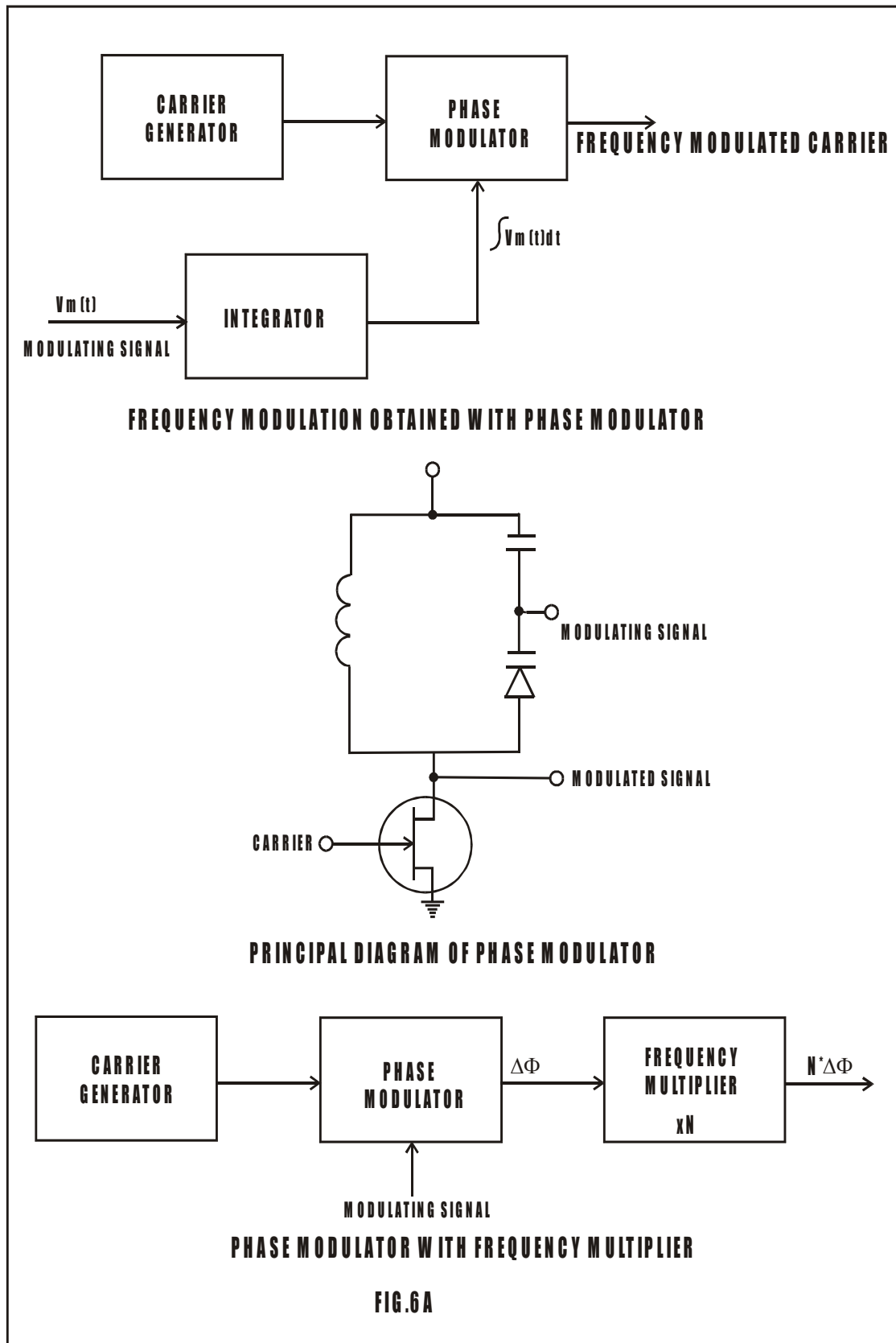
$$\phi(t) = \Omega t + K \cdot V_m(t)$$

The phase modulated signal $V_M(t)$ becomes:

$$V_M(t) = A \sin [\Omega t + K \cdot V_m(t)]$$

The instantaneous pulse of the signal modulated in phase is obtained detecting the instantaneous angle $\phi(t)$ in respect to time

$$\Omega(t) = \Omega + K \cdot \frac{dV_m(t)}{dt}$$



From which you detect the instantaneous frequency $F(t)$ of the phase-modulated signal:

$$F(t) = \frac{\Omega(t)}{2\pi} = \frac{\Omega}{2\pi} + \frac{K}{2\pi} \cdot \frac{dv_m(t)}{dt}$$

From the last relation you observe that $F(t)$ is proportional to the derivative of the modulating signal (in frequency modulation the instantaneous frequency of the modulated signal is proportional to the instantaneous value of the modulating signal). In case of the sine wave modulating signal we have $v_m(t) = B \cdot \sin(w \cdot t)$, and the last relations becomes

$$\phi(t) = \Omega \cdot t + K \cdot B \cdot \sin(w \cdot t)$$

$$v_M(t) = A \cdot \sin[\Omega \cdot t + K \cdot B \cdot \sin(w \cdot t)]$$

$$F(t) = \frac{\Omega}{2\pi} = K \cdot B \cdot \frac{w}{2\pi} \cdot \cos(w \cdot t) = F + K \cdot B \cdot f \cdot \cos(w \cdot t)$$

From the last relation you can note that the frequency shift ΔF of the phase modulated signal is:

$$\Delta F = K \cdot B \cdot f$$

And depends on the amplitude ($K \cdot B$) as well as on the frequency (f) of the modulating signal

Fig. 6.1 shows the waveforms related to the carrier, the modulating signal and phase-modulated signals.

Conversion Of The Phase Modulation Into Frequency Modulation

We have seen before that the instantaneous frequency of the phase-modulated signal is:

$$F_{PM}(t) = \frac{\Omega(t)}{2\pi} = \frac{\Omega}{2\pi} + \frac{K}{2\pi} \cdot \frac{dv_m(t)}{dt}$$

In the experiment 5 we have seen that the instantaneous frequency of the frequency-modulated signal is:

$$F_{FM}(t) = \frac{\Omega(t)}{2\pi} = \frac{\Omega}{2\pi} + \frac{K}{2\pi} \cdot v_m(t)$$

From the two last relations, you can detect that the instantaneous frequency of a signal modulated in signal $v_m(t)$ is equivalent to a signal modulated in frequency by modulating signal $v_m(t) = dv_m(t)/dt$.

This propriety can be used to obtain a frequency modulation from a phase modulator. It is sufficient to integrate the modulating signal, before applying it to the phase modulator. The following is obtained:

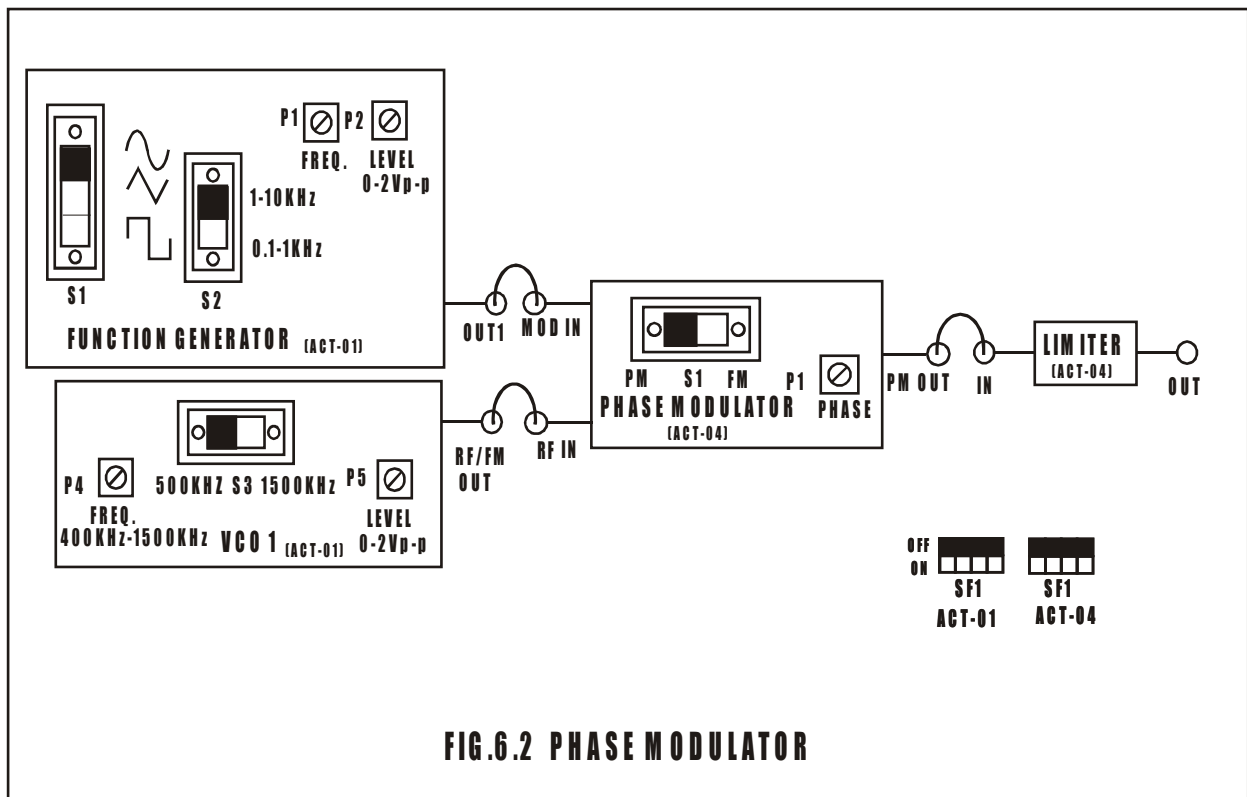
$$F_{PM}(t) = \frac{\Omega}{2\pi} + \frac{K}{2\pi} \cdot \frac{d}{dt} v_m(t) dt = \frac{\Omega(t)}{2\pi} = \frac{\Omega}{2\pi} + \frac{K}{2\pi} \cdot v_m(t) = F_{FM}(t)$$

Phase Modulator

Phase Modulator is an amplifier with resonant load, whose resonance frequency is varied, with the Varicap diode, by the modulating signal.

If the circuit is tuned to the input frequency, the output signal results in phase with the input one; on the contrary the signal shows lag or lead. By varying the resonance frequency of the circuit you can introduce a shift between one fixed input signal (carrier) and the output signal (phase modulated signal). The maximum phase shift $\Delta\theta$, which can be obtained with a phase modulator, is very low (fraction of radiance), to increase it the modulated signal can be frequency multiplied. The phase shift results equal to N times the starting deviation:

$$\Delta\theta N = N \cdot \Delta\theta.$$



A. To study and observe the waveforms of the Phase Modulated Signal

EQUIPMENT:-

- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

Calibration of the Modulator

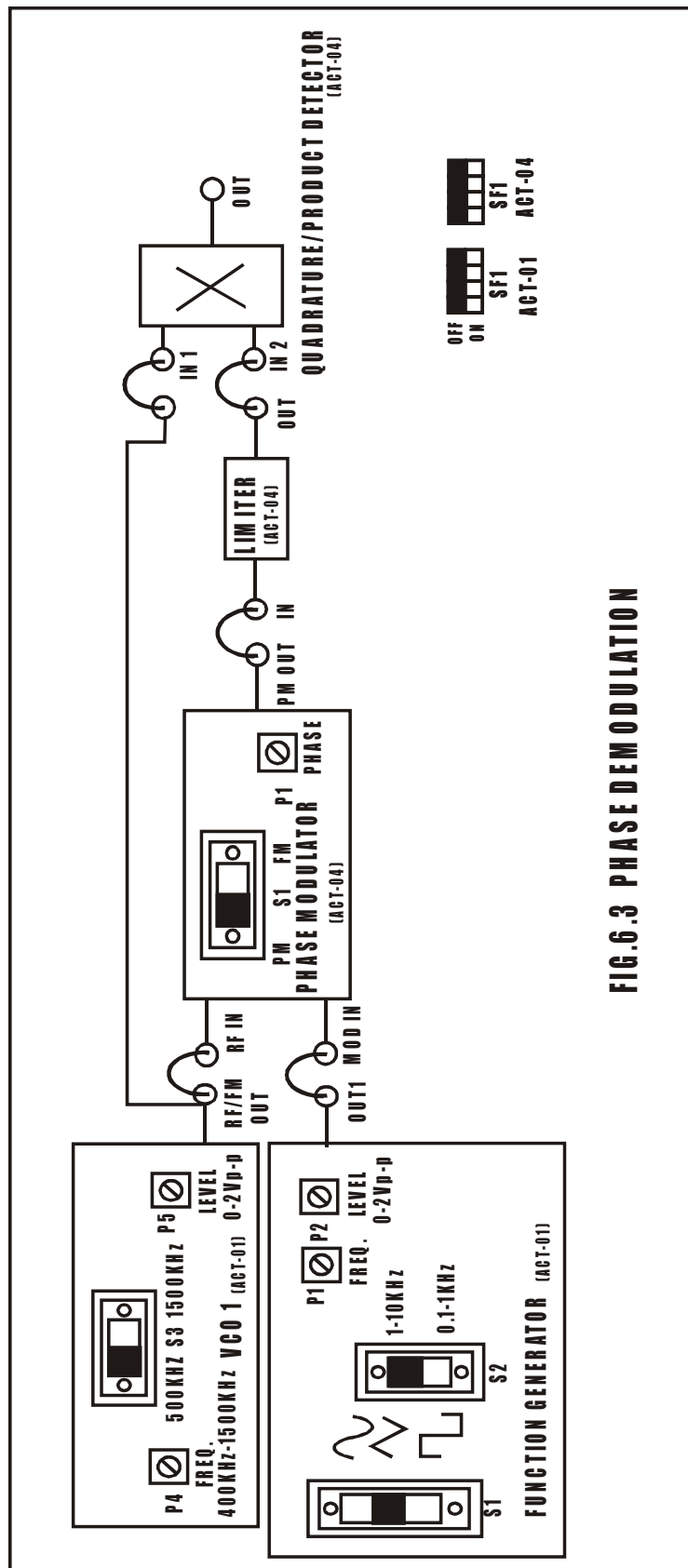
- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.6.2 & Carry out the following connections.
- Carry out the following presetting in ACT-01:
 - Keep switch **S3** on 500 KHz position.
 - Keep VCO1 **FREQ.** about 450 KHz using pot **P4**.
 - Keep VCO1 **LEVEL** about 1.5 Vpp using pot **P5**.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to input of Phase modulator (ACT-04) **RF IN** post.
- Connect the oscilloscope (probes 10:1) to the input and output of the modulator, and synchronize it to the input signal. In this condition the resonant circuit is centered to the input carrier frequency (450 KHz). Check that the output signal is shifted by about 90° in respect to the input signal. This shift is the global shift introduced by the modulator in absence of modulation.
- In ACT-01:
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in lower position (**0.1-1KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
- Connect sine wave signal output of FUNCTION GENERATOR post **OUT1** of ACT-01 to MOD IN post of PHASE MODULATOR of ACT-04.
- Keep the oscilloscope (probes 10:1) connected to the input of the modulator and synchronize it to the input signal.
- Gradually increase the amplitude of the modulating signal. If the modulator is calibrated to the carrier frequency, the oscilloscope detects wave forms as shown in fig. 6.1
- As the output waveform posts out, the width of the trace obtained by adding more phase shifted sine waves keeps constant. This does not occur in the case of frequency-modulated signal, where the trace widens moving from right to left (see figure 4.5).

- The amplitude variation of the phase-modulated signal is due to the analogous variation introduced by the resonant circuit used as modulator.
- The amplitude variation of the modulated signal can be removed by inserting a limiter Circuit.
 - Connect the output of Phase modulator (ACT-04) **PM OUT** post to the input Of limiter (ACT-04) **IN** post. As a result of this the amplitude variation is removed across the output of the limiter.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-04 Switch fault section to ON position. This will open R38 from Limiter. Limiter action is disabled.
- Put switch 4 of SF 1 in ACT-04 Switch fault section to ON position. This will open DV1's GND side in phase modulator, no variation at phase modulator output and amplitude is very low.

**FIG.6.3 PHASE DEMODULATION**

B. To study and observe the waveforms of Phase Demodulated Signal

EQUIPMENT:-

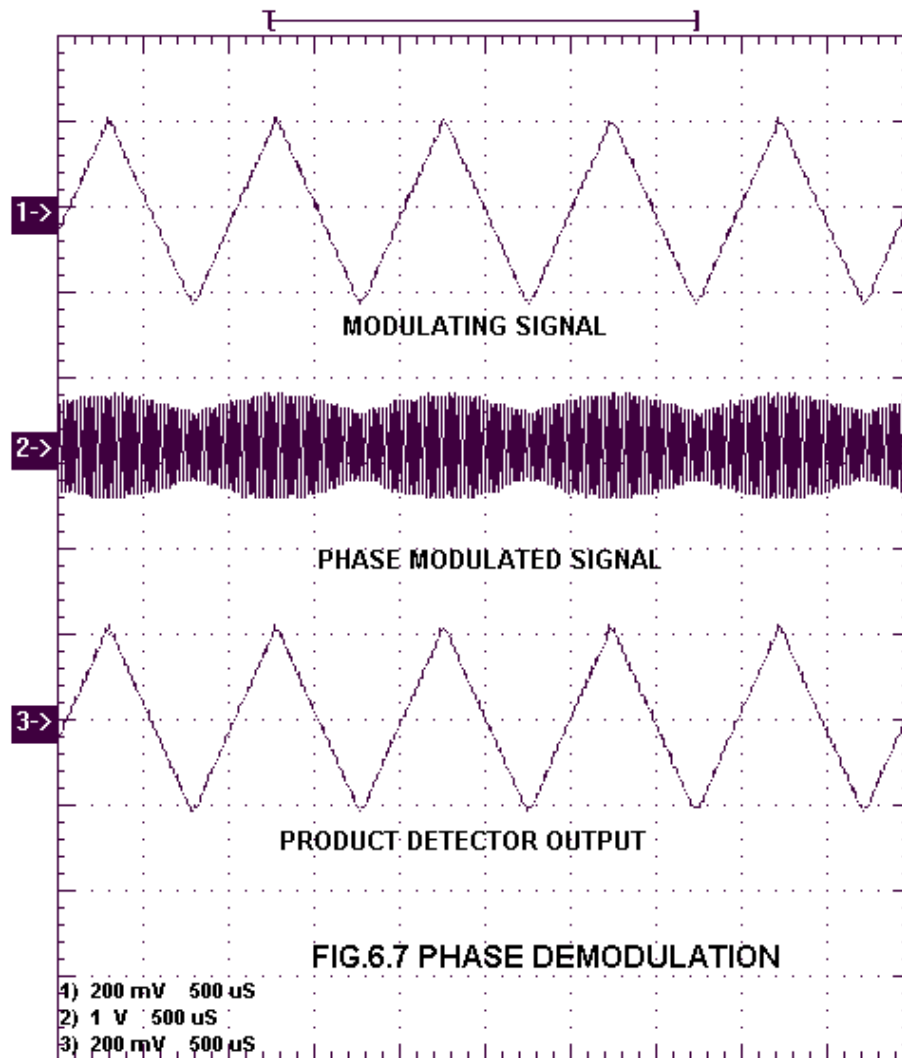
- Modules ACT-01, ACT-04.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

The product detector generates the output with the product of the two input signals. Consider the carrier signal and the phase-modulated signal are to be applied to the two inputs. The multiplication of the two signals produces different components, among which a low frequency component proportional to the phase difference between the same signals. This signal, which corresponds to the detected signal, is separated via the low pass filter.

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.6.3 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S1** in triangular wave position.
 - Keep switch **S2** in upper position (**1-10KHz**).
 - Using pot **P1** keep triangular wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** on 500 KHz position.
 - Keep VCO1 **FREQ.** about 450 KHz using pot **P4**.
 - Keep VCO1 **LEVEL** about 1.5 Vpp using pot **P5**.
- In ACT-04 keep switch **S1** at **PM** position.
- Connect the output of Function Generator (ACT-01) **OUT1** post to the input of PHASE MODULATOR **MOD IN** post of (ACT-04).
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Phase modulator (ACT-04) **RF IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of product detector (ACT-04) **IN1** post.
- Connect the output of Phase modulator (ACT-04) **PM OUT** post to the input of limiter (ACT-04) **IN** post.
- Connect the output of limiter **OUT** post to **IN2** post of product detector (ACT-04).
- Connect the oscilloscope to the modulating signal and to the detected signal and observe the waveforms.



SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-04 Switch fault section to ON position. This will open R38 from Limiter. Limiter action is disabled.
- Put switch 3 of SF 1 in ACT-04 Switch fault section to ON position. This will open TR3 from Product detector, 90° output disturbed.
- Put switch 4 of SF 1 in ACT-04 Switch fault section to ON position. This will open DV1's GND side in phase modulator, no variation at phase modulator output and amplitude is very low.

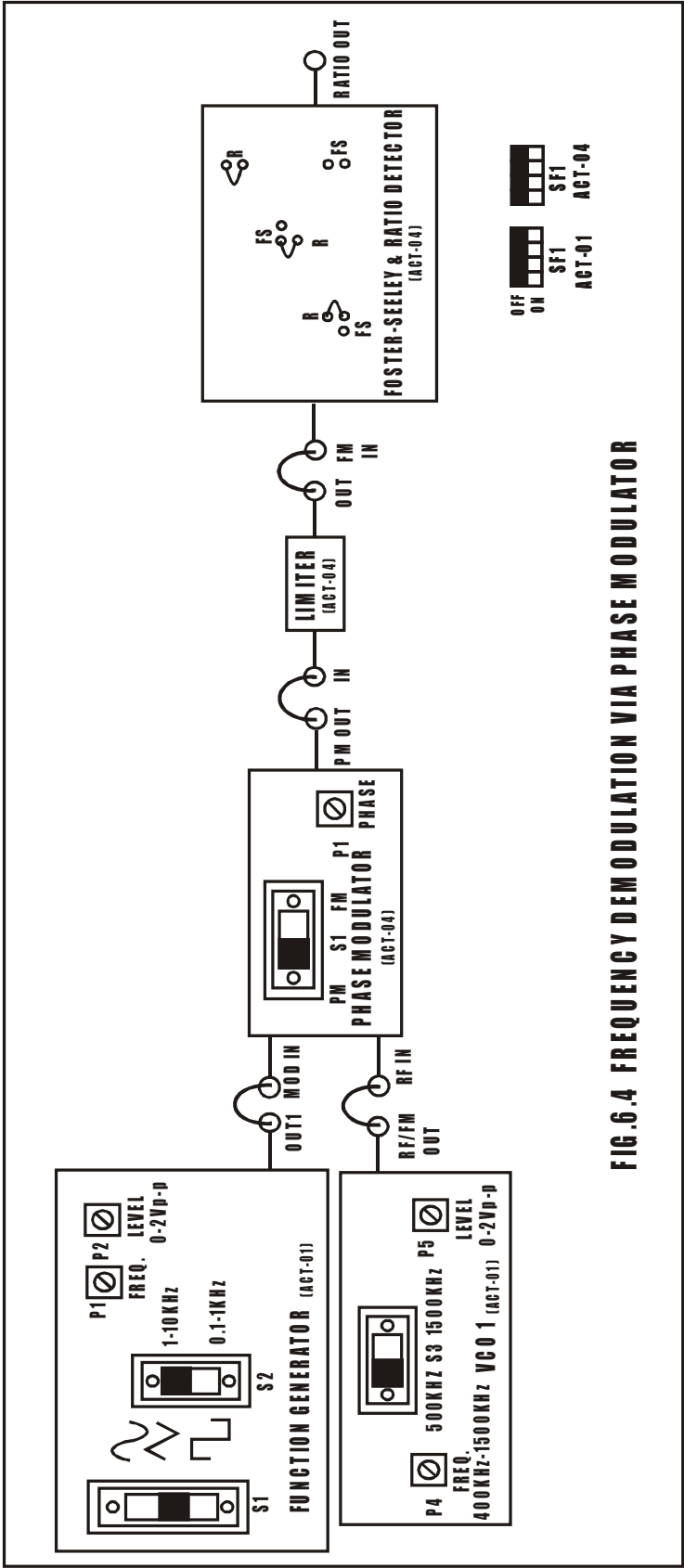


FIG.6.4 FREQUENCY DEMODULATION VIA PHASE MODULATOR

C. To Study Frequency Demodulation via Phase Modulator

EQUIPMENT:-

- Modules ACT-01, ACT-04
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

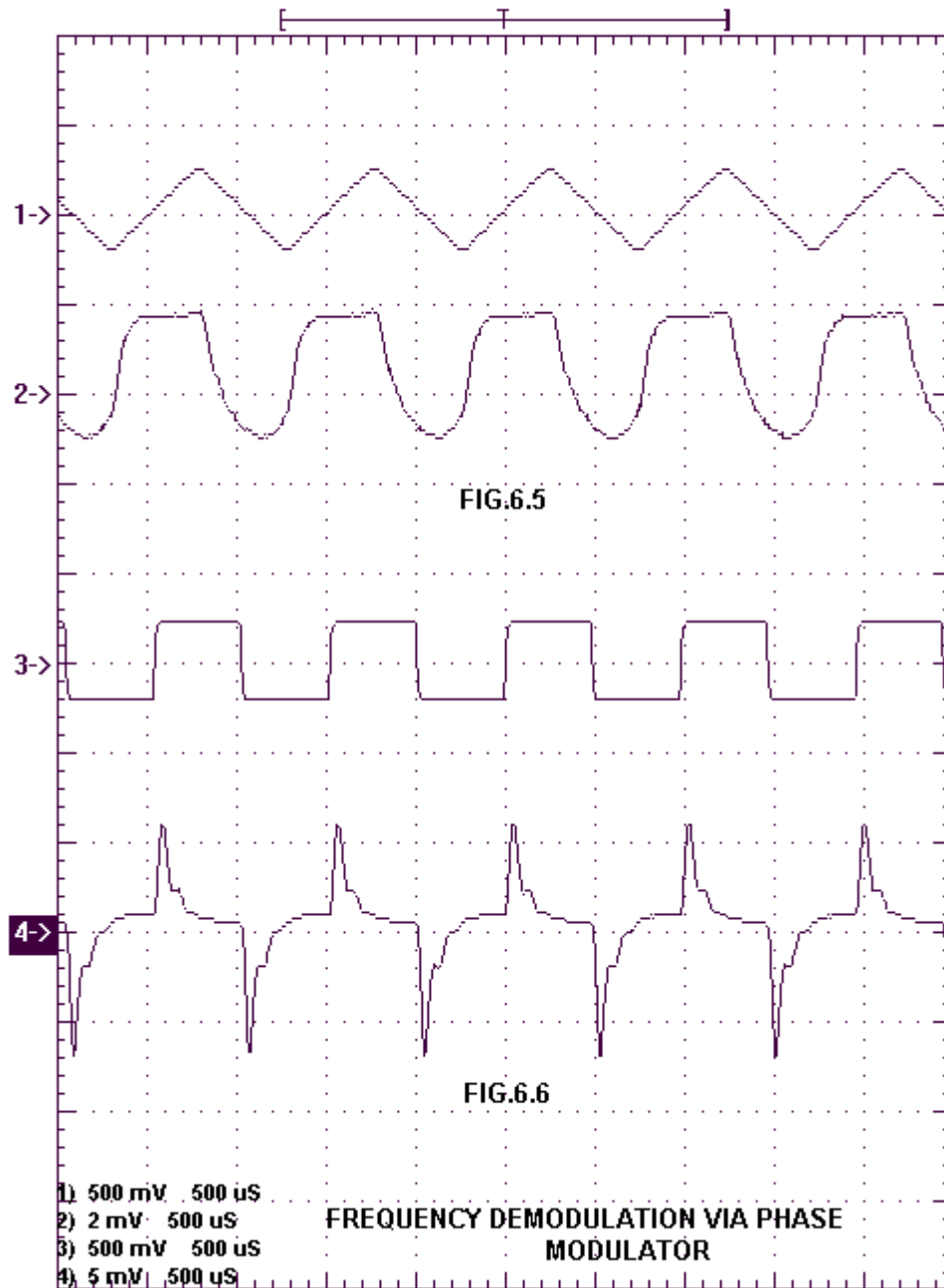
Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-04 and switch it on.
- Refer to the fig.6.4 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S1** in triangular wave position.
 - Keep switch **S2** in lower position (**0.1-1KHz**).
 - Using pot **P1** keep triangular wave signal frequency at **1 KHz**.
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** on 500 KHz position.
 - Keeps VCO1 **FREQ.** about 450 KHz using pot **P4**
 - Keep VCO1 **LEVEL** about 1.5 Vpp using pot **P5**
- In ACT-04 keep switch **S1** at **PM** position and Set the Frequency Demodulator in Ratio mode (Jumper in R position).
- Connect the output of Function Generator (ACT-01) **OUT1** post to the input of PHASE MODULATOR **MOD IN** post of (ACT-04).
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Phase modulator (ACT-04) **RF IN** post.
- Connect the output of Phase modulator (ACT-04) **PM OUT** post to the input of limiter (ACT-04) **IN** post.
- Connect the output of limiter **OUT** post to the input of Foster Seeley Demodulator (ACT-04) **FM IN** post.
- Be sure that the Phase modulator and Ratio DETECTOR are tuned to 450 KHz i.e. follow the modulator calibration and demodulator calibration.
- Connect the oscilloscope to the modulating signal and to the detected signal and examine the waveforms. As the phase shift (and consequently the frequency deviation) introduced by the modulator is very small also the amplitude of the detected signal is very small (about 10 mVpp)
- Observe the following:
 - If the modulator is set to FM (Inserted integrator the FM detector supplies the same wave form of the modulating signal this means that the signal generated by the modulator is effectively and FM signal.
 - If the modulator is set to PM (Disconnected integrator the FM detector supplies derivative of the same wave form of the modulating signal if the

modulating signal is a triangular wave the detected signal will be a square wave as shown in fig. 6.5.

- If the modulating signal is the square wave the detected signal will be a wave with positive and negative pulses as shown in fig. 6.6.



SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.
- Put switch 2 of SF 1 in ACT-04 Switch fault section to ON position. This will open R38 from Limiter. Limiter action is disabled.
- Put switch 4 of SF 1 in ACT-04 Switch fault section to ON position. This will open DV1's GND side in phase modulator, no variation at phase modulator output and amplitude is very low.

EXPERIMENT NO.7

EXPERIMENT NO.7

NAME:-

To Study of Frequency Conversion

OBJECTIVE:-

- A1 To Study the frequency relations with the Mixer.
- A2 To Study the effect of Image Frequency.
- B1 To observe the frequency response of the IF filter on CRO.
- B2 To plot the frequency response of the IF filter.

THEORY:-

In many applications it is convenient to operate a frequency translation of a signal: in radio electronics this technique is largely used in super heterodyne receiver.

The radio frequency signal (RF) detected by the antenna is amplified by an RF amplifier and transferred, via a frequency mixer and a local oscillator (LO), to a lower and fixed frequency (Intermediate Frequency IF). The translation to a fixed and lower frequency than the received one enables to:

- Use a fixed band amplifier for the amplification of all the received signals, independently from their frequency.
- Operate on lower frequency signals.

The frequency translation is sometimes used in communication systems. Where for example, the modulation is carried out at an Intermediate frequency, having value lower to that of the effectively transmitted frequency

As already said, to carry out the frequency translation a locally generated translation is necessary, called local oscillation, which interacts with the radio frequency signal so as to produce the new frequency.

If the oscillation is produced by the same operating circuit translation device, it is normally called as Converter; in the reverse case we speak respectively of Local Oscillator and Mixer. The frequency translation is differently indicated as frequency conversion, mixing, beating, heterodyne process. The techniques used are many, and depend on the device used and on the operating frequencies. During the exercises we use a frequency mixer carried out with Dual Gate MOSFET.

Frequency Mixer with Dual Gate MOSFET

The RF and LO signals are separately applied to two gates of the MOSFET. The operation of the MOSFET can be analyzed by presuming that, when the signal LO has a higher amplitude than the signal RF, the trans-conductance g_m of the device is determined only by the amplitude of signal LO and is independent of RF. Consider the characteristic curves of a MOSFET fig. 7.1 the trans-conductance g_m as function of the Gate₂- Source voltage (V_{G2}), for V_{G1} included into a narrow range of values, can be defined as;

$$g_m = \frac{\Delta I_D}{\Delta V_{G1}}$$

With D as Drain current. If for example V_{G1} (signal RF) is limited to a small value around ($\pm 100\text{mV}$) of -0.75V , and V_{G2} (signal LO) takes values ranging between 1 and 3V, the trans-conductance practically depends only on V_{G2} . It is possible to detect then for g_m the representation of fig. 7.2.

The heterodyne action of the MOSFET depends on its biasing: if V_{G2} acts in a zone where g_m is linear for example between A & B of fig. 7.2, g_m can be expressed as:

$$g_m \approx a + b \cdot V_{G2}$$

Set:

$$V_{G1} = A_{RF} \cdot \cos(\omega_{RF} \cdot t)$$

$$V_{G2} = A_{LO} \cdot \cos(\omega_{LO} \cdot t)$$

The output current I_D becomes:

$$\begin{aligned} I_D &= g_m \cdot V_{G1} = a \cdot A_{RF} \cdot \cos(\omega_{RF} \cdot t) + b \cdot A_{RF} \cdot A_{LO} \cdot \cos(\omega_{LO} \cdot t) \cdot \cos(\omega_{RF} \cdot t) = \\ &= a \cdot A_{RF} \cdot \cos(\omega_{RF} \cdot t) + \frac{b \cdot A_{RF} \cdot A_{LO}}{2} \cdot [\cos(\omega_{LO} - \omega_{RF}) \cdot t + \cos(\omega_{LO} + \omega_{RF}) \cdot t] \end{aligned}$$

Using the circuit tuned on the Drain you extract the intermediate frequency component, with pulse:

$$\omega_{IF} = \omega_{LO} - \omega_{RF}$$

If V_{G2} do not acts in a linear segment of the trans-conductance g_m , then g_m won't be expressed anymore with a linear relation and the result of the conversion operated by the MOSFET will contain also the harmonics of the frequency of LO. A frequency IF can be obtained equal to:

$$\omega_{IF} = 3 \cdot \omega_{LO} - \omega_{RF}$$

We define Conversion Trans-conductance g_c as the ration between the pulse output current i_{IF} and the pulse input voltage v_{RF} :

$$g_c = \frac{i_{IF}}{v_{RF}}$$

The mixer used in the exercise is composed of three section :

- The MOSFET, which carries out the interaction between the LO and the RF signal.
 - A band pass filter centered to 450 KHz, consisting of two tuned circuits coupled between them.
- An output amplifier stage

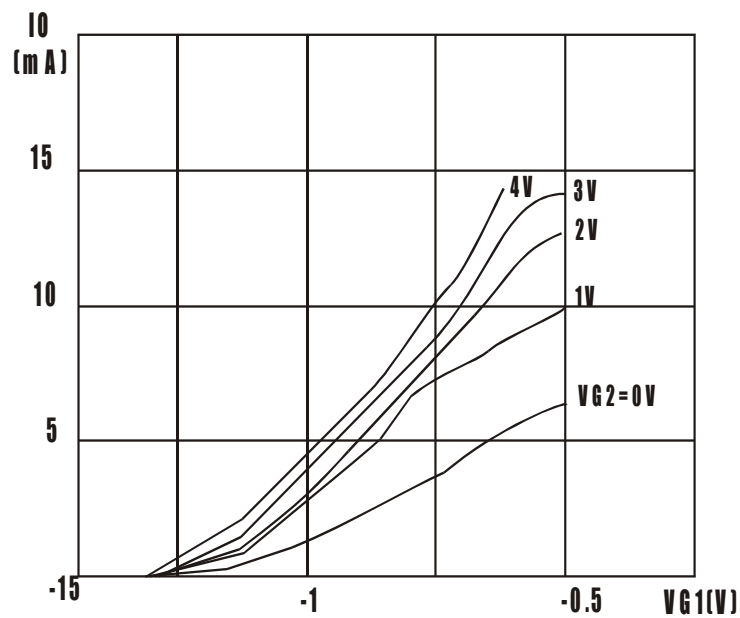


FIG.7.1 CHARACTERISTIC CURVE

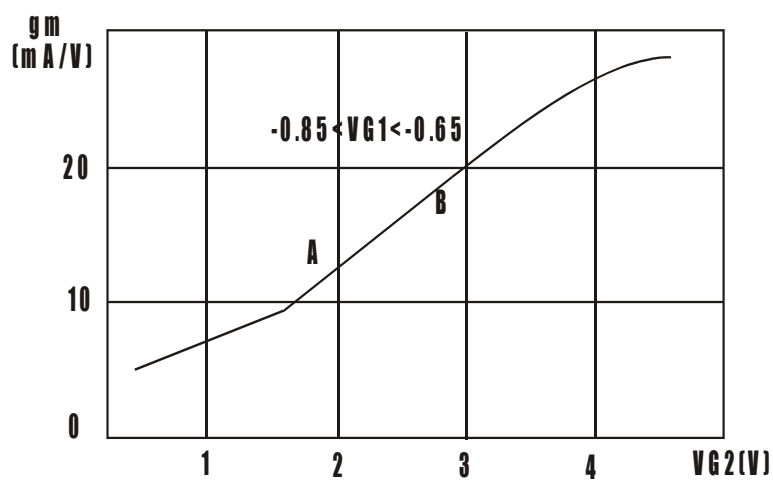
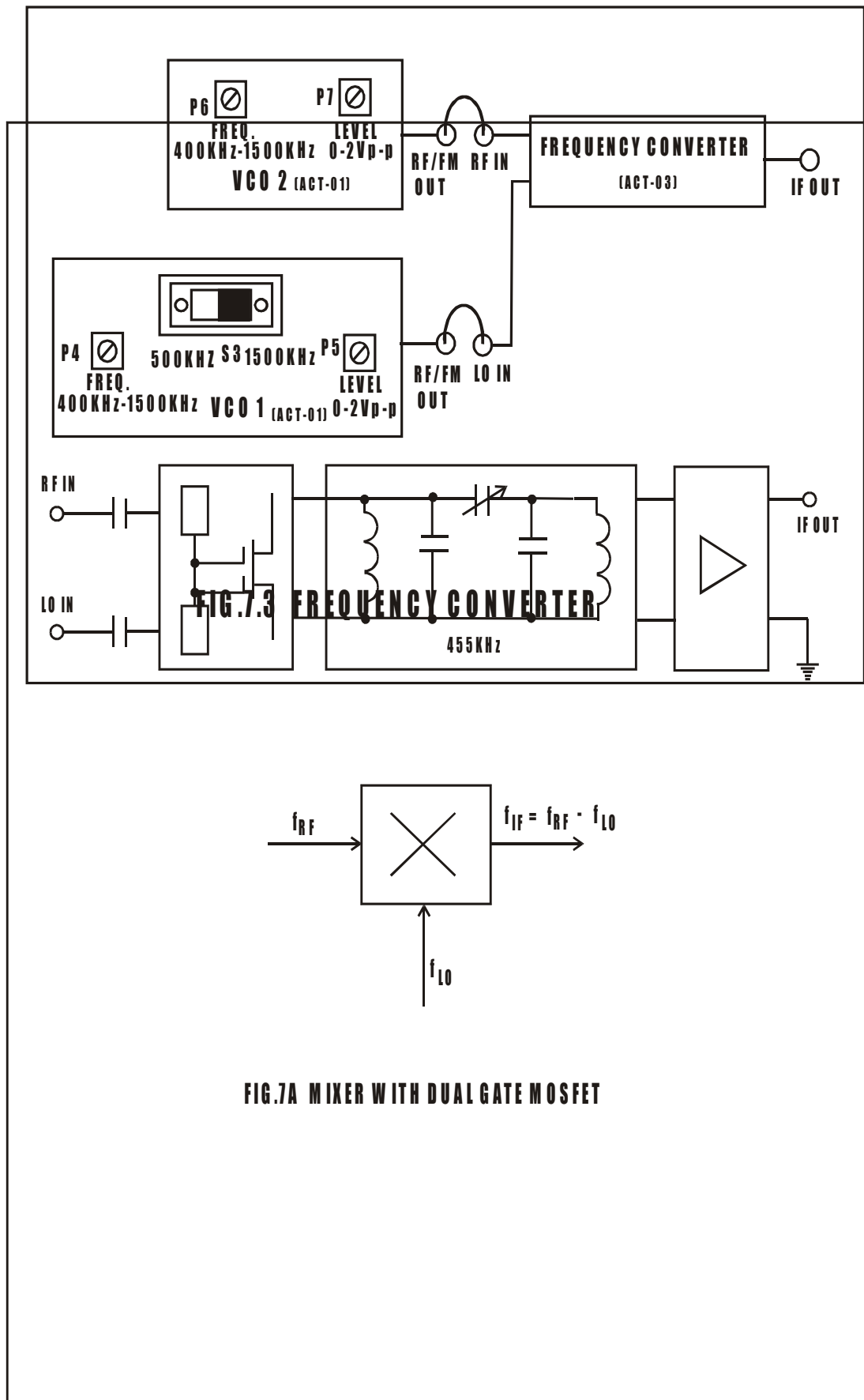


FIG.7.2 TRANSCONDUCTANCE g_m



A1.To study the Frequency Relations with the Mixer

EQUIPMENT:-

- Modules ACT-01, ACT-03.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-03 and switch it on.
- Refer to the fig.7.3 & Carry out the following connections and switch settings.
- Carry out the following presetting in ACT-01:
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **1000 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1Vpp**.
 - Using pot **P6** keep VCO2 frequency at **550 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **100mVpp**.
- Connect the output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **RF IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the other input of Frequency Converter (ACT-03) **LO IN** post.
- Connect the oscilloscope to the output of the mixer **IF OUT** post, and accurately vary the frequency LO using pot **P4** until you detect a sine wave signal.
- Measure the frequencies of the two input signals and the output signal, and check that the last is the difference of the first two: $f_{IF} = f_{LO} - f_{RF}$.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.

- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 1 of SF 1 in ACT-03 Switch fault section to ON position. This will open C17 from FET from frequency converter section. Frequency converter output gets reduced this will disturb the next section output.

A2. To study the effect of Image Frequency

EQUIPMENT:-

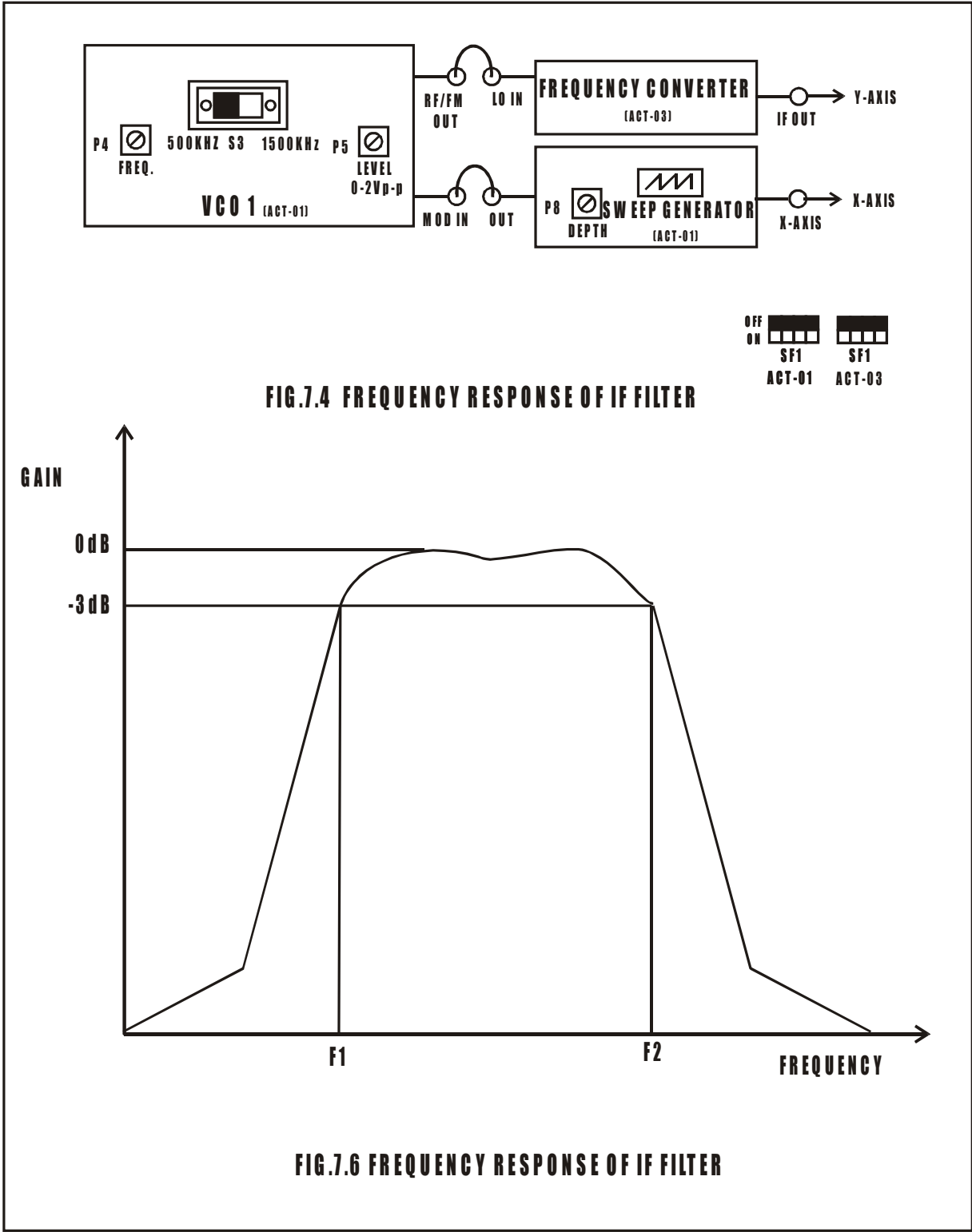
- Modules ACT-01, ACT-03
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Perform the operation as done in A (1) with the following presetting:
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **1000 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1Vpp**.
 - Using pot **P6** keep VCO2 frequency at **1450 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **100mVpp**.
- Accurately adjust the above frequency until the output is crossed again by a sine wave form. Measure the frequencies and check that now:

$$f_{IF} = f_{LO} - f_{RF}$$
- The last frequency relation indicates that the two signals with different frequency RF are converted to the same frequency IF. If they were contemporarily present, there would be interference between the two and this would make a proper reception of the signal impossible. The unwished frequency is called image frequency. To prevent this inconvenience it is necessary to prevent that the image signal reaches the input of the mixer, and this is carried out interposing selective filters between the input of the signal RF and the mixer.



B1. To observe the Frequency Response of the if Filter on oscilloscope

EQUIPMENT:-

- Modules ACT-01, ACT-03.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

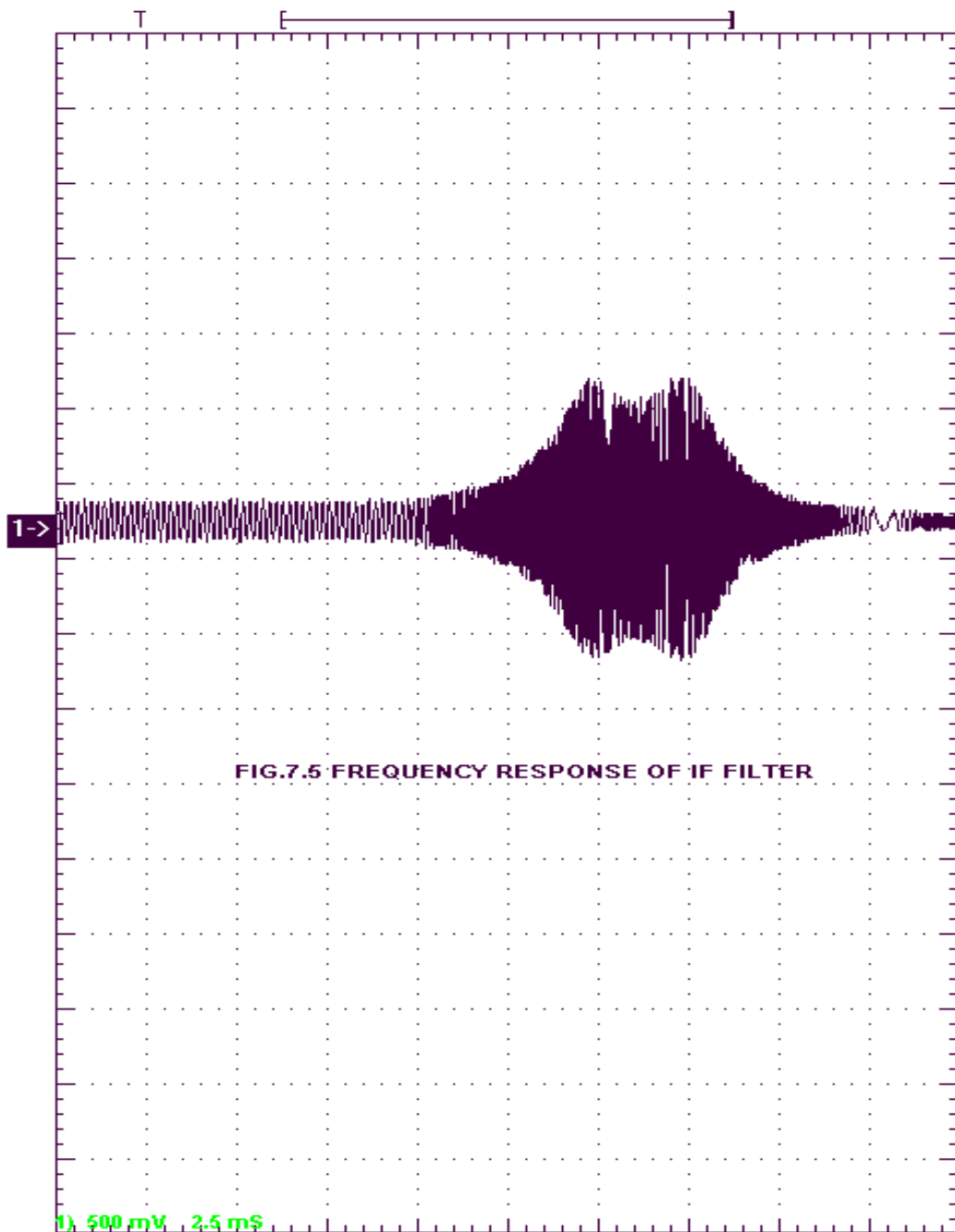
- Connect the power supply with proper polarity to the kits ACT-01 & ACT-03 and switch it on.
- Refer to the fig.7.4 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **450 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1Vpp**.
 - Using pot **P8** keep SWEEP **DEPTH** almost completely counter clockwise.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the other input of Frequency Converter (ACT-03) **LO IN** post.
- Connect the output of Frequency Converter (ACT-03) **IF OUT** post to the Y-axis of Oscilloscope.
- Connect the **MOD IN** post of VCO1 (ACT-01) to the **OUT** post of SWEEP Circuit (ACT-01).
- Connect the **X-AXIS** post of SWEEP Circuit (ACT-01) to the X-axis of Oscilloscope.
- Adjust the central frequency of the VCO1 and the amplitude of the Sweep (DEPTH) to obtain a curve similar to the ones of fig. 7.5 on the oscilloscope.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 In ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in

- 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 In ACT-03 Switch fault section to ON position. This will open C17 from FET from frequency converter section. Frequency converter output gets reduced this will disturb the next section output.



B2. To plot the Frequency Response of the if Filter

EQUIPMENT:-

- Modules ACT-01, ACT-03.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-03 and switch it on.
- Refer to the fig.7.4 & Carry out the following connections and switch settings.
- Do the following presetting in ACT-01:
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **455 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **1Vpp**.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **LO IN** post.
- Be V_o and V_i the peak-to-peak voltages measured across the output and input of the mixer. The gain G of the 455 KHz circuit is given by $G = V_o/V_i$ and, in decibel:

$$G_{dB} = 20 \cdot \log (V_o/V_i)$$
- Repeat the measurements carried out in the last test "4" by varying the frequency from 430 to 480 KHz with steps of 5 KHz, calculate G_{dB} in correspondence to each frequency and report everything on a table.
- With the data in the table plot a graph setting G_{dB} on the Y-axis (vertical) and the frequency on the X-axis (horizontal): you obtain the frequency response curve of the circuit, similar to the one of fig. 7.6.
- Evaluate the band pass B :

$$B = f_2 - f_1.$$
 With f_2 and f_1 as frequencies in correspondence to which the gain G_{dB} diminishes of 3dB in respect to its minimum value (fig. 7.6) You will find a bandwidth of about 20 KHz.

Input Conversion

- The frequency mixer can be used also to convert a signal to higher frequencies. Connect the output of the mixer to the 455 KHz ceramic filter mounted on ACT-02, so that the band of the total IF filter is restricted. Across the input RF IN apply a sine wave signal with frequency of about 10 KHz and amplitude about 100 mVpp (use the FUNCTION GENERATOR of ACT-01). Take the frequency LO to about 445 KHz, the output of the ceramic filter.

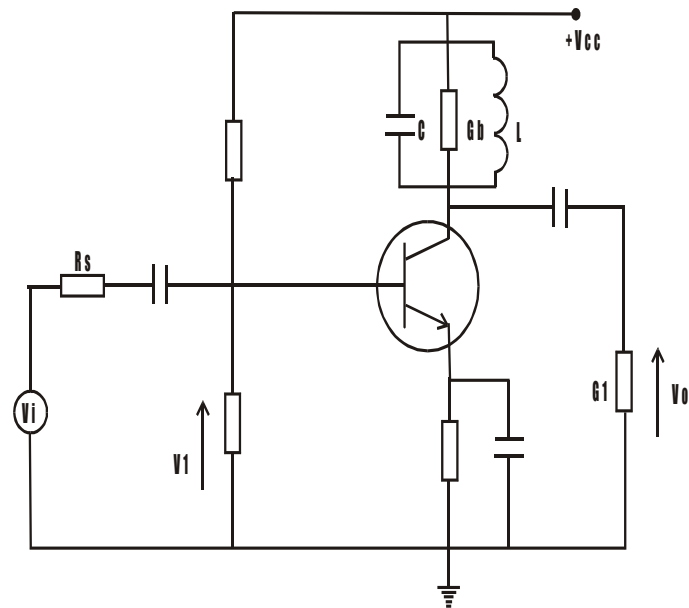
- Measure the frequencies of the two input signals and the output signal, and check that the last now is the sum of the first two:
 $f_{IF} = f_{LO} + f_{RF}$.

SWITCH FAULTS:-

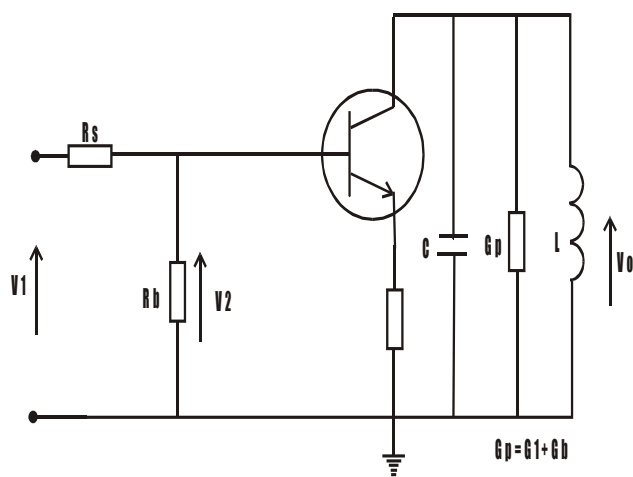
Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 1 of SF 1 in ACT-03 Switch fault section to ON position. This will open C17 from FET from frequency converter section. Frequency converter output gets reduced this will disturb the next section output.

EXPERIMENT NO.8



A) ELECTRICAL DIAGRAM



B) EQUIVALENT CIRCUIT

FIG.8.1 SELECTIVE TRANSISTOR AMPLIFIER

EXPERIMENT NO. 8

NAME:-

To Study of Selective Amplifiers

OBJECTIVE:-

- A1 To find the Central frequency and Bandwidth of an Amplifier.
- A2 Impedance transformation with transformer.
- A3 Impedance transformation with Autotransformer.
- B1. Calibration gain of 2-Stage IF Amplifier.
- B2. Automatic gain control
- B3. AM Detector

THEORY:-

Selective Amplifiers for Small Signals

The selective amplifier is used in all applications, in which it is necessary to amplify the signals belonging to a narrow frequency band. The main parameters defining the properties of a selective amplifier are:

- Central operating frequency f_0
- Bandwidth around f_0 .
- Center band gain.

The electrical diagram of the generic selective amplifier with transistor is shown in fig. 8.1. The dynamic load of the active element consists of a resonant circuit RLC. The central operating frequency f_0 is determined by the resonance frequency of circuit RLC, which is obviously fixed to be coincident with the frequency range of the signal which must be amplified.

G_b is the parallel conductance of the coil and G_1 is the load. Varying L and C it is possible to choose the central frequency and the band width. The center band gain of the amplifier is detected using the Giacoletto equivalent circuit for the transistor, which is:

$$A_o = V_o/V_1 = -g_m \bullet R_{eq}.$$

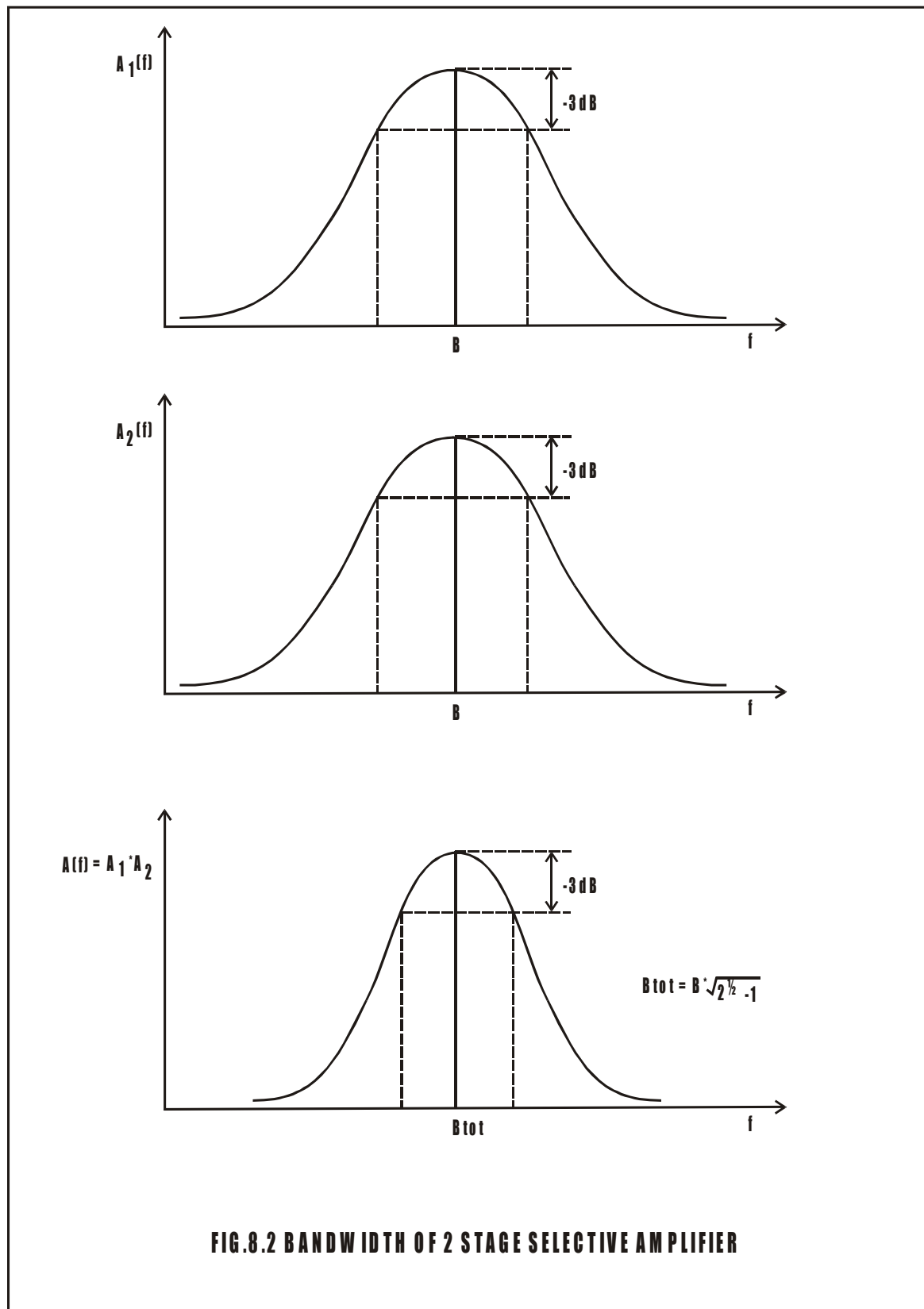
Where:

g_m = transistor trans-conductance.

R_{eq} = R_p / r_o .

$1/R_p = G_b + G_1 = G_p$.

R_o = output resistance of the transistor.



Response Curve

The response curve of the amplifier has the same behavior of the resonance curve of the tuned circuit GLC.

The gain A of the amplifier as function of frequency, the resonance frequency and the band pass are equal to:

$$A = \frac{-g_m \bullet R_{eq}}{V[1 + (\overline{R_{eq} \bullet \omega_0 \bullet C}) \bullet (\omega/\omega_0 - \omega_0/\omega)^2]}$$

$$\omega_0 = 1/\sqrt{L \bullet C} = > f_0 = \omega_0/2\pi$$

$$B = 1 / (2\pi \bullet R_{eq} \bullet C)$$

As you can see from the last relations, once the necessary band pass and the load resistance are determined, the value of C keeps univocally determined by the 3rd relation and, consequently, also by the value of L of the 2nd. If we consider more selective stages in cascade, with equal central frequency and bandwidth, the total band pass of the amplifier is obviously reduced in respect to the one of the single stages. If B is the band of the single stages and n the number of stages, the band B_{tot} results (fig. 8.2):

$$B_{tot} = B \bullet \sqrt[n]{2^{1/n} - 1}$$

Impedance Transformation

It may happen that, in the frequency range under operation, difficulties rise about the choice of the capacity or in obtaining a coil with adequate coefficient of merit Q . Suppose, for example, to design an amplifier with intermediate frequency $f_0 = 455$ KHz, with a pass band of 10 KHz and a load $r_1 = 1K\Omega$. Neglecting r_o and R_b , the value of C is:

$$C = 1 / (2\pi \bullet R_1 \bullet B) \approx 16 \text{ nF}$$

$$\text{The inductance } L \text{ is: } L = 1 / (2\pi \bullet f)^2 \bullet C \approx 8\mu\text{H.}$$

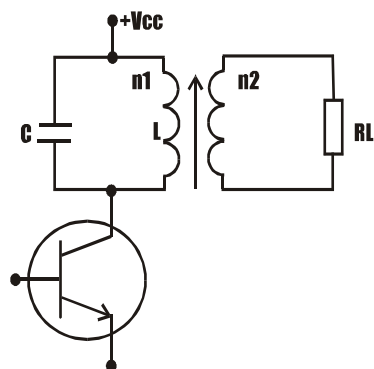
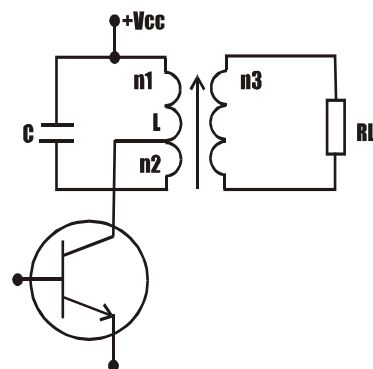
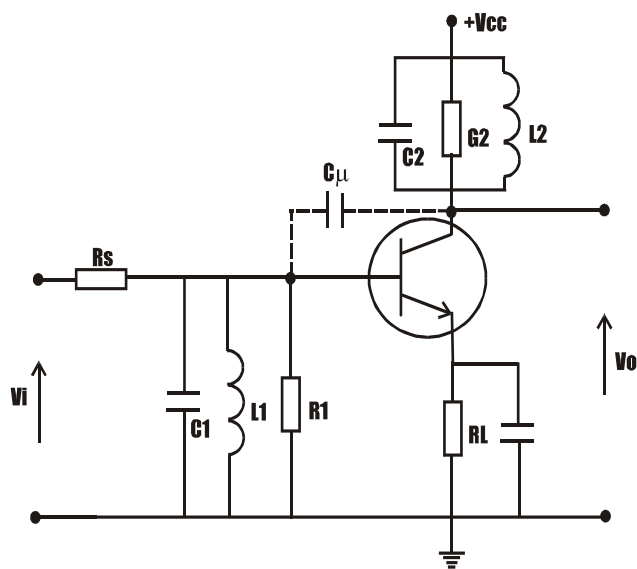
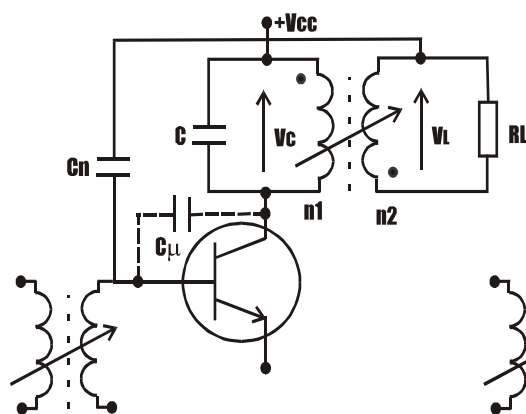
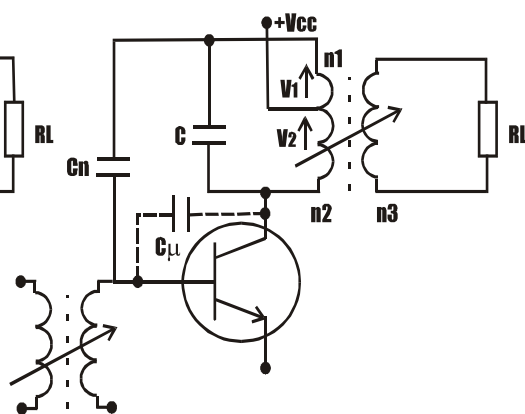
The value of the inductance is very small for the operating frequency chosen, i.e. it is difficult to carry out such a coil with high coefficient of merit Q . It is better to carry out the load impedance matching so that the values of L and C can be chosen with a wide margin. The solution adopted uses a transformer and an autotransformer.

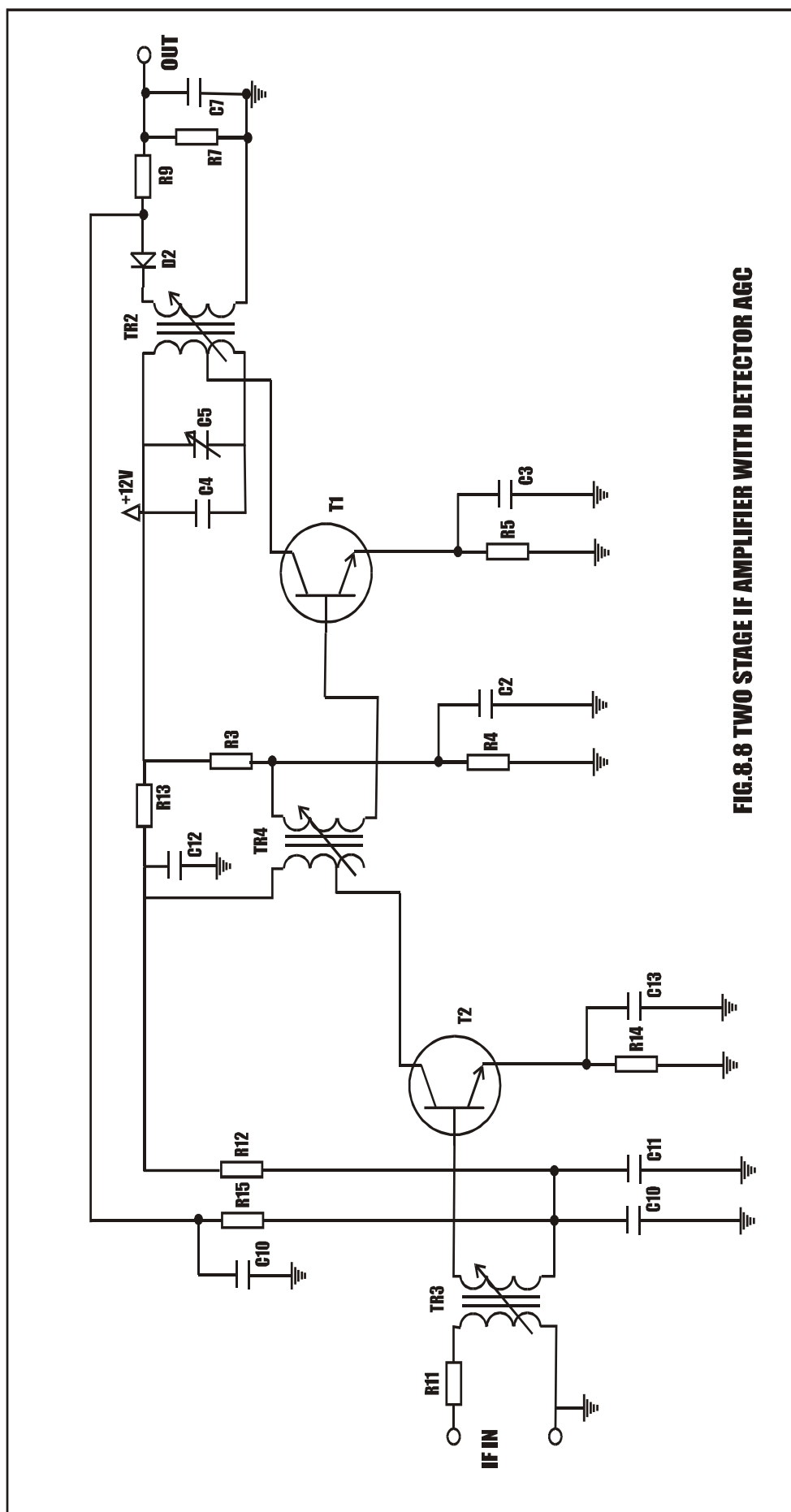
Impedance Matching With Transformer

Refer to fig. 8.3. As the core of the transformer is ferrite, the coupling coefficient can be considered practically unitary and the transformer can be ideal. It follows that the resistance seen on the primary equivalent to the resonance frequency is to;

$$B = \frac{G_{tot}}{2\pi \bullet C_{eq}}$$

In conclusion we can say that: once the total conductance G_{tot} seen from the collector (to ensure the stability) is fixed, the transformation of C is used to obtain the desired band, the transformation of R to obtain the set G_{tot} .

**FIG.8.3 MATCHING WITH TRANSFORMER****FIG.8.4 MATCHING WITH AUTOTRANSFORMER****FIG.8.5 SELECTIVE AMPLIFIER WITH TUNED INPUT STAGE****FIG.8.6 NEUTRALIZATION TYPE 1****FIG.8.7 NEUTRALIZATION TYPE 2**



Stability and Alignment

Sometimes the selective amplifiers operate at frequencies, for which it is not possible to ensure the reactive effects of the transistor, and in particular the inner feedback between the collectors and base due to the capacity C_μ . This capacity creates two problems:

- The resonance frequency of the tuned circuit does not depend only on the passive components L and C , but also on function C_μ , i.e. parameter of the transistor (whose value is hardly reliable).
- When a second tuned circuit is added to the input of the transistor, C_μ supplies a bilateral coupling between the input and output resonant circuits. Due to this coupling it may be difficult to align the amplifier, i.e. to tune the two resonant circuits to obtain a regular and symmetrical response centered on f_0 . In some cases the circuit becomes unstable and starts oscillating.

Figure 8.5 shows a selective amplifier with tuned circuits on the input as well as the output. The condition for the stability of the amplifier is:

$$G_1 \cdot G_2 > 2\pi \cdot f_0 \cdot C_\mu \cdot g_m$$

In practice it is sufficient that $G_1 \cdot G_2$ is 4 or 5 times higher than $2\pi \cdot f_0 \cdot C_\mu \cdot g_m$

Neutralization Type 1

To ensure the stability of the amplifier, sometimes you can add a proper external feedback, which cancels the inner feedback due to C_μ . This technique is known as neutralization of the amplifier stage. The neutralization is carried out mainly in two ways. When the different stages are coupled to a transformer, the diagram of figure 8.6 can be used.

The base of the transistor is reached by current proportional to v_c (due to the inner feedback) and a current proportional to v_L (due to the external feedback). As v_L is in phase opposition with respect to v_c , the sum of the currents is annulled by properly choosing C_n as follows:

$$C_n = \frac{n_1}{n_2} \cdot C_\mu$$

Neutralization Type 2

Another neutralization technique is shown in fig 8.7. The voltage v_2 is in phase opposition with respect to v_1 ; by properly choosing n_1/n_2 and C_n it is possible to annul the currents on the base of the transistor due to the inner (C_μ) and external (C_n) feedback.

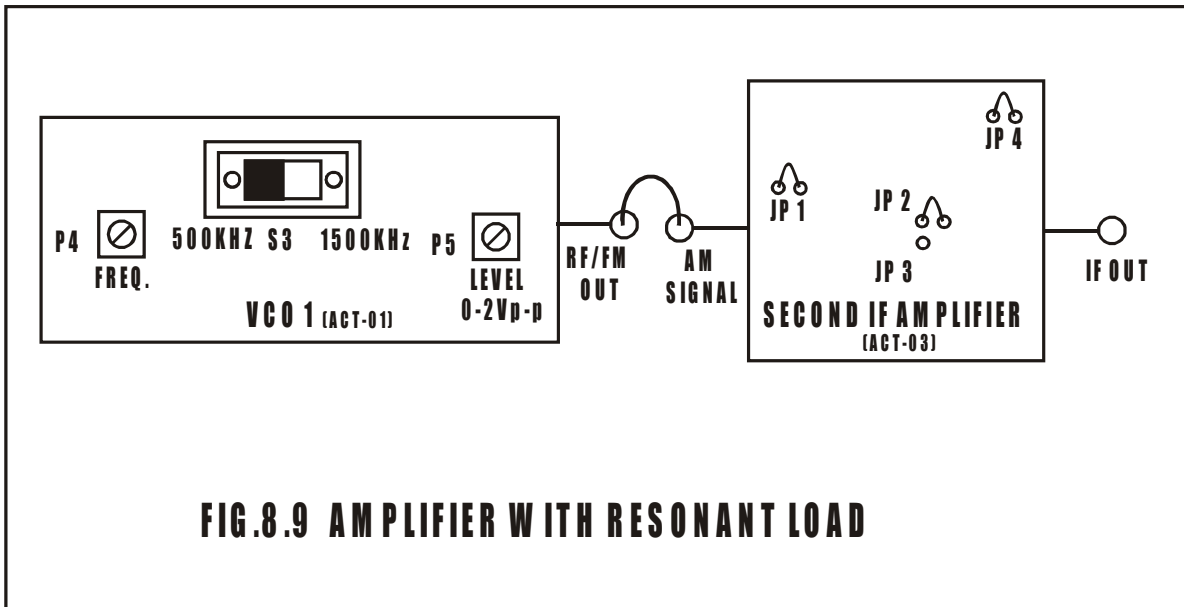
2-Stage IF Amplifier with AM and AGC Detector

Refer to fig. 8.8, which reports the electrical diagram of a 2-stage intermediate Frequency Amplifier, with AM diode detector and AGC (Automatic gain Control). This circuit is typically used in super heterodyne AM receivers.

The IF signal is applied to the first transformer IF (TR3) and by this to the first stage (transistor T2). The biasing of T2 depends on the AGC voltage coming from the detector, and in this way the amplification of the first stage is varied. Across the transformer TR4 the signal is taken to the second IF stage (transistor T1) and

by this, with the transformer TR2, to the detector diode.

The diode is connected so that the negative envelope of the modulated signal is detected. The detected signal seems to be composed of the low frequency modulating signal and a negative continuous component proportional to the amplitude of the IF signals. The d.c. Component is separated by the signal (via the low pass circuit R-C), and constitutes the AGC voltage used to vary the amplification of the first stage IF.



A1. To find the Central Frequency and Bandwidth of an Amplifier

EQUIPMENT:-

- Modules ACT-01, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-03 and switch it on.
- In ACT-01:
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency between 400 – 500 KHz.
 - Using pot **P5** keep VCO1 amplitude at **50mVpp**.

- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to post **AM SIGNAL** of ACT-03.
- Set the IF second stage as in fig. 8.9 (short only the jumpers: JP1, JP2 & JP4 and keep jumpers JP3, JP5, JP6, JP7, JP8, JP9, JP10 & JP11 open).
- Connect the oscilloscope (probes 10:1) to the input and the output of the amplifier (jumper JP2 and post **AM SIGNAL**). Vary the input frequency to obtain the maximum output: this frequency is the central frequency of the amplifier, and corresponds to the resonance frequency of the load L-C of the same amplifier.
- Rotate the variable capacitor C5, and note that the central frequency varies. Calibrate C5 to obtain a central frequency of 455 KHz.
- Suppose V_o and V_i are the output and input peak-to-peak voltages. The gain G of the amplifier is given by $G = V_o/V_i$ and, in decibel:

$$G_{dB} = 20 \bullet \log (V_o/V_i)$$
- Vary the frequency from 430 to 480 KHz with steps of 5 KHz, calculate G_{dB} in correspondence to each frequency and report it in a table
- With the data of the table plot a graph setting G_{dB} on the Y-axis (vertical) and the frequency on the X-axis (horizontal)
- Evaluate the band pass $B1$:

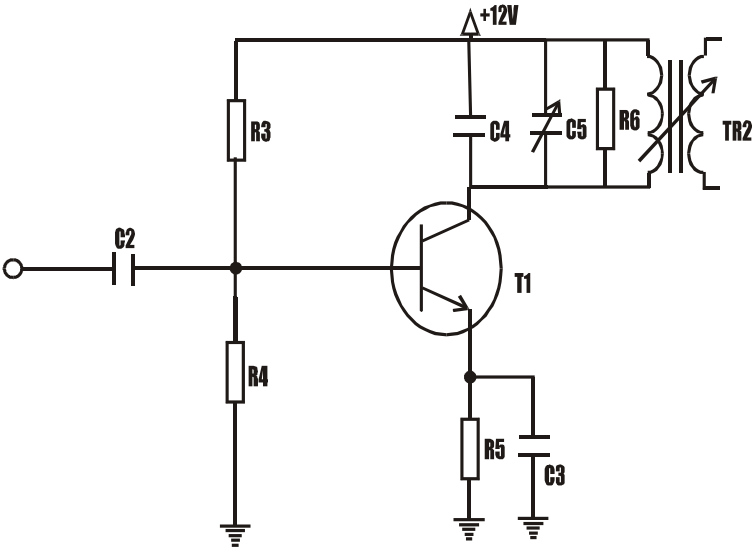
$$B1 = f2 - f1.$$

With frequency $f2$ and $f1$ in correspondence to which the gain G_{dB} diminishes of 3dB in respect to its minimum value you will find a bandwidth of about 50 KHz.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600KHz



AMPLIFIER WITH RESONANT LOAD

FREQUENCY (KHz)	430	435	440	445	450	455	460	465	470	475	480	490	500
INPUT VOLTAGE (mV _{pp})	50	50	50	50	50	50	50	50	50	50	50	50	50
OUTPUT VOLTAGE (V _{pp})	2.4	2.6	2.8	2.9	3.0	2.9	2.8	2.7	2.7	2.6	2.6	2.5	2.1
GAIN (dB)	33.6	34.3	34.9	35.2	35.5	35.2	34.9	34.6	34.6	34.3	34.3	33.9	32.4

GAIN OF AN AMPLIFIER (INDICATIVE VALUES)

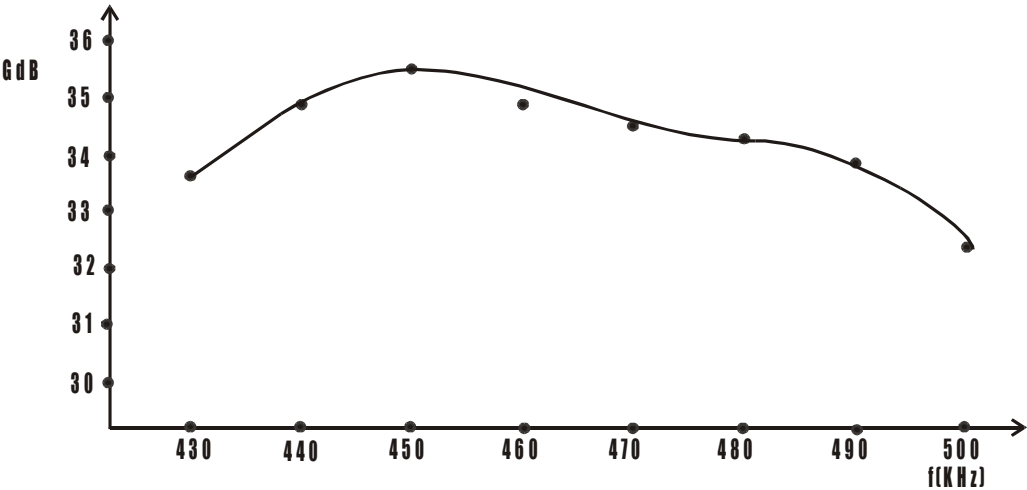


FIG.4.9A FREQUENCY RESPONSE OF AN AMPLIFIER

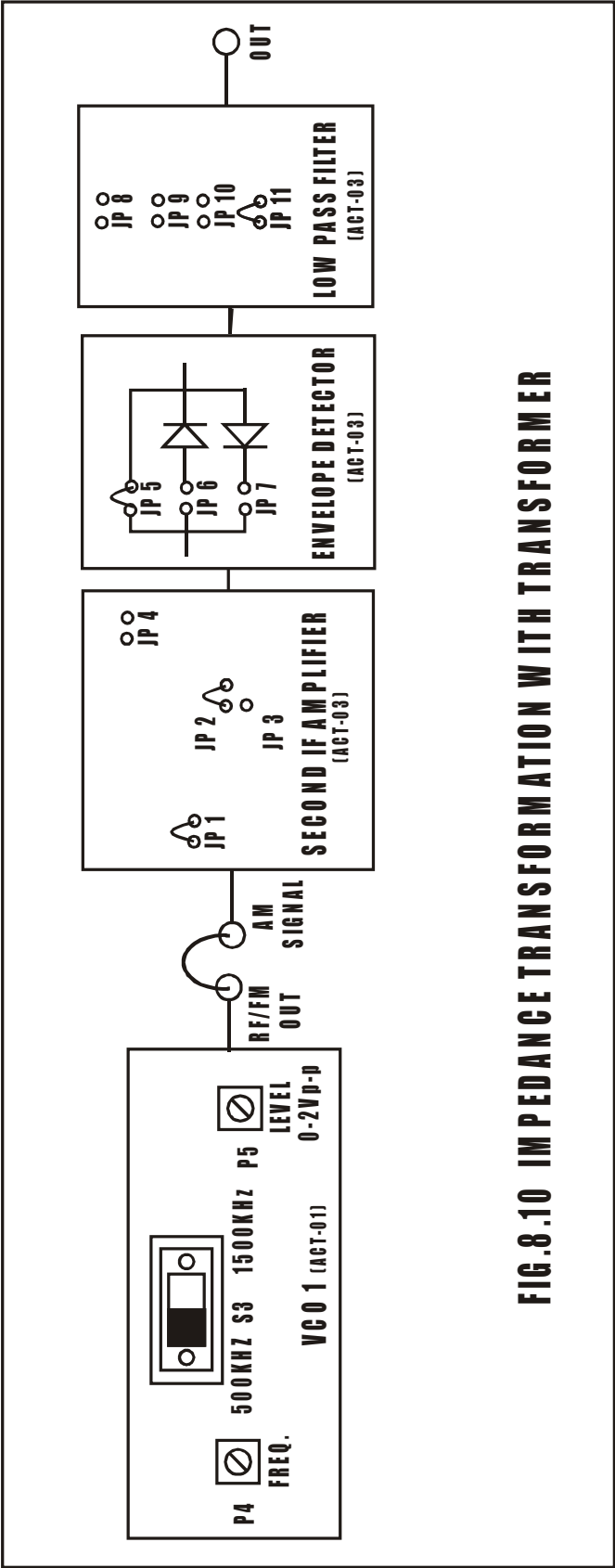


FIG.8.10 IMPEDANCE TRANSFORMATION WITH TRANSFORMER

A2. Impedance Transformation with Transformer

EQUIPMENT:-

- Modules ACT-01, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-03 and switch it on.
- In ACT-01:
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **455 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **30mVpp**.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to post **AM SIGNAL** of ACT-03.
- Set the second stage IF as in fig. 8.10 (short only the jumpers: JP1, JP2, JP5, & JP11 and keep jumpers JP3, JP4, JP6, JP7, JP8, JP9 & JP10 open).
- Connect the oscilloscope (probes 10:1) to the input of the amplifier post **AM SIGNAL** and on the load post **OUT**.
- Across the input of the amplifier apply a signal with 50mVpp-amplitude and 455 KHz-frequency. Calibrate it to 455 KHz by adjusting the capacity C5 varies the central frequency of the resonant circuit.
- Vary the frequency of the input signal and evaluate the new bandwidth B2 of the amplifier. You will find a value of about 30 KHz.
- Although the load is a resistance lower then the one of the last exercise, the bandwidth B2 is smaller to the impedance B1 measured before. This is due to the impedance transformation operated by the transformer TR2. The winding ratio of TR2 is equal to about 12:

$$(n_1 + n_2) / n_3 \approx 12$$

The value of the equivalent resistance seen from the collector of the transistor is equal to:

$$R_{eq} = \frac{(n_1 + n_2)^2}{n_3} \cdot R_8 \approx 144 \text{ K}\Omega$$

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will Open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-03 Switch fault section to ON position. This will open C7 from 2nd IF section low pass filter, filter output gets distorted.
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600KHz

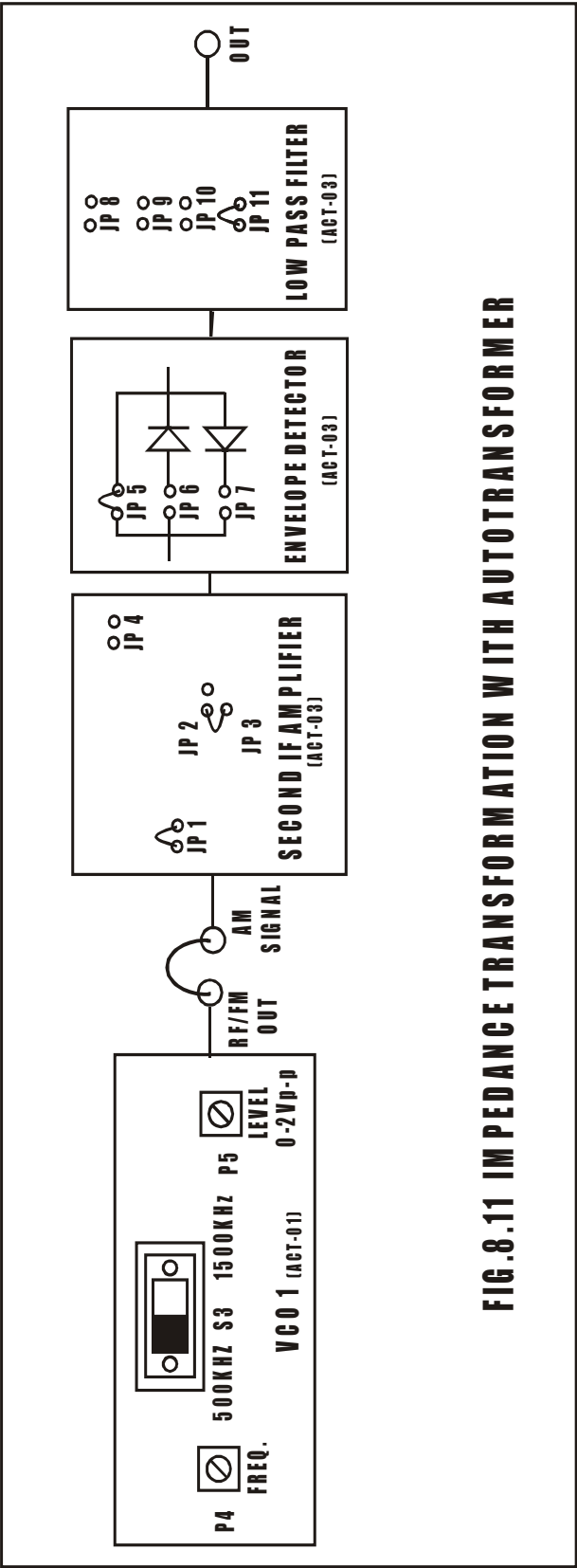


FIG.8.11 IMPEDANCE TRANSFORMATION WITH AUTOTRANSFORMER

A3. Impedance Transformation with Autotransformer

EQUIPMENT:-

- Modules ACT-01, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE

- Connect the power supply with proper polarity to the kits ACT-01 & ACT-03 and switch it on.
- In ACT-01:
 - Keep switch **S3** in **500 KHz** position.
 - Using pot **P4** keep VCO1 frequency between 400 – 500 KHz.
 - Using pot **P5** keep VCO1 amplitude at **30mVpp**.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to post **AM SIGNAL** of ACT-03.
- Set the second IF stage as in fig. 8.11 (short only the jumpers: JP1, JP3, JP5, & JP11 and keep jumpers JP2, JP4, JP6, JP7, JP8, JP9 & JP10 open).
- Connect the oscilloscope (probes 10:1) to the input of the amplifier post **AM SIGNAL** and on the load post **IF OUT**.
- Across the input post **AM SIGNAL** apply a signal with 50mVpp amplitude and a frequency ranging between 400 and 500 KHz.
- Evaluate the new bandwidth B3 of the Amplifier. You will find a bandwidth of about 20 KHz, smaller than the bandwidth B2 of the last Experiment. This result is explained by considering that:
 - The transformer coupling transforms the resistance R8 into a higher equivalent resistance, as already seen in the last exercise
 - The autotransformer transforms the tuning capacity ($C=C4//C5$) into a higher equivalent capacity. The equivalent capacity seen from the transistor collector is equal to:

$$C_{eq} = (1 + \frac{n_2}{n_1})^2 \cdot C_{n1}$$

Where the ratio n_2/n_1 of the transformer used is about 0.4
Increasing the equivalent capacity you make the band B3 of the amplifier Diminish. This is evident observing the relation which expresses B3:

$$B3 = \frac{1}{2\pi \cdot R_{eq} \cdot C_{eq}}$$

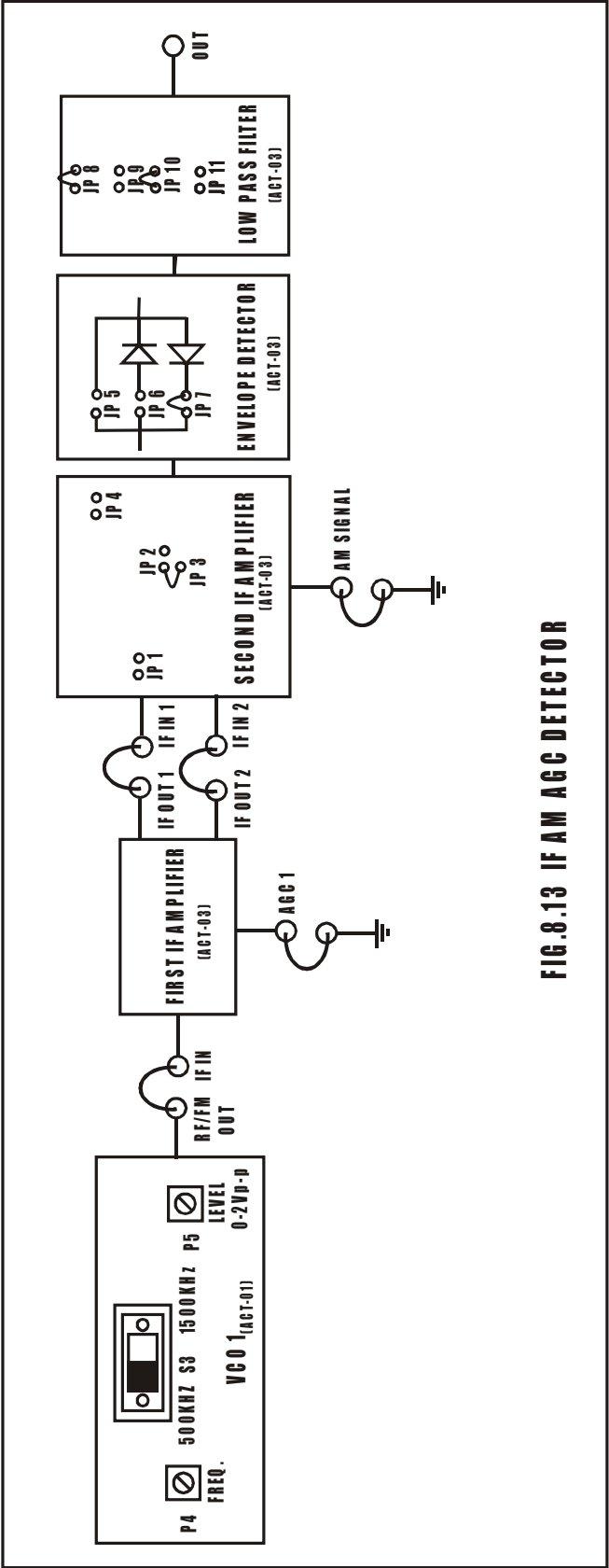


FIG.8.13 IF AM AGC DETECTOR

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-03 Switch fault section to ON position. This will open C7 from 2nd IF section low pass filter, filter output gets distorted.
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600KHz

B1. Calibration Gain Of 2-Stage IF Amplifier

EQUIPMENT:-

- Modules ACT-01, ACT-02, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01, ACT-02 & ACT-03 and switch it on.
- Refer to the fig.8.13 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S3** in **500 KHz** position. Using pot **P4** keep VCO1 frequency at **455 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **20mVpp**.
- Connect the output of 1st IF (ACT-03) **IF OUT 1** post to post **IF IN 1** of 2nd IF (ACT-03).
- Connect the output of 1st IF (ACT-03) **IF OUT 2** post to post **IF IN 2** of 2nd IF (ACT-03).
- Connect post **AM SIGNAL** of 2nd IF (ACT-03) to **GND** Post.
- Connect post **IF OUT 2** of 1st IF to ground post.
- Set the circuit as in fig. 8.13, to obtain a 2-Stage IF amplifier with amplitude detector and Automatic Gain Control (connect jumpers JP3, JP7, JP8 and JP10 and keep jumpers JP1, JP2, JP4, JP5, JP6, JP9 & JP11 open). The

connection between posts **IF OUT 2** and ground sets a fixed gain on the first stage.

- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to input **IF IN** post of ACT-03.
- Connect the oscilloscope (probe 10:1) to the output of the amplifier post **IF OUT**. Keep both the probes away from each other to avoid noise or distortion.

Adjust the variable capacity C5 of the second IF stage, and eventually the IF Transformers if necessary to obtain the maximum amplitude of the sine wave form detected (about 3 Vpp).

- Calculate the center band gain of the first stage (measure the output across post **AGC1**) and of the complete amplifier post **IF OUT**.
- Plot point by point the frequency response of the first stage and of the complete amplifier (vary the frequency of the input signal and measure the corresponding gain).

Repeat the measurement using a wobbled signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency
- Put switch 2 of SF 1 in ACT-03 Switch fault section to ON position. This will open C7 from 2nd IF section low pass filter, filter output gets distorted.
- Put switch 3 of SF 1 in ACT-03 Switch fault section to ON position. This will open R41 Bias resistor of Transistor T2 from GND side in 1st IF section. 1st IF section output reduces.
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600KHz

B2. Automatic Gain Control

EQUIPMENT:-

- Modules ACT-01, ACT-02, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Perform the procedure as done in B (1)
- Connect post **AGC 1** to post **AGC 2**, connecting the AGC in this way. Apply the central 455 KHz-frequency to IF IN. Vary the amplitude of the input IF signal and measure with a voltmeter the voltage variation of AGC across post **AGC 2**. Note that when the input signal increases the negative value of the AGC voltage increases and then the gain of the first amplifier stage diminishes.

B3. AM Detector**EQUIPMENT:-**

- Modules ACT-01, ACT-02, ACT-03.
- Power supply.
- Oscilloscope.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Perform the procedure as done in B (1).
- Apply an AM signal, with carrier frequency at 455 KHz across IF IN. Examine the waveforms in the different posts of the circuit, and in particular the output of the envelope detector post **AGC 2**.

EXPERIMENT NO.9

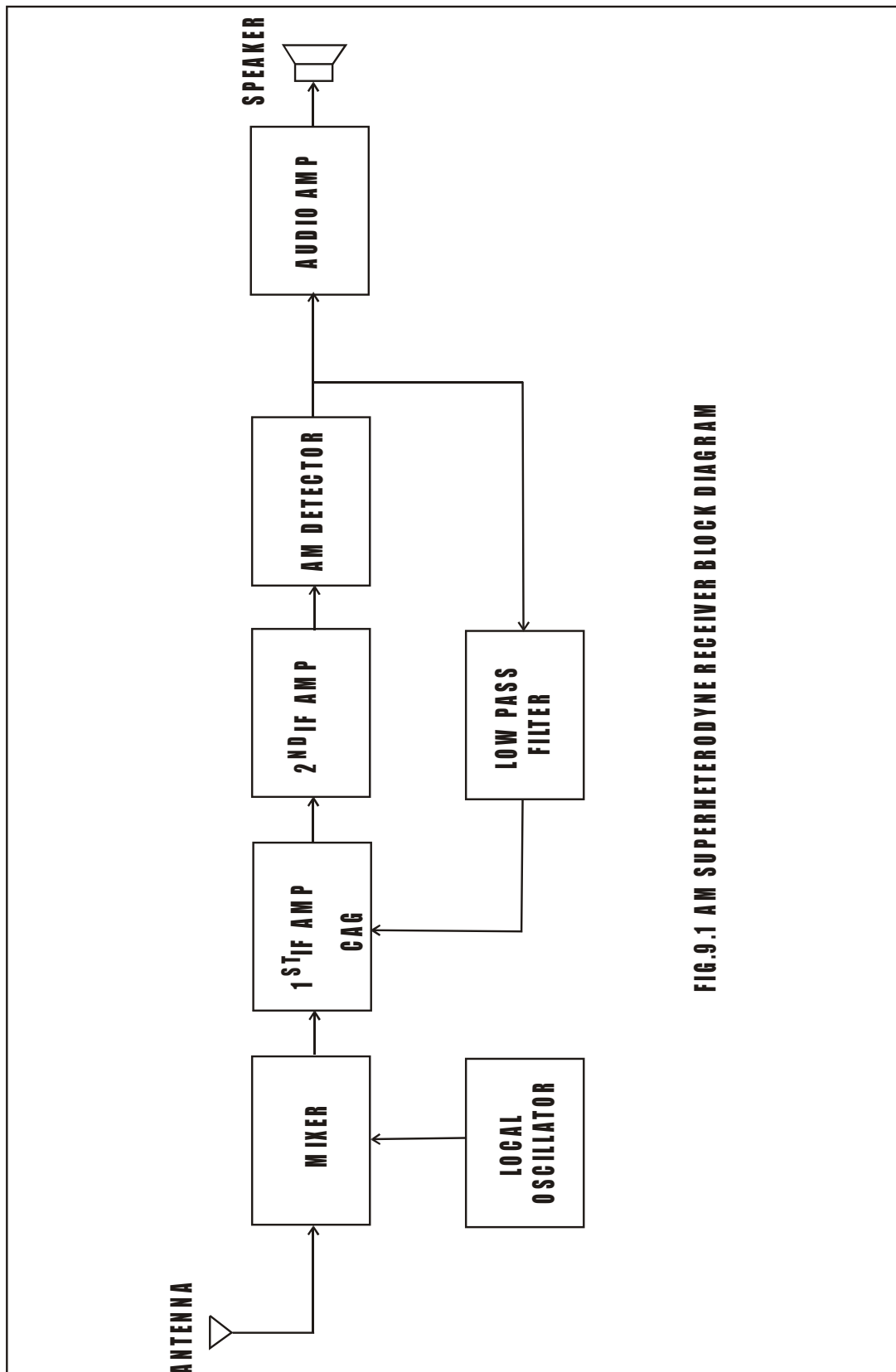


FIG.9.1 AM SUPERHETERODYNE RECEIVER BLOCK DIAGRAM

EXPERIMENT NO.9

NAME:-

AM Super Heterodyne Receiver

OBJECTIVE:-

To study an AM Super-heterodyne Receiver

THEORY:-

The block diagram of a simple superheterodyne AM receiver is reported in fig. 9.1. The RF signal modulated in amplitude enters from the antenna and is applied to a mixer stage, which is also fed by the oscillation generated by the local oscillator. The signal is converted to a lower frequency (intermediate frequency IF) and amplified by a 2-stage selective amplifier. The next detector demodulates the AM signal, supplying the audio information associated to it. The detected signal seems to be composed of the low frequency modulating signal and a continuous component proportional to the amplitude of the IF signals. Only the low frequency component is sent to the next audio amplifier. The only continuous component, obtained by filtering the detected signal, constitutes the AGC voltage (Automatic Gain Control) and is used to vary the amplification of the first IF stage. If the IF signal has a high amplitude the amplification is reduced, if the IF signal is low the amplification is increased.

EQUIPMENT:-

- Modules ACT-01, ACT-02, ACT-03.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01, ACT-02 & ACT-03 and switch it on.
- Refer to the fig.9.2 & Carry out the following connections and switch settings.

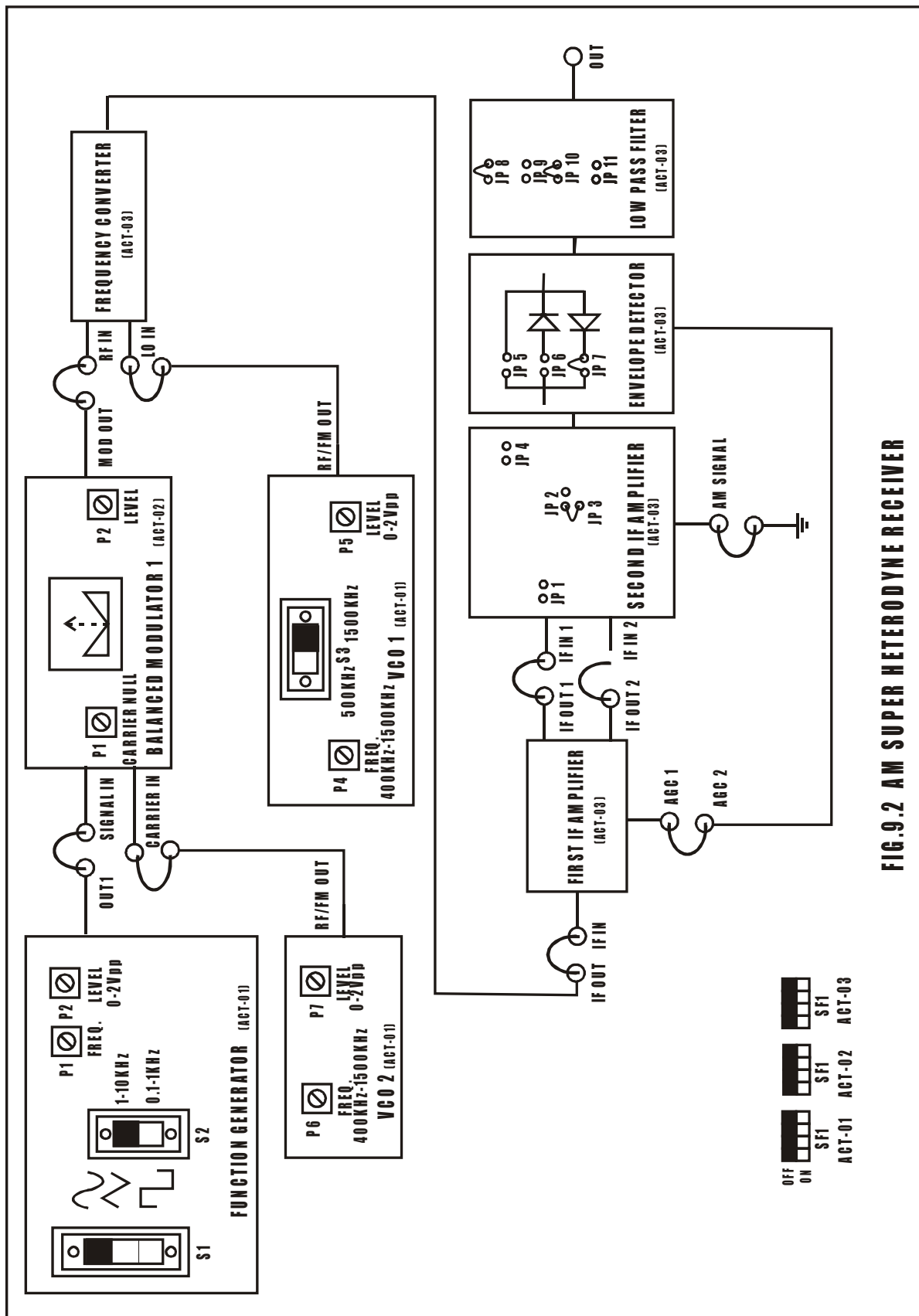


FIG.9.2 AM SUPER HETERODYNE RECEIVER

- In ACT-01
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).
 - Using pot **P1** keep sine wave signal frequency at **1 KHz**
 - Using pot **P2** keep sine wave signal amplitude at **0.5Vpp**.
 - Keep switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **1355 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **0.5Vpp**.
 - Using pot **P6** keep VCO2 frequency at **900 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **200mVpp**.
- In ACT-02
 - Keep BALANCED MODULATOR1 **CARRIER NULL** completely clockwise or counter clockwise position using pot **P1** to unbalance the modulator.
 - Keeps BALANCED MODULATOR1 **LEVEL** at center using pot **P2**.
- Connect the Jumpers JP3, JP7, JP8, and JP10 and keep jumpers JP1, JP2, JP4, JP5, JP6, JP9 & JP11 open of ACT-03 (To obtain IF Amplifier/Detector as shown in circuit 9.3).
- Connect the output of Function Generator (ACT-01) **OUT** post to the input of Balance modulator 1 (ACT-02) **SIGNAL IN** post.
- Connect the output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Balance modulator 1 **CARRIER IN** post.
- Connect the output of Balance modulator 1 (ACT-02) **OUT** post to the input of Frequency Converter (ACT-03) **RF IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **LO IN** post.
- Connect the output of Frequency Converter (ACT-03) **IF OUT** post to the input of 1st IF (ACT-03) **IF IN** post.
- Connect the output of 1st IF (ACT-03) **IF OUT 1** post to the input of 2nd IF (ACT-03) **IF IN 1** post and connect **AM SIGNAL** post to **GND** post.
- Connect post **IF OUT 2** of 1st IF (ACT-03) to post **IF IN 2** of 2nd IF (ACT-03).
- Connect post **AGC 1** of 1st IF (ACT-03) to post **AGC 2** of 2nd IF (ACT-03). This establishes Automatic Gain Control.

Calibration of the IF Stages

- Set the amplitude of the modulating signal to zero. Connect the oscilloscope to the output of the mixer (post **IF OUT** of ACT-03). Eventually adjust the variable capacity of the IF filter to obtain the maximum amplitude (about 50mVpp) of the detected sine wave form.
- Connect the oscilloscope to the output of the IF amplifier (post **IF OUT** of ACT-03). Adjust the variable capacity C5 of the second IF stage, and eventually the IF transformers (accessible from the front side) to obtain the maximum amplitude of the detected sine wave form (about 2 Vpp).

Automatic Gain Control

- Connect the oscilloscope to posts **RF IN** and **IF OUT** of ACT-03.
- Vary the amplitude of the RF input signal to the receiver (post **RF IN** of ACT-03) from 10 to about 100 mVpp, and check that the amplitude of the IF signal (post **IF OUT**) keeps almost constant.

- Note that when the RF signal increases the CAG voltage across post **IF OUT1** diminishes. Consequently there is an amplification of the first IF stage, and the output amplitude of the same IF signal is in this way kept almost constant.
- Disconnect the CAG (chord **AGC 1** to **AGC 2**) and connect post **AGC 1** to ground. Check that now amplitude of the IF signal follows the variations of the RF input signal.

Analysis of the waveforms

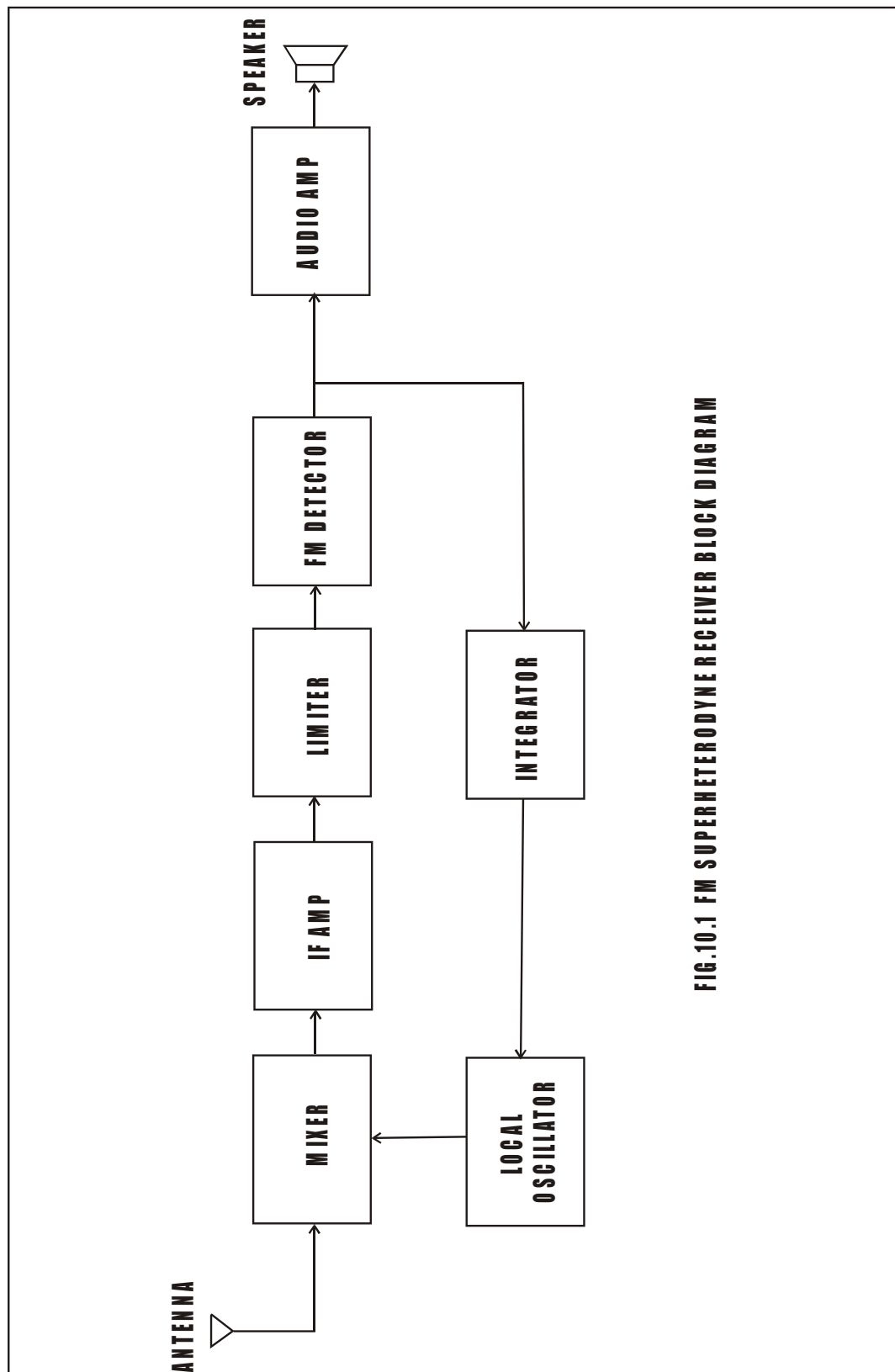
- Introduce a modulation in the RF carrier (increase the LEVEL of the function generator of ACT-01).
- Observe the AM signal before and after the mixer (posts **RF IN** and **IF OUT** of ACT-03). Check that the carrier frequencies are different.
- Connect the oscilloscope before and after the detector diode (posts **IF OUT** and **OUT** of ACT-03) and measure the AM signal and the detected signal.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 2 of SF 1 in ACT-01 Switch fault section to ON position. This will open the resistor R56A in VCO 1. When frequency selector switch S3 is in 500 KHz position, 400KHz to 500KHz variation disabled and output gives fixed frequency.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 2 of SF 1 in ACT-02 Switch fault section to ON position. This will open the center point of carrier null pot in balance modulator 1. Balance modulator 1 output is double sideband with suppressed carrier only.
- Put switch 1 of SF 1 in ACT-03 Switch fault section to ON position. This will open C17 from FET from frequency converter section. Frequency converter output gets reduced this will disturb the next section output.
- Put switch 2 of SF 1 in ACT-03 Switch fault section to ON position. This will open C7 from 2nd IF section low pass filter, filter output gets distorted.
- Put switch 3 of SF 1 in ACT-03 Switch fault section to ON position. This will open R41 Bias resistor of Transistor T2 from GND side in 1st IF section. 1st IF section output reduces.
- Put switch 4 of SF 1 in ACT-03 Switch fault section to ON position. This will open C4 of 2nd IF section resonance frequency changes from 450 KHz to 600KHz.

EXPERIMENT NO.10

**FIG.10.1 FM SUPERHETERODYNE RECEIVER BLOCK DIAGRAM**

EXPERIMENT NO.10

NAME:-

FM Super Heterodyne Receiver

OBJECTIVE:-

To study FM Super-heterodyne Receiver

THEORY:-

The block diagram of a simple superheterodyne FM receiver is shown in fig. 10.1. The RF signal modulated in frequency received from the antenna is applied to a mixer stage where another signal is local oscillator output. The signal is converted to lower frequency (intermediate frequency IF, 455 KHz in this case) and amplified. It is then applied to a limiter stage, which removes the amplitude variations contained in the FM signal. The next Detector demodulates the FM signal, supplying the audio information associated to it.

The detected signal result is composed of low frequency modulating signal and continuous component, proportional to the shift between the carrier frequency of the FM signal (after the intermediate frequency conversion) and the frequency to which the Detector is calibrated. Only the low frequency component is integrated and used to control the frequency of the local oscillator, so that an intermediate frequency equal to the central frequency of the Detector is obtained.

EQUIPMENT:-

- Modules ACT-01, ACT-03, ACT-04.
- Power supply.
- Oscilloscope.
- Frequency meter.
- Connecting Links.

Note: keep all the switches in switch fault section in off position

PROCEDURE:-

- Connect the power supply with proper polarity to the kits ACT-01, ACT-03 & ACT-04 and switch it on.
- Refer to the fig.10.2 & Carry out the following connections and switch settings.
- In ACT-01:
 - Keep switch **S1** in sine wave position.
 - Keep switch **S2** in upper position (**1-10 KHz**).

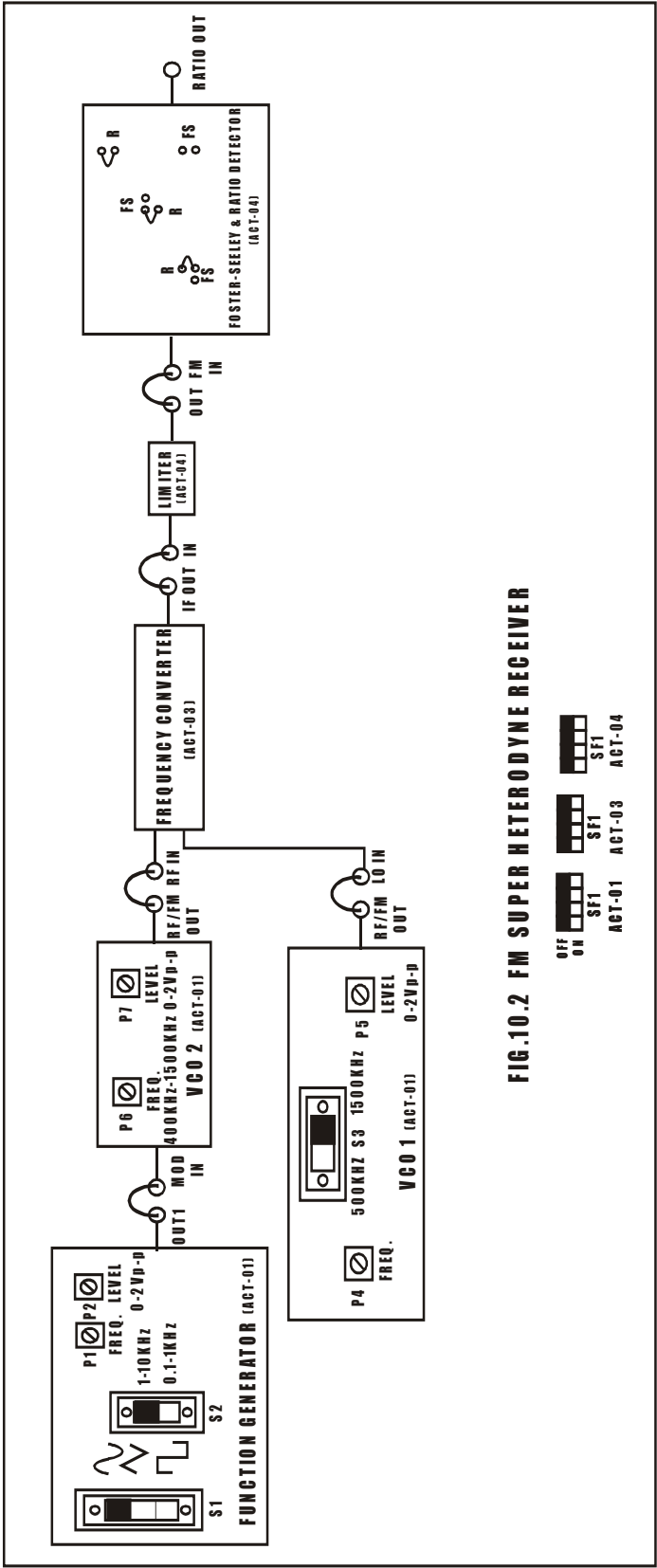


FIG.10.2 FM SUPER HETERODYNE RECEIVER

- Using pot **P1** keep sine wave signal frequency at **1 KHz**
 - Using pot **P2** keep sine wave signal amplitude at **200mVpp**.
 - Keep Switch **S3** in **1500 KHz** position.
 - Using pot **P4** keep VCO1 frequency at **900 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
 - Using pot **P6** keep VCO2 frequency at **450 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **2Vpp**.
- Connect the output of Function Generator (ACT-01) **OUT1** post to the input of VCO2 **MOD IN** post.
- Connect the output of VCO2 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **RF IN** post.
- Connect the output of VCO1 (ACT-01) **RF/FM OUT** post to the input of Frequency Converter (ACT-03) **LO IN** post.
- Connect the output of Frequency Converter (ACT-03) **IF OUT** post to the input of Limiter (ACT-04) **IN** post.
- Connect the ground post of Frequency Converter (ACT-03) to the ground post of Limiter (ACT-04).
- Connect the output of Limiter (ACT-04) **OUT** post to the input of Foster-Seeley & Ratio Detector (ACT-04) **FM IN** post.
- Set the Frequency DETECTOR (ACT-04) in Ratio mode (Jumpers in R position).

Calibration of the Intermediate Frequency Stages

- Connect the oscilloscope (probes 10:1) to the output of the mixer (post **IF OUT** of ACT-03). Adjust the variable capacity of the IF filter to obtain the larger response curve. Eventually adjust the coils (accessible on the front side) to center the IF filter to 455 KHz. Examine the waveform across the output of the limiter (post **OUT** of ACT-04) and observe that the amplitude of the FM signal now is uniform.
- Connect the oscilloscope to the output of the Detector (between **RATIO OUT** and ground). Reset the modulating signal (FUNCTION GENERATOR) and eventually adjust the variable capacity of the Detector to obtain a null voltage.

Examination of the waveforms of the Receiver

- Carry out the following presetting:
 - Using pot **P4** keep VCO1 frequency at **900 KHz**.
 - Using pot **P5** keep VCO1 amplitude at **2Vpp**.
 - Using pot **P6** keep VCO2 frequency at **450 KHz**.
 - Using pot **P7** keep VCO2 amplitude at **2Vpp**.
- Introduce a modulation into the RF signal (increase the LEVEL of the function generator of ACT-01 to about 0.2 Vpp.)
- Examine the FM signal before and after the mixer (posts **RF IN** and **IF OUT** of ACT-03). Check that the carrier frequencies are different.
- Examine the FM signal before and after the limiter (posts **IN** and **OUT** of ACT-04). Check that the amplitude of the FM signal is uniform across the output.
- Connect the oscilloscope to the output of the Detector (post **RATIO OUT** of ACT-04) and detect the signal.
- Note that the detected signal is distorted if the frequency of the RF signal is varied of about +/- 5 KHz in respect to the nominal value. This distortion is due to the fact that also the IF signal is moved in respect to the central frequency of the Detector

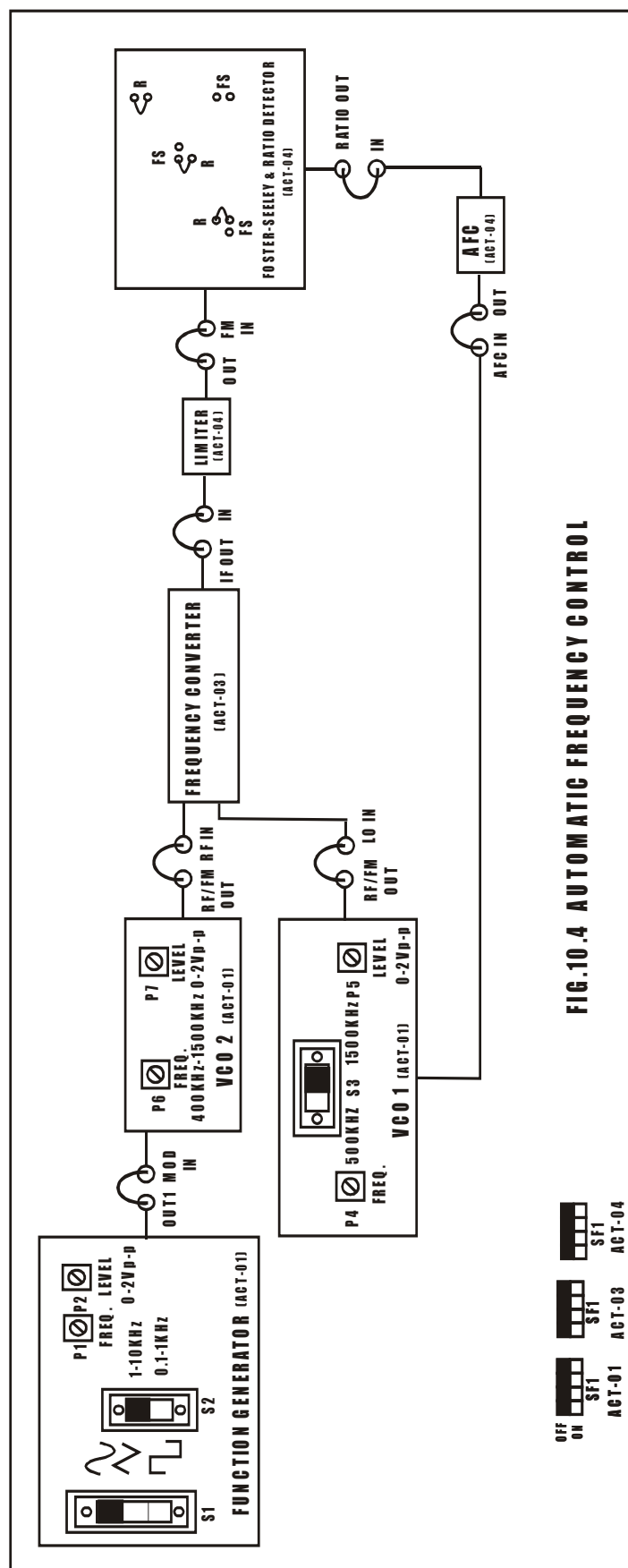


FIG.10.4 AUTOMATIC FREQUENCY CONTROL

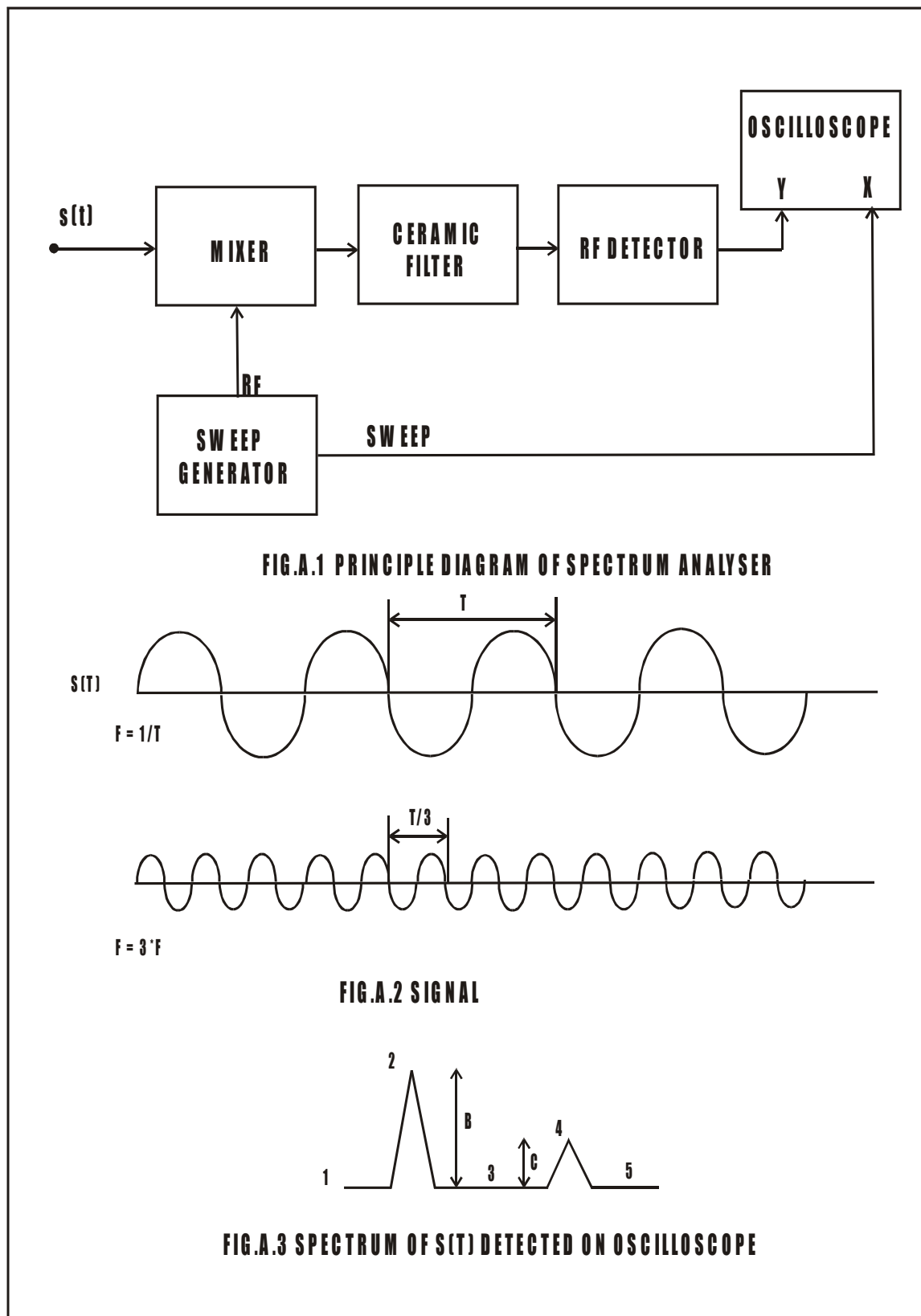
Automatic Frequency Control

- Introduce the Automatic Frequency Control (AFC) carrying out the connections of fig. 10.4. The AFC operation can be explained in the following way:
 - The Detector supplies d. c. voltage proportional to the difference between the Intermediate Nominal Frequency to which the Detector is centered (455 KHz) and the effective FI.
 - If the receiver is tuned to a slightly superior frequency than the proper one (local oscillator with higher frequency in respect to the nominal one), results superior to 455 KHz also the FI.
 - The d. c. voltage supplied by the Detector increases and so the output of the Detector diminishes.
 - The output of the integrator and the tuning voltage of VCO1 are added, and the result constitutes the effective biasing of the Varicap of the same VCO.
 - If the bias voltage tends to drop, consequently the frequency of the VCO1 drops too carrying out the tuning automatic regulation.
 - A similar operation occurs if the receiver is tuned to a slightly lower frequency to the correct one: the AFC acts so that the frequency of the local oscillator increases so that FI takes the nominal value.
- Note that the detected signal is with distortion although the frequency of the RF signal (VCO2) is varied of about +/-100KHz in respect to the nominal value. This means that the AFC automatically moves the frequency of this local oscillator VCO1, to keep almost constant (455 KHz of FI) the difference of frequency between RF signal and local oscillator.
- Connect the frequency meter to the local oscillator (post **LO IN** of ACT-03) and check how its frequency varies.

SWITCH FAULTS:-

Note: Keep the connections as per the procedure. Now put corresponding switch fault button in ON position & observe the different effect on the output. The faults are normally used one at a time.

- Put switch 1 of SF 1 in ACT-01 Switch fault section to ON position. This will open pin 1 of U2 (function generator), symmetry of sine wave disturbs Upper shape gets distorted.
- Put switch 3 of SF 1 in ACT-01 Switch fault section to ON position. This will open R41 bias resistor of U8 from GND side in VCO2. We get fix frequency at the output.
- Put switch 4 of SF 1 in ACT-01 Switch fault section to ON position. This will open R52 Bias resistor of transistor T4 from GND side in VCO2. We get limited amplitude variation at the output.
- Put switch 1 of SF 1 in ACT-03 Switch fault section to ON position. This will open C17 from FET from frequency converter section. Frequency converter output gets reduced this will disturb the next section output.
- Put switch 1 of SF 1 in ACT-04 Switch fault section to ON position. This will open C31 in Foster Seeley section, filtering effect is removed, and output is distorted.
- Put switch 2 of SF 1 in ACT-04 Switch fault section to ON position. This will open R38 from Limiter. Limiter action is disabled.



APPENDIX “A”

APPENDIX “A”

SPECTRUM ANALYZER

The block diagram of the circuit enabling the analysis of the spectrum of a signal is shown in fig. A.1

The signal $s(t)$, whose spectrum is to be analyzed, is sent to a mixer circuit together with a wobbled signal. Only when the difference of the frequencies present across the input of the mixer is equal to 455 KHz the resulting signal passes across the narrow band IF filter (Intermediate Frequency), and is applied to the diode envelope detector.

The output of the detector is taken to the Y-axis of the oscilloscope, so that each spectral component of $s(t)$ causes a vertical deviation on the screen.

The scan signal (SWEEP) of the wobbulator is applied to the X-axis of the oscilloscope, so a link is created between the horizontal movement of the trace and the instantaneous frequency of the wobbulator.

An example will better explain the operating principle. Suppose you want analyze the spectrum of signal $s(t)$ as the one of fig. A.2), where $F=100$ KHz and $F' = 300$ KHz. First of all, we must know the frequency interval occupied by the spectrum of $s(t)$, so that the operating frequencies of the wobbulator are known. In this case you will choose for the wobbulator a brushing of frequencies ranging from 500 to 800 KHz, so that a beating at 455 KHz is obtained with the components F and F' .

This occurs only when the instantaneous frequency F_v of the wobbulator takes the following values:

$$F_{v1} = 555 \text{ KHz.} \qquad F_{v2} = 755 \text{ KHz.}$$

In the first case you obtain an intermediate frequency $F_{if} = 455$ KHz by means of the beating with F :

$$F_{if} = F_{v1} - F \qquad 455 = 555 - 100 \text{ KHz}$$

In the second case you obtain an intermediate frequency F_{if} by means of the beating with F' :

$$F_{if} = F_{v2} - F' \qquad 455 = 755 - 300 \text{ KHz.}$$

The waveform detected on the oscilloscope (fig. A.3) is produced as follows:

- Consider that starting from the lower frequency of the wobbulator, equal to 500 KHz. The signal across the output of the IF filter is null, so also the signal applied to the Y-axis of the oscilloscope will be null. The scan signal (SWEEP) applied to the X-axis takes a minimum value, so we are in zone 1.
- Be $F_v = 555$ KHz. The beating caused by F_v and F supplies a signal with the detector. The Y-axis of the oscilloscope is reached by a voltage value proportional to the amplitude of the spectral component with frequency F . The scan signal is increased, by moving the trace to the right: we are in correspondence of peak 2.
- The zone 3 is analogous to 1; the scan signal is further increased by moving the trace more to the right.

-
- Be $F_v = 755$ KHz. The beating caused by F_v and f' supplies a signal with frequency of 455 KHz, which crosses the IF filter and is applied to the detector. The Y-axis of the oscilloscope is reached by a voltage level proportional to the amplitude of the spectral component with frequency F' . The scan has almost reached its maximum amplitude. We are in correspondence to peak 4.
 - The zone 5 is analogous to the 1, the scan signal reaches its maximum value and the trace concerns the extreme right on the screen.
 - The scan signal returns to its minimum value (to which corresponds $F_v = 500$ KHz) and the process starts from the beginning again.

A proper evaluation of the frequency and the amplitude of the spectral components detected on the screen, an external set-post signal must be available to calibrate the Y-axis in amplitude and the X-axis in frequency on the oscilloscope.